

1205.5935v1: formula evidence

1 row shows a REFINEMENT below, not the OCR reading: 1 verified against the source. Each is marked [refined] beside its identifier. The model is unchanged — the original reading is still what ‘latex’ holds; this report chose the refinement for it. Rows: 1205.5935v1_FO0422

The page, the confidence and the picture of an inline formula are its HOST LINE’s — a formula has none of its own. A line’s confidence is not a formula’s.

Inline formulas (first occurrence)

Identifier	Page	Conf.	LaTeX source	Rendered	Image
1205.5935v1_F00001	3	■ 1.000	u	u	objects that represent all the higher-dimensional subspaces. Given two vectors u and v , traditional vector
1205.5935v1_F00002	3	■ 1.000	v	v	objects that represent all the higher-dimensional subspaces. Given two vectors u and v , traditional vector
1205.5935v1_F00003	3	■ 1.000	$u \cdot v$	$u \cdot v$	where $u \cdot v$ is the inner product and $ u ^2 = u \cdot u$ is the square of the length of u . The cross product represents
1205.5935v1_F00004	3	■ 1.000	$ u ^2 = u \cdot u$	$ u ^2 = u \cdot u$	where $u \cdot v$ is the inner product and $ u ^2 = u \cdot u$ is the square of the length of u . The cross product represents
1205.5935v1_F00005	3	■ 1.000	$u \, v$	uv	I begin with a formal product of vectors uv that obeys the usual rules for multiplication; for example,
1205.5935v1_F00006	4	■ 1.000	$u^2 = u \cdot u = u ^2$	$u^2 = u \cdot u = u ^2$	I can immediately do something interesting with this: notice that $u^2 = u \cdot u = u ^2$, so the square of any
1205.5935v1_F00007	4	■ 1.000	$u \, u^{-1} = u^2 / u^2 = 1$	$uu^{-1} = u^2 / u^2 = 1$	is the multiplicative inverse of u , since obviously $uu^{-1} = u^2 / u^2 = 1$. So in a certain sense we can divide by
1205.5935v1_F00008	4	■ 0.997	$\wedge$	$\wedge$	with the symbol $\wedge$, so now the geometric product can be decomposed as
1205.5935v1_F00009	4	■ 1.000	$u \wedge v$	$u \wedge v$	To get some idea of what $u \wedge v$ is, I'll use the fact that it's antisymmetric in u and v , while $u \cdot v$ is symmetric,
1205.5935v1_F00010	4	■ 0.997	$v \, v = v ^2$	$vv = v ^2$	Now $vv = v ^2$, and the same is true for u , while $u \cdot v = u \, v \cos \theta$, so
1205.5935v1_F00011	4	■ 0.997	$u \cdot v = u \, v \cos \theta$	$u \cdot v = u \, v \cos \theta$	Now $vv = v ^2$, and the same is true for u , while $u \cdot v = u \, v \cos \theta$, so
1205.5935v1_F00012	4	■ 1.000	λ	λ	you change the sign of $u \wedge v$. There's no unique parallelogram associated with $u \wedge v$ because for any λ ,
1205.5935v1_F00013	5	■ 1.000	r	r	attributes of attitude, weight, and orientation. These r -blades and their sums, called <i>multivectors</i> , make up
1205.5935v1_F00014	5	■ 1.000	a	a	I'll start by solving two standard linear algebra problems. Let's suppose a plane is spanned by vectors a and
1205.5935v1_F00015	5	■ 1.000	b	b	b , and you have a known vector x in the plane that you want to expand in terms of a and b . Therefore you
1205.5935v1_F00016	5	■ 1.000	x	x	b , and you have a known vector x in the plane that you want to expand in terms of a and b . Therefore you
1205.5935v1_F00017	5	■ 1.000	α	α	want scalars α and β such that
1205.5935v1_F00018	5	■ 1.000	β	β	want scalars α and β such that
1205.5935v1_F00019	5	■ 1.000	$a \wedge a = 0$	$a \wedge a = 0$	To solve this, take the outer product of both sides with a ; since $a \wedge a = 0$, you get
1205.5935v1_F00020	5	■ 1.000	$a \wedge x$	$a \wedge x$	This makes sense geometrically: both $a \wedge x$ and $a \wedge b$ are bivectors in the same plane, so one should be a
1205.5935v1_F00021	5	■ 1.000	$a \wedge b$	$a \wedge b$	This makes sense geometrically: both $a \wedge x$ and $a \wedge b$ are bivectors in the same plane, so one should be a
1205.5935v1_F00022	5	■ 0.948	$(a \wedge x)(a \wedge b)^{-1}$	$(a \wedge x)(a \wedge b)^{-1}$	The problem with this is that it could mean either $(a \wedge x)(a \wedge b)^{-1}$ or $(a \wedge b)^{-1}(a \wedge x)$; but in this case
1205.5935v1_F00023	5	■ 0.948	$(a \wedge b)^{-1}(a \wedge x)$	$(a \wedge b)^{-1}(a \wedge x)$	The problem with this is that it could mean either $(a \wedge x)(a \wedge b)^{-1}$ or $(a \wedge b)^{-1}(a \wedge x)$; but in this case
1205.5935v1_F00024	5	■ 1.000	$\alpha = (x \wedge b) / (a \wedge b)$	$\alpha = (x \wedge b) / (a \wedge b)$	$\alpha = (x \wedge b) / (a \wedge b)$, so now we know that
1205.5935v1_F00025	5	■ 1.000	$a \wedge b \neq 0$	$a \wedge b \neq 0$	defined iff $a \wedge b \neq 0$, which is exactly the condition that a and b span the plane.
1205.5935v1_F00026	5	■ 0.960	a, b	a, b	The generalization from planes to volumes is straightforward; if a , b , and c span the space then
1205.5935v1_F00027	5	■ 0.960	c	c	The generalization from planes to volumes is straightforward; if a , b , and c span the space then
1205.5935v1_F00028	5	■ 1.000	p	p	plane: The first passes through point p and points in direction a , while the second passes through point q
1205.5935v1_F00029	5	■ 1.000	q	q	plane: The first passes through point p and points in direction a , while the second passes through point q
1205.5935v1_F00030	6	■ 1.000	$a \wedge x = a \wedge p$	$a \wedge x = a \wedge p$	for some λ . That means that $a \wedge x = a \wedge p$. And the fact that x lies on the line through q pointing along b
1205.5935v1_F00031	6	■ 0.999	$x \wedge b = q \wedge b$	$x \wedge b = q \wedge b$	tells me that $x \wedge b = q \wedge b$, so when I put all this in Eq. (16) I find that the intersection point x is
1205.5935v1_F00032	6	■ 1.000	n	n	Let v be the vector we want to reflect and let n be a vector along the reflection axis. Then
1205.5935v1_F00033	6	■ 1.000	v'	v'	of v from n . (I'll bet you've never heard that term before.) Now let v' be the reflected vector; its component
1205.5935v1_F00034	6	■ 1.000	n^{-1}	n^{-1}	The appearance of both n and n^{-1} in Eq. (22) guarantees that the result depends neither on the weight
1205.5935v1_F00035	6	■ 1.000	θ	θ	Hamilton discovered a great way to perform rotations: To rotate through angle θ in a plane, perform two
1205.5935v1_F00036	6	■ 1.000	$\theta / 2$	$\theta / 2$	reflections in succession along any two axes in the plane, as long as (a) the angle between the axes is $\theta / 2$
1205.5935v1_F00037	7	■ 1.000	ϕ	ϕ	Figure 1: The vector v makes an angle ϕ with axis n . It is then reflected along n , producing vector v' , then
1205.5935v1_F00038	7	■ 1.000	m	m	along axis m , producing vector v'' . Notice that the angle between vectors v and v'' is θ , twice the angle
1205.5935v1_F00039	7	■ 1.000	v''	v''	along axis m , producing vector v'' . Notice that the angle between vectors v and v'' is θ , twice the angle
1205.5935v1_F00040	7	■ 0.516	$R = mn$	$R = mn$	where $R = mn$. R is an object called a <i>rotor</i> . Typically a rotor is the product of unit vectors, in which case
1205.5935v1_F00041	7	■ 0.516	R	R	where $R = mn$. R is an object called a <i>rotor</i> . Typically a rotor is the product of unit vectors, in which case
1205.5935v1_F00042	7	■ 1.000	$m = m^{-1}$	$m = m^{-1}$	$m = m^{-1}$ and $n = n^{-1}$, which means $R^{-1} = nm$.
1205.5935v1_F00043	7	■ 1.000	$n = n^{-1}$	$n = n^{-1}$	$m = m^{-1}$ and $n = n^{-1}$, which means $R^{-1} = nm$.
1205.5935v1_F00044	7	■ 1.000	$R^{-1} = nm$	$R^{-1} = nm$	$m = m^{-1}$ and $n = n^{-1}$, which means $R^{-1} = nm$.
1205.5935v1_F00045	7	■ 1.000	$\theta / 2, m \cdot n = \cos(\theta / 2)$	$\theta / 2, m \cdot n = \cos(\theta / 2)$	$\theta / 2, m \cdot n = \cos(\theta / 2)$ and $(n \wedge m)^2 = -\sin^2(\theta / 2)$. Therefore the bivector $B = (n \wedge m) / \sin(\theta / 2)$ is a unit

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1285.5935v1_F08046	7	■1.000	(n \wedge m)^{2}=-\sin ^{2}(\theta / 2)	$(n \wedge m)^2 = -\sin^2(\theta/2)$	$\theta/2$, $m \cdot n = \cos(\theta/2)$ and $(n \wedge m)^2 = -\sin^2(\theta/2)$. Therefore the bivector $B = (n \wedge m)/\sin(\theta/2)$ is a unit
1285.5935v1_F08047	7	■1.000	B=(n \wedge m) / \sin (\theta / 2)	$B = (n \wedge m)/\sin(\theta/2)$	$\theta/2$, $m \cdot n = \cos(\theta/2)$ and $(n \wedge m)^2 = -\sin^2(\theta/2)$. Therefore the bivector $B = (n \wedge m)/\sin(\theta/2)$ is a unit
1285.5935v1_F08048	7	■1.000	B^{2}=-1	$B^2 = -1$	bivector: $B^2 = -1$. So now
1285.5935v1_F08049	8	■1.000	B	B	where the exponential is defined by its power series; the scalar terms and the terms proportional to B can
1285.5935v1_F08050	8	■1.000	z^{\prime }=e^{i \theta } z	$z' = e^{i\theta} z$	formula for rotating a vector that looks a lot like the formula for rotating a complex number, $z' = e^{i\theta} z$,
1285.5935v1_F08051	8	■1.000	(a \wedge b)^{-1}	$(a \wedge b)^{-1}$	1. In my first two examples in Section 1.2, I used the inverse of a 2-blade, $(a \wedge b)^{-1}$, and I mentioned
1285.5935v1_F08052	8	■1.000	n v n^{-1}	nvn^{-1}	2. In Eq. (22), I multiplied three vectors, nvn^{-1} , and the result was also a vector. However, you could
1285.5935v1_F08053	8	■1.000	(v \wedge n) n^{-1}	$(v \wedge n)n^{-1}$	3. Then I multiplied a bivector by a vector, $(v \wedge n)n^{-1}$. What's that? In this case the result was another
1285.5935v1_F08054	10	■1.000	\mathbb{R}	$\mathbb{R}$	in either case. For example, all applications of geometric algebra I'm familiar with take the scalars to be $\mathbb{R}$,
1285.5935v1_F08055	10	■1.000	i	i	when you multiply by i ? In fact, the two notions of weight and orientation make sense only for real vector
1285.5935v1_F08056	10	■1.000	\mathcal{G}	$\mathcal{G}$	A <i>geometric algebra</i> is a set $\mathcal{G}$ with two composition laws, addition and multiplication (also called the
1285.5935v1_F08057	10	■1.000	A, B	A, B	A generic element of $\mathcal{G}$ is denoted by a capital Roman letter (A, B , etc.) and is called a <i>multivector</i> .
1285.5935v1_F08058	10	■0.874	\mathcal{G}_0	$\mathcal{G}_0$	Axiom 2. $\mathcal{G}$ contains a field $\mathcal{G}_0$ of characteristic zero which includes 0 and 1.
1285.5935v1_F08059	11	■0.558	\left(\lambda , \mu	$(\lambda, \mu$	denoted by lower case Greek letters (λ, μ , etc.). Being a field means that $\mathcal{G}_0$ is closed under addition and
1285.5935v1_F08060	11	■0.991	1+1=0	$1 + 1 = 0$	haven't ruled out $1 + 1 = 0$, which would be bad when I try to divide by 2. (Which I'll be doing frequently;
1285.5935v1_F08061	11	■1.000	\mathcal{G}_1	$\mathcal{G}_1$	Axiom 3. $\mathcal{G}$ contains a subset $\mathcal{G}_1$ closed under addition, and $\lambda \in \mathcal{G}_0, v \in \mathcal{G}_1$ implies $\lambda v = v\lambda \in \mathcal{G}_1$.
1285.5935v1_F08062	11	■1.000	\lambda \in \mathcal{G}_0, v \in \mathcal{G}_1	$\lambda \in \mathcal{G}_0, v \in \mathcal{G}_1$	Axiom 3. $\mathcal{G}$ contains a subset $\mathcal{G}_1$ closed under addition, and $\lambda \in \mathcal{G}_0, v \in \mathcal{G}_1$ implies $\lambda v = v\lambda \in \mathcal{G}_1$.
1285.5935v1_F08063	11	■1.000	\lambda v=v \lambda \in \mathcal{G}_1	$\lambda v = v\lambda \in \mathcal{G}_1$	Axiom 3. $\mathcal{G}$ contains a subset $\mathcal{G}_1$ closed under addition, and $\lambda \in \mathcal{G}_0, v \in \mathcal{G}_1$ implies $\lambda v = v\lambda \in \mathcal{G}_1$.
1285.5935v1_F08064	11	■0.592	\left(a, b, u, v\right.	$(a, b, u, v$	denoted by lower case Roman letters (a, b, u, v , etc.). The axioms imply that $\mathcal{G}_1$ obeys all the rules of a
1285.5935v1_F08065	11	■0.969	\mathbb{R}^3	$\mathbb{R}^3$	vector space $\mathbb{R}^3$ with the real numbers as scalars; the zero scalar is the number 0, but the zero vector is the
1285.5935v1_F08066	11	■0.992	0 v=0	$0v = 0$	2. $0v = 0$ for any vector v by Axiom 1.
1285.5935v1_F08067	11	■0.979	u\rfloor v	$u\rfloor v$	is called their <i>inner product</i> and is denoted either $u\rfloor v$ or $u\lfloor v$. It is symmetric and linear in both terms,
1285.5935v1_F08068	11	■0.979	u\lfloor v	$u\lfloor v$	is called their <i>inner product</i> and is denoted either $u\rfloor v$ or $u\lfloor v$. It is symmetric and linear in both terms,
1285.5935v1_F08069	11	■0.896	u \downharpoonleft v=0, u	$u\downharpoonleft v = 0, u$	Vectors u and v are said to be <i>orthogonal</i> if $u\rfloor v = 0$, u is a <i>unit</i> vector if $u^2 = \pm 1$, and u is <i>null</i> if $u^2 = 0$.
1285.5935v1_F08070	11	■0.896	u^2=\pm 1	$u^2 = \pm 1$	Vectors u and v are said to be <i>orthogonal</i> if $u\rfloor v = 0$, u is a <i>unit</i> vector if $u^2 = \pm 1$, and u is <i>null</i> if $u^2 = 0$.
1285.5935v1_F08071	11	■0.896	u^2=0	$u^2 = 0$	Vectors u and v are said to be <i>orthogonal</i> if $u\rfloor v = 0$, u is a <i>unit</i> vector if $u^2 = \pm 1$, and u is <i>null</i> if $u^2 = 0$.
1285.5935v1_F08072	11	■1.000	v^{-1}=v / v^2	$v^{-1} = v/v^2$	comment that if v is non-null, then v is invertible and $v^{-1} = v/v^2$.
1285.5935v1_F08073	11	■0.401	\lfloor	$\lfloor$	I have two inner products, $\rfloor$ and $\lfloor$, instead of just the usual $\cdot$, for reasons that won't be clear until
1285.5935v1_F08074	12	■1.000	r>1	$r > 1$	Now I'll name other elements of $\mathcal{G}$. Let $r > 1$; then an <i>r-blade</i> or <i>simple r-vector</i> is a product of r
1285.5935v1_F08075	12	■1.000	\mathcal{G}_r	$\mathcal{G}_r$	<i>trivectors</i> . The set of r -vectors is called $\mathcal{G}_r$. Notice that this definition of simple r -vectors uses the geometric
1285.5935v1_F08076	12	■1.000	A \in \mathcal{G}_r	$A \in \mathcal{G}_r$	From these definitions and Axiom 3 it follows that that if $A \in \mathcal{G}_r$ and λ is a scalar,
1285.5935v1_F08077	12	■0.998	0 \in \mathcal{G}_r	$0 \in \mathcal{G}_r$	is a vector space with $\mathcal{G}_0$ as scalars and (b) $0 \in \mathcal{G}_r$ for every r . So all the results I gave right after Axiom 3
1285.5935v1_F08078	12	■1.000	\mathcal{G}_0=\mathcal{G}_1	$\mathcal{G}_0 = \mathcal{G}_1$	Axiom 6. If $\mathcal{G}_0 = \mathcal{G}_1$, then $\mathcal{G} = \mathcal{G}_0$. Otherwise, $\mathcal{G}$ is the direct sum of all the $\mathcal{G}_r$.
1285.5935v1_F08079	12	■1.000	\mathcal{G}=\mathcal{G}_0	$\mathcal{G} = \mathcal{G}_0$	Axiom 6. If $\mathcal{G}_0 = \mathcal{G}_1$, then $\mathcal{G} = \mathcal{G}_0$. Otherwise, $\mathcal{G}$ is the direct sum of all the $\mathcal{G}_r$.
1285.5935v1_F08080	12	■1.000	A \in \mathcal{G}	$A \in \mathcal{G}$	algebra. When there are vectors around, the axiom says that every $A \in \mathcal{G}$ may be expressed one and only
1285.5935v1_F08081	12	■1.000	A=\sum_r A_r	$A = \sum_r A_r$	one way as $A = \sum_r A_r$ where $A_r \in \mathcal{G}_r$ and all but finitely many A_r vanish. Therefore, every $A \neq 0$ is either
1285.5935v1_F08082	12	■1.000	A_r \in \mathcal{G}_r	$A_r \in \mathcal{G}_r$	one way as $A = \sum_r A_r$ where $A_r \in \mathcal{G}_r$ and all but finitely many A_r vanish. Therefore, every $A \neq 0$ is either
1285.5935v1_F08083	12	■1.000	A_r	A_r	one way as $A = \sum_r A_r$ where $A_r \in \mathcal{G}_r$ and all but finitely many A_r vanish. Therefore, every $A \neq 0$ is either
1285.5935v1_F08084	12	■1.000	A \neq 0	$A \neq 0$	one way as $A = \sum_r A_r$ where $A_r \in \mathcal{G}_r$ and all but finitely many A_r vanish. Therefore, every $A \neq 0$ is either
1285.5935v1_F08085	12	■1.000	\left\langle r\right\rangle :\mathcal{G} \rightarrow \mathcal{G}_r	$\rangle_r : \mathcal{G} \rightarrow \mathcal{G}_r$	For each r , let the grade operator $\langle \rangle_r : \mathcal{G} \rightarrow \mathcal{G}_r$ project each $A \in \mathcal{G}$ onto its unique grade- r component.
1285.5935v1_F08086	12	■1.000	A	A	(a) A is an r -vector iff $A = \langle A \rangle_r$.
1285.5935v1_F08087	12	■1.000	A=\langle A\rangle_r	$A = \langle A \rangle_r$	(a) A is an r -vector iff $A = \langle A \rangle_r$.
1285.5935v1_F08088	12	■1.000	\langle A+B\rangle_r=\langle A\rangle_r+\langle B\rangle_r	$\langle A + B \rangle_r = \langle A \rangle_r + \langle B \rangle_r$	(b) $\langle A + B \rangle_r = \langle A \rangle_r + \langle B \rangle_r$.
1285.5935v1_F08089	12	■1.000	\langle \lambda A\rangle_r=\langle A \lambda\rangle_r=\lambda \langle A\rangle_r	$\langle \lambda A \rangle_r = \langle A \lambda \rangle_r = \lambda \langle A \rangle_r$	(c) $\langle \lambda A \rangle_r = \langle A \lambda \rangle_r = \lambda \langle A \rangle_r$.
1285.5935v1_F08090	12	■1.000	\left\langle A\right\rangle _s=\left\langle A\right\rangle _r\delta _{rs}	$\langle \langle A \rangle_r \rangle_s = \langle A \rangle_r \delta_{rs}$	(d) $\langle \langle A \rangle_r \rangle_s = \langle A \rangle_r \delta_{rs}$. (Thus the $\langle \rangle_r$ are independent projection operators.)
1285.5935v1_F08091	12	■1.000	\left\langle r\right\rangle	$\rangle_r$	(d) $\langle \langle A \rangle_r \rangle_s = \langle A \rangle_r \delta_{rs}$. (Thus the $\langle \rangle_r$ are independent projection operators.)
1285.5935v1_F08092	12	■1.000	\sum_r \langle A\rangle_r=A	$\sum_r \langle A \rangle_r = A$	(e) $\sum_r \langle A \rangle_r = A$ for any $A \in \mathcal{G}$. (Thus the $\langle \rangle_r$ are a complete set of projection operators.)
1285.5935v1_F08093	12	■1.000	\langle A\rangle_r=0	$\langle A \rangle_r = 0$	(f) $\langle A \rangle_r = 0$ if $r < 0$ for all $A \in \mathcal{G}$.
1285.5935v1_F08094	12	■1.000	r<0	$r < 0$	(f) $\langle A \rangle_r = 0$ if $r < 0$ for all $A \in \mathcal{G}$.

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1205.5935v1_F00095	12	■ 1.000	<code>\left\langle\right\rangle</code>	$\rangle$	Because we take the scalar part of multivectors so often, I will let $\langle \rangle$ mean $\langle \rangle_0$.
1205.5935v1_F00096	12	■ 1.000	<code>\left\langle\right\rangle_{\{0\}}</code>	$\rangle_0$	Because we take the scalar part of multivectors so often, I will let $\langle \rangle$ mean $\langle \rangle_0$.
1205.5935v1_F00097	12	■ 1.000	<code>a_{\{1\}}, a_{\{2\}}</code>	a_1, a_2	The notation A_r will usually mean that A_r is an r -vector. The exception is vectors: a_1, a_2 , etc., are
1205.5935v1_F00098	12	■ 1.000	<code>\langle A \rangle_{\langle A \rangle_r}</code>	$\langle A \rangle_r$	grade- r component of multivector A , which is more properly denoted $\langle A \rangle_r$, but that notation is cumbersome
1205.5935v1_F00099	13	■ 1.000	<code>\boldsymbol{A}_{\boldsymbol{r}}</code>	$\mathbf{A}_r$	$\mathbf{A}_r$ is an r -blade. The exceptions are scalars (0-blades) and vectors (1-blades). I want a special notation for
1205.5935v1_F00100	13	■ 1.000	<code>\lambda A=A \lambda</code>	$\lambda A = A \lambda$	Axiom 6 tells us that the relation $\lambda A = A \lambda$, which I proved above for any homogeneous multivector A ,
1205.5935v1_F00101	13	■ 1.000	<code>\mathcal{G}_{\{+\}}</code>	$\mathcal{G}_+$	of subspaces $\mathcal{G}_+$ and $\mathcal{G}_-$ consisting of the even-grade and odd-grade multivectors respectively. Since many
1205.5935v1_F00102	13	■ 1.000	<code>\mathcal{G}_{\{-}}</code>	$\mathcal{G}_-$	of subspaces $\mathcal{G}_+$ and $\mathcal{G}_-$ consisting of the even-grade and odd-grade multivectors respectively. Since many
1205.5935v1_F00103	13	■ 1.000	<code>(-1)^{\{r\}}</code>	$(-1)^r$	identities will contain factors of $(-1)^r$, they will only depend on whether the multivectors are even or odd.
1205.5935v1_F00104	13	■ 1.000	<code>A_{\{+\}}</code>	A_+	some of my notation to cover even and odd cases; the notations A_+ and A_- will mean that these objects
1205.5935v1_F00105	13	■ 1.000	<code>A_{\{-}}</code>	A_-	some of my notation to cover even and odd cases; the notations A_+ and A_- will mean that these objects
1205.5935v1_F00106	13	■ 1.000	<code>A,\langle A \rangle_{\{+\}}</code>	$A, \langle A \rangle_+$	have only even-grade or odd-grade terms, respectively, and for any multivector A , $\langle A \rangle_+$ (resp. $\langle A \rangle_-$) is the
1205.5935v1_F00107	13	■ 1.000	<code>\langle A \rangle_{\{-}}</code>	$\langle A \rangle_-$	have only even-grade or odd-grade terms, respectively, and for any multivector A , $\langle A \rangle_+$ (resp. $\langle A \rangle_-$) is the
1205.5935v1_F00108	13	■ 1.000	<code>\left\{a_{\{i\}}\right\}_{\{i=1,\ldots,r\}}</code>	$\{a_i\}_{i=1,\dots,r}$	2. To do this, I define the outer product of vectors $\{a_i\}_{i=1,\dots,r}$ to be their fully antisymmetrized product, or
1205.5935v1_F00109	13	■ 0.870	<code>\sigma</code>	σ	where σ is a permutation of 1 through r , $\text{sgn } \sigma$ is the sign of the permutation (1 for even and -1 for odd),
1205.5935v1_F00110	13	■ 0.870	<code>r,\operatorname{sgn}\sigma</code>	$r, \text{sgn } \sigma$	where σ is a permutation of 1 through r , $\text{sgn } \sigma$ is the sign of the permutation (1 for even and -1 for odd),
1205.5935v1_F00111	13	■ 0.996	<code>r!</code>	$r!$	and the sum is over all $r!$ possible permutations. If $r = 2$ this reduces to the outer product of two vectors
1205.5935v1_F00112	13	■ 0.996	<code>r=2</code>	$r = 2$	and the sum is over all $r!$ possible permutations. If $r = 2$ this reduces to the outer product of two vectors
1205.5935v1_F00113	13	■ 1.000	<code>a_{\{i\}}</code>	a_i	<i>Proof.</i> To begin, I'll show that if the a_i all anticommute then the outer product reduces to the
1205.5935v1_F00114	13	■ 1.000	<code>\left\{e_{\{i\}}\right\}_{\{i=1,\ldots,r\}}</code>	$\{e_i\}_{i=1,\dots,r}$	geometric product, so the result is an r -blade. Let $\{e_i\}_{i=1,\dots,r}$ anticommute with one another,
1205.5935v1_F00115	13	■ 1.000	<code>e_{\{i\}}</code>	e_i	In each term, the e_i can be reordered so they're in ascending numerical order, and each interchange
1205.5935v1_F00116	13	■ 1.000	<code>\operatorname{sgn}\sigma</code>	$\text{sgn } \sigma$	of two e_i introduces a minus sign. The end result is a factor of the form $\text{sgn } \sigma$, which cancels the
1205.5935v1_F00117	14	■ 0.358	<code>e_{\{1\}}\cdot e_{\{2\}}=0</code>	$e_1 \cdot e_2 = 0$	was already obvious from Eq. (6) since $e_1 \cdot e_2 = 0$, or $e_1 \rfloor e_2 = 0$ as I would say it now.
1205.5935v1_F00118	14	■ 0.358	<code>\left.e_{\{1\}}\right\rfloor e_{\{2\}}=0</code>	$e_1 \rfloor e_2 = 0$	was already obvious from Eq. (6) since $e_1 \cdot e_2 = 0$, or $e_1 \rfloor e_2 = 0$ as I would say it now.
1205.5935v1_F00119	14	■ 0.553	<code>M</code>	M	M with entries $M_{ij} = a_i \rfloor a_j$. This is a real symmetric matrix, so it can be diagonalized by an
1205.5935v1_F00120	14	■ 0.553	<code>M_{\{i\}j}=a_{\{i\}}\downharpoonleft a_{\{j\}}</code>	$M_{ij} = a_i \downharpoonleft a_j$	M with entries $M_{ij} = a_i \rfloor a_j$. This is a real symmetric matrix, so it can be diagonalized by an
1205.5935v1_F00121	14	■ 0.679	<code>\operatorname{det}(R)=\pm 1</code>	$\det(R) = \pm 1$	Now $\det(R) = \pm 1$, and if it equals -1 I interchange e_1 and e_2 and relabel them, with the result
1205.5935v1_F00122	14	■ 0.679	<code>e_{\{1\}}</code>	e_1	Now $\det(R) = \pm 1$, and if it equals -1 I interchange e_1 and e_2 and relabel them, with the result
1205.5935v1_F00123	14	■ 0.679	<code>e_{\{2\}}</code>	e_2	Now $\det(R) = \pm 1$, and if it equals -1 I interchange e_1 and e_2 and relabel them, with the result
1205.5935v1_F00125	14	■ 1.000	<code>a_{\{1\}}\wedge a_{\{2\}}\wedge\cdots\wedge a_{\{r\}}=0</code>	$a_1 \wedge a_2 \wedge \cdots \wedge a_r = 0$	Theorem 2. The simple r -vector $a_1 \wedge a_2 \wedge \cdots \wedge a_r = 0$ iff the vectors $\{a_i\}_{i=1,\dots,r}$ are linearly dependent.
1205.5935v1_F00126	14	■ 1.000	<code>a\wedge\boldsymbol{A}_{\boldsymbol{r}}</code>	$a \wedge \mathbf{A}_r$	To solidify the connection between subspaces and r -blades, here's a really cool result. It uses $a \wedge \mathbf{A}_r$,
1205.5935v1_F00127	14	■ 1.000	<code>r\geq 1</code>	$r \geq 1$	Theorem 3. If $\mathbf{A}_r$ is a nonzero r -blade with $r \geq 1$, then vector a lies in the span of the factors of $\mathbf{A}_r$ iff
1205.5935v1_F00128	14	■ 1.000	<code>a\wedge\boldsymbol{A}_{\boldsymbol{r}}=0</code>	$a \wedge \mathbf{A}_r = 0$	$a \wedge \mathbf{A}_r = 0$.
1205.5935v1_F00129	15	■ 1.000	<code>a\wedge\boldsymbol{A}_{\boldsymbol{r}}\neq 0</code>	$a \wedge \mathbf{A}_r \neq 0$	The proof of this theorem actually shows a bit more. If $a \wedge \mathbf{A}_r \neq 0$, then it's an $r + 1$ -blade, and it
1205.5935v1_F00130	15	■ 1.000	<code>r+1</code>	$r + 1$	The proof of this theorem actually shows a bit more. If $a \wedge \mathbf{A}_r \neq 0$, then it's an $r + 1$ -blade, and it
1205.5935v1_F00131	15	■ 0.998	<code>\boldsymbol{B}_{\boldsymbol{r}}</code>	$\mathbf{B}_r$	Theorem 4. Two nonzero r -blades $\mathbf{A}_r$ and $\mathbf{B}_r$ define the same subspace iff each is a nonzero multiple of
1205.5935v1_F00132	15	■ 0.999	<code>\boldsymbol{A}_{\boldsymbol{r}}=\lambda\boldsymbol{B}_{\boldsymbol{r}}</code>	$\mathbf{A}_r = \lambda \mathbf{B}_r$	<i>Proof.</i> If $\mathbf{A}_r = \lambda \mathbf{B}_r$ for some nonzero λ , then clearly $a \wedge \mathbf{A}_r = 0$ iff $a \wedge \mathbf{B}_r = 0$, so they
1205.5935v1_F00133	15	■ 0.999	<code>a\wedge\boldsymbol{B}_{\boldsymbol{r}}=0</code>	$a \wedge \mathbf{B}_r = 0$	<i>Proof.</i> If $\mathbf{A}_r = \lambda \mathbf{B}_r$ for some nonzero λ , then clearly $a \wedge \mathbf{A}_r = 0$ iff $a \wedge \mathbf{B}_r = 0$, so they
1205.5935v1_F00134	15	■ 1.000	<code>\boldsymbol{A}_{\boldsymbol{r}}=a_1\wedge a_2\wedge\cdots\wedge a_r</code>	$\mathbf{A}_r = a_1 \wedge a_2 \wedge \cdots \wedge a_r$	$\mathbf{A}_r = a_1 \wedge a_2 \wedge \cdots \wedge a_r$ and $\mathbf{B}_r = b_1 \wedge b_2 \wedge \cdots \wedge b_r$ for some linearly independent sets of vectors
1205.5935v1_F00135	15	■ 1.000	<code>\boldsymbol{B}_{\boldsymbol{r}}=b_1\wedge b_2\wedge\cdots\wedge b_r</code>	$\mathbf{B}_r = b_1 \wedge b_2 \wedge \cdots \wedge b_r$	$\mathbf{A}_r = a_1 \wedge a_2 \wedge \cdots \wedge a_r$ and $\mathbf{B}_r = b_1 \wedge b_2 \wedge \cdots \wedge b_r$ for some linearly independent sets of vectors
1205.5935v1_F00136	15	■ 1.000	<code>\left\{b_{\{j\}}\right\}_{\{j=1,\ldots,r\}}</code>	$\{b_j\}_{j=1,\dots,r}$	$\{a_i\}_{i=1,\dots,r}$ and $\{b_j\}_{j=1,\dots,r}$, and each of the b_j is a linear combination of the a_i . Substituting
1205.5935v1_F00137	15	■ 1.000	<code>b_{\{j\}}</code>	b_j	$\{a_i\}_{i=1,\dots,r}$ and $\{b_j\}_{j=1,\dots,r}$, and each of the b_j is a linear combination of the a_i . Substituting
1205.5935v1_F00138	15	■ 1.000	<code>\boldsymbol{A}_{\boldsymbol{s}}</code>	$\mathbf{A}_s$	Theorem 5. If $\mathbf{A}_r$ represents a proper subspace of $\mathbf{A}_s$, then $\mathbf{A}_r$ can be factored out of $\mathbf{A}_s$; that is, there
1205.5935v1_F00139	15	■ 1.000	<code>\boldsymbol{A}_{\boldsymbol{s}-\boldsymbol{r}}</code>	$\mathbf{A}_{s-r}$	exists a blade $\mathbf{A}_{s-r}$ such that $\mathbf{A}_s = \mathbf{A}_r \wedge \mathbf{A}_{s-r}$.
1205.5935v1_F00140	15	■ 1.000	<code>\boldsymbol{A}_{\boldsymbol{s}}=\boldsymbol{A}_{\boldsymbol{r}}\wedge\boldsymbol{A}_{\boldsymbol{s}-\boldsymbol{r}}</code>	$\mathbf{A}_s = \mathbf{A}_r \wedge \mathbf{A}_{s-r}$	exists a blade $\mathbf{A}_{s-r}$ such that $\mathbf{A}_s = \mathbf{A}_r \wedge \mathbf{A}_{s-r}$.
1205.5935v1_F00141	15	■ 1.000	<code>\boldsymbol{A}_{\boldsymbol{r}}=a_1\wedge\cdots\wedge a_r</code>	$\mathbf{A}_r = a_1 \wedge \cdots \wedge a_r$	<i>Proof.</i> Let $\mathbf{A}_r = a_1 \wedge \cdots \wedge a_r$; then $\{a_j\}_{j=1,\dots,r}$ is a linearly independent set lying in $\mathbf{A}_s$, so it
1205.5935v1_F00142	15	■ 1.000	<code>\left\{a_{\{j\}}\right\}_{\{j=1,\ldots,r\}}</code>	$\{a_j\}_{j=1,\dots,r}$	<i>Proof.</i> Let $\mathbf{A}_r = a_1 \wedge \cdots \wedge a_r$; then $\{a_j\}_{j=1,\dots,r}$ is a linearly independent set lying in $\mathbf{A}_s$, so it
1205.5935v1_F00143	15	■ 1.000	<code>\left\{a_{\{j\}}\right\}_{\{j=1,\ldots,s\}}</code>	$\{a_j\}_{j=1,\dots,s}$	can be extended to a basis $\{a_j\}_{j=1,\dots,s}$ of $\mathbf{A}_s$. That means $a_1 \wedge \cdots \wedge a_s$ defines the same subspace
1205.5935v1_F00144	15	■ 1.000	<code>a_{\{1\}}\wedge\cdots\wedge a_{\{s\}}</code>	$a_1 \wedge \cdots \wedge a_s$	can be extended to a basis $\{a_j\}_{j=1,\dots,s}$ of $\mathbf{A}_s$. That means $a_1 \wedge \cdots \wedge a_s$ defines the same subspace
1205.5935v1_F00145	15	■ 0.996	<code>A_{\{r\}}\wedge B_{\{s\}}=(-1)^{\{r\}s}B_{\{s\}}\wedge A_{\{r\}}</code>	$A_r \wedge B_s = (-1)^{rs} B_s \wedge A_r$	I'll show later that $A_r \wedge B_s = (-1)^{rs} B_s \wedge A_r$ for any r - and s -vectors, so $\mathbf{A}_r$ can be factored out of $\mathbf{A}_s$ from
1205.5935v1_F00146	15	■ 0.996	<code>s</code>	s	I'll show later that $A_r \wedge B_s = (-1)^{rs} B_s \wedge A_r$ for any r - and s -vectors, so $\mathbf{A}_r$ can be factored out of $\mathbf{A}_s$ from

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1205.5935v1_F00147	15	■ 0.537	<code>a\rfloor \boldsymbol{A}_{\{\boldsymbol{r}\}}</code>	$a \rfloor \mathbf{A}_r$	subspace. This theorem uses $a \rfloor \mathbf{A}_r$, the left inner product of a vector and an r -blade, which once again I
1205.5935v1_F00148	15	■ 0.713	<code>r \geq 1, a</code>	$r \geq 1, a$	Theorem 6. If $\mathbf{A}_r$ is a nonzero r -blade with $r \geq 1$, a is orthogonal to the subspace $\mathbf{A}_r$ iff $a \rfloor \mathbf{A}_r = 0$.
1205.5935v1_F00149	15	■ 0.713	<code>\rfloor \boldsymbol{A}_{\{\boldsymbol{r}\}}=0</code>	$\rfloor \mathbf{A}_r = 0$	Theorem 6. If $\mathbf{A}_r$ is a nonzero r -blade with $r \geq 1$, a is orthogonal to the subspace $\mathbf{A}_r$ iff $a \rfloor \mathbf{A}_r = 0$.
1205.5935v1_F00150	15	■ 1.000	<code>a_{\{j\}}</code>	a_j	where I used Eq. (77) in the second line. (I'll derive it later. The check over a_j means it's
1205.5935v1_F00151	15	■ 0.528	<code>a\rfloor \boldsymbol{A}_{\{\boldsymbol{r}\}}=0</code>	$a \rfloor \mathbf{A}_r = 0$	and $a \rfloor \mathbf{A}_r = 0$. If instead a is orthogonal to all the a_j but one, then $a \rfloor \mathbf{A}_r$ is the product of
1205.5935v1_F00152	15	■ 0.528	<code>a \downharpoonleft \boldsymbol{A}_{\{\boldsymbol{r}\}}</code>	$a \downharpoonleft \mathbf{A}_r$	and $a \rfloor \mathbf{A}_r = 0$. If instead a is orthogonal to all the a_j but one, then $a \rfloor \mathbf{A}_r$ is the product of
1205.5935v1_F00153	15	■ 0.986	<code>a \downharpoonleft \boldsymbol{A}_{\{\boldsymbol{r}\}} \neq 0</code>	$a \downharpoonleft \mathbf{A}_r \neq 0$	linearly independent. So $a \rfloor \mathbf{A}_r \neq 0$. The remaining case is a nonorthogonal to multiple a_j , in
1205.5935v1_F00154	15	■ 0.973	<code>1< s \leq r</code>	$1 < s \leq r$	which case let $\{a_j\}_{j=1,\dots,s}$ where $1 < s \leq r$ be the vectors for which $a \rfloor a_j \neq 0$, and for $j = 2, \dots, s$
1205.5935v1_F00155	15	■ 0.973	<code>a \leq a_{\{j\}} \neq 0</code>	$a \leq a_j \neq 0$	which case let $\{a_j\}_{j=1,\dots,s}$ where $1 < s \leq r$ be the vectors for which $a \rfloor a_j \neq 0$, and for $j = 2, \dots, s$
1205.5935v1_F00156	15	■ 0.973	<code>j=2, \ldots, s</code>	$j = 2, \dots, s$	which case let $\{a_j\}_{j=1,\dots,s}$ where $1 < s \leq r$ be the vectors for which $a \rfloor a_j \neq 0$, and for $j = 2, \dots, s$
1205.5935v1_F00157	16	■ 0.978	<code>a \downharpoonleft b_{\{j\}}=0</code>	$a \downharpoonleft b_j = 0$	None of the b_j equal 0 because the a_j are linearly independent, and $a \rfloor b_j = 0$; better yet, because
1205.5935v1_F00158	16	■ 1.000	<code>a_{\{1\}}</code>	a_1	each of the b_j is just a_j with a multiple of a_1 added to it, the outer product is unchanged by
1205.5935v1_F00159	16	■ 0.999	<code>a \downharpoonleft \boldsymbol{A}_{\{\boldsymbol{r}\}}=0</code>	$a \downharpoonleft \mathbf{A}_r = 0$	$a \rfloor \mathbf{A}_r = 0$.
1205.5935v1_F00160	16	■ 0.728	<code>r-1</code>	$r - 1$	$r - 1$ -blade, and it represents the subspace of $\mathbf{A}_r$ that is orthogonal to a . I'll use this fact later when I
1205.5935v1_F00161	16	■ 0.679	<code>\boldsymbol{A}_{\{r\}}</code>	$\mathbf{A}_r$	So not only do we know that r -blade $\mathbf{A}_r$ represents an r -dimensional subspace, we have an easy way to
1205.5935v1_F00162	16	■ 1.000	<code>-a</code>	$-a$	moving along a , then b , then $-a$, then $-b$ back to the beginning. A little experimentation shows that
1205.5935v1_F00163	16	■ 1.000	<code>-b</code>	$-b$	moving along a , then b , then $-a$, then $-b$ back to the beginning. A little experimentation shows that
1205.5935v1_F00164	17	■ 0.512	<code>{ }^{\text{v}}</code>	$\mathbf{v}$	vector anyway with a check over it. For example, if $\{a_i\}_{i=1,\dots,r}$ is a collection of vectors, then $a_1 a_2 \cdots \tilde{a}_j \cdots a_r$
1205.5935v1_F00165	17	■ 0.512	<code>a_{\{1\}} a_{\{2\}} \cdots \check{a}_{\{j\}} \cdots a_{\{r\}}</code>	$a_1 a_2 \cdots \check{a}_j \cdots a_r$	vector anyway with a check over it. For example, if $\{a_i\}_{i=1,\dots,r}$ is a collection of vectors, then $a_1 a_2 \cdots \tilde{a}_j \cdots a_r$
1205.5935v1_F00166	17	■ 1.000	<code>r=1</code>	$r = 1$	calculate it the other way.) When $r = 1$, I recover the inner product of vectors defined earlier. An equivalent
1205.5935v1_F00167	17	■ 0.939	<code>a \downharpoonleft A_{\{r\}}</code>	$a \downharpoonleft A_r$	Theorem 7. $a \rfloor A_r$ and $A_r \rfloor a$ are both $r - 1$ -vectors, so the left or right inner product with a vector is a
1205.5935v1_F00168	17	■ 0.939	<code>A_{\{r\}} \lfloor a</code>	$A_r \rfloor a$	Theorem 7. $a \rfloor A_r$ and $A_r \rfloor a$ are both $r - 1$ -vectors, so the left or right inner product with a vector is a
1205.5935v1_F00169	17	■ 1.000	<code>a, a_{\{1\}}, a_{\{2\}}, \ldots, a_{\{r\}}</code>	$a, a_1, a_2, \dots, a_r$	<i>Proof.</i> To show this, I start by proving this relation: if $a, a_1, a_2, \dots, a_r$ are vectors, then
1205.5935v1_F00170	18	■ 1.000	<code>e_{\{1\}} e_{\{2\}} \cdots e_{\{r\}}</code>	$e_1 e_2 \cdots e_r$	a sum of r -blades, and any r -blade can be written $e_1 e_2 \cdots e_r$, so it follows that for a vector a and
1205.5935v1_F00171	18	■ 0.999	<code>\frac{1}{2}\left[a A_{\{r\}}-(-1)^{\{r\}} A_{\{r\}} a\right]</code>	$\frac{1}{2}[a A_r - (-1)^r A_r a]$	r -vector A_r the quantity $\frac{1}{2}[a A_r - (-1)^r A_r a]$, which is the left inner product, is an $r - 1$ -vector.
1205.5935v1_F00172	18	■ 1.000	<code>r=0</code>	$r = 0$	$r = 0$ (so $A_r = \lambda$) both inner products reduce to
1205.5935v1_F00173	18	■ 1.000	<code>A_{\{r\}}=\lambda</code>	$A_r = \lambda$	$r = 0$ (so $A_r = \lambda$) both inner products reduce to
1205.5935v1_F00174	18	■ 0.963	<code>a \wedge A_{\{r\}}</code>	$a \wedge A_r$	Theorem 8. $a \wedge A_r$ is an $r + 1$ -vector, so the outer product with a vector is a grade raising operation.
1205.5935v1_F00175	18	■ 1.000	<code>a_{\{1\}} \wedge a_{\{2\}} \wedge \cdots \wedge a_{\{r\}}=e_{\{1\}} e_{\{2\}} \cdots e_{\{r\}}</code>	$a_1 \wedge a_2 \wedge \cdots \wedge a_r = e_1 e_2 \cdots e_r$	of two vectors, so suppose it's true for $r - 1$. Let $a_1 \wedge a_2 \wedge \cdots \wedge a_r = e_1 e_2 \cdots e_r$ where any two e_i
1205.5935v1_F00176	18	■ 0.959	<code>(r+1)</code>	$(r + 1)$	sum of $(r + 1)!$ permutations. I want to regroup it into $r + 1$ terms, each of which consists of
1205.5935v1_F00177	19	■ 0.998	<code>\pi</code>	π	permutations that put a first be called π , and the ones that put e_i first be called π_i ; then
1205.5935v1_F00178	19	■ 0.998	<code>\pi_{\{i\}}</code>	π_i	permutations that put a first be called π , and the ones that put e_i first be called π_i ; then
1205.5935v1_F00179	19	■ 0.999	<code>a_{\{\pi_{\{j\}}(i)\}}</code>	$a_{\pi_j(i)}$	where one of the $a_{\pi_j(i)}$ is a and the others are the e_i other than e_j . Now the $e_{\pi(i)}$ in the first
1205.5935v1_F00180	19	■ 0.999	<code>e_{\{j\}}</code>	e_j	where one of the $a_{\pi_j(i)}$ is a and the others are the e_i other than e_j . Now the $e_{\pi(i)}$ in the first
1205.5935v1_F00181	19	■ 0.999	<code>e_{\{\pi(i)\}}</code>	$e_{\pi(i)}$	where one of the $a_{\pi_j(i)}$ is a and the others are the e_i other than e_j . Now the $e_{\pi(i)}$ in the first
1205.5935v1_F00182	19	■ 0.672	<code>\operatorname{sgn} \pi</code>	$\operatorname{sgn} \pi$	term on the right hand side can be rearranged, canceling the $\operatorname{sgn} \pi$ factors just as before, so
1205.5935v1_F00183	19	■ 1.000	<code>\pi_{\{j\}}</code>	π_j	As for the other terms, π_j as a permutation of a and all the e_i that puts e_j in the first spot
1205.5935v1_F00184	19	■ 1.000	<code>(-1)^{\{j\}}</code>	$(-1)^j$	$(-1)^j$, because this is the factor gained by moving e_j from its original $j + 1$ position to the
1205.5935v1_F00185	19	■ 1.000	<code>j+1</code>	$j + 1$	$(-1)^j$, because this is the factor gained by moving e_j from its original $j + 1$ position to the
1205.5935v1_F00186	19	■ 1.000	<code>j-1 e_{\{i\}}</code>	$j - 1 e_i$	term on the third line I moved e_j past the first $j - 1 e_i$. Now since
1205.5935v1_F00187	20	■ 1.000	<code>a_{\{1\}} \wedge a_{\{2\}} \wedge \cdots \wedge a_{\{r\}}</code>	$a_1 \wedge a_2 \wedge \cdots \wedge a_r$	and any r -blade is of the form $a_1 \wedge a_2 \wedge \cdots \wedge a_r$, so it follows that for a vector a and r -vector A_r
1205.5935v1_F00188	20	■ 1.000	<code>\frac{1}{2}\left[a A_{\{r\}}+(-1)^{\{r\}} A_{\{r\}} a\right]</code>	$\frac{1}{2}[a A_r + (-1)^r A_r a]$	the quantity $\frac{1}{2}[a A_r + (-1)^r A_r a]$, which is just the outer product, is an $r + 1$ -vector. And that's
1205.5935v1_F00189	20	■ 1.000	<code>A_{\{r\}} a</code>	$A_r a$	and similar results hold for $A_r a$:
1205.5935v1_F00190	20	■ 1.000	<code>B=\sum_{\{s\}} B_{\{s\}}</code>	$B = \sum_s B_s$	Let $A = \sum_r A_r$ and $B = \sum_s B_s$; then $AB = \sum_{r,s} A_r B_s$, so I'll consider each term separately.
1205.5935v1_F00191	20	■ 1.000	<code>A B=\sum_{\{r, s\}} A_{\{r\}} B_{\{s\}}</code>	$AB = \sum_{r,s} A_r B_s$	Let $A = \sum_r A_r$ and $B = \sum_s B_s$; then $AB = \sum_{r,s} A_r B_s$, so I'll consider each term separately.
1205.5935v1_F00192	20	■ 0.993	<code>A_{\{r\}} B_{\{s\}}</code>	$A_r B_s$	Theorem 9. $A_r B_s$ consists of $\min\{r, s\} + 1$ terms of grades $ r - s , r - s + 2, r - s + 4, \dots, r + s$, or
1205.5935v1_F00193	20	■ 0.993	<code>\min \{r, s\}+1</code>	$\min\{r, s\} + 1$	Theorem 9. $A_r B_s$ consists of $\min\{r, s\} + 1$ terms of grades $ r - s , r - s + 2, r - s + 4, \dots, r + s$, or
1205.5935v1_F00194	20	■ 0.993	<code> r-s , r-s +2, r-s +4, \ldots, r+s</code>	$ r - s , r - s + 2, r - s + 4, \dots, r + s$	Theorem 9. $A_r B_s$ consists of $\min\{r, s\} + 1$ terms of grades $ r - s , r - s + 2, r - s + 4, \dots, r + s$, or
1205.5935v1_F00195	20	■ 1.000	<code>s=0</code>	$s = 0$	<i>Proof.</i> If $r = 0$ or $s = 0$ this expression is obviously true, so next I'll consider the case
1205.5935v1_F00196	20	■ 1.000	<code>0< r \leq s</code>	$0 < r \leq s$	$0 < r \leq s$, assume A_r is an r -blade $\mathbf{A}_r$ (if it's true for a blade it's true for sums of blades), and
1205.5935v1_F00197	20	■ 0.988	<code>a \boldsymbol{A}_{\{\boldsymbol{r}\}}-\mathbf{1}</code>	$a \mathbf{A}_{r-1}$	so assume it's true for $r - 1$. $\mathbf{A}_r$ can be written $a \mathbf{A}_{r-1}$ where a is a vector and $\mathbf{A}_{r-1}$ is an
1205.5935v1_F00198	20	■ 0.988	<code>\boldsymbol{A}_{\{\boldsymbol{r}\}}-\mathbf{1}</code>	$\mathbf{A}_{r-1}$	so assume it's true for $r - 1$. $\mathbf{A}_r$ can be written $a \mathbf{A}_{r-1}$ where a is a vector and $\mathbf{A}_{r-1}$ is an
1205.5935v1_F00199	21	■ 1.000	<code>r \leq s</code>	$r \leq s$	follows from the inequality $r \leq s$. Now applying Eq. (64),

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1205.5935v1_F00200	21	■ 1.000	$s-r= r-s $, $\boldsymbol{A}_{\boldsymbol{r}}\boldsymbol{B}_s$	$s-r= r-s $, $\boldsymbol{A}_r\boldsymbol{B}_s$	Since $s-r= r-s $, $\boldsymbol{A}_r\boldsymbol{B}_s$ is now expressed as a sum of terms of grade $ r-s $, $ r-s +2$,
1205.5935v1_F00201	21	■ 1.000	$ r-s , r-s +2$	$ r-s , r-s +2$	Since $s-r= r-s $, $\boldsymbol{A}_r\boldsymbol{B}_s$ is now expressed as a sum of terms of grade $ r-s $, $ r-s +2$,
1205.5935v1_F00202	21	■ 1.000	$ r-s +4$, $\ldots$, $r+s$	$ r-s +4, \ldots, r+s$	$ r-s +4, \ldots, r+s$, which proves the result for r . The remaining case is $0<s\leq r$, which is
1205.5935v1_F00203	21	■ 1.000	$0<s\leq r$	$0<s\leq r$	$ r-s +4, \ldots, r+s$, which proves the result for r . The remaining case is $0<s\leq r$, which is
1205.5935v1_F00204	21	■ 1.000	$\boldsymbol{A}_{\mathbf{2}}=\boldsymbol{e}_1\boldsymbol{e}_2$	$\boldsymbol{A}_2=e_1e_2$	consider Eqs. (71) for the special case $r=2$, so $\boldsymbol{A}_2=e_1e_2$:
1205.5935v1_F00205	21	■ 0.999	$\boldsymbol{A}_{\mathbf{2}}\boldsymbol{B}_s=\boldsymbol{e}_1\boldsymbol{e}_2\boldsymbol{B}_s$	$\boldsymbol{A}_2\boldsymbol{B}_s=e_1e_2\boldsymbol{B}_s$	I could have arrived at the same result by writing $\boldsymbol{A}_2\boldsymbol{B}_s=e_1e_2\boldsymbol{B}_s$, using Eq. (64) to expand $e_2\boldsymbol{B}_s$, and using
1205.5935v1_F00206	21	■ 0.999	$\boldsymbol{e}_2\boldsymbol{B}_s$	$e_2\boldsymbol{B}_s$	I could have arrived at the same result by writing $\boldsymbol{A}_2\boldsymbol{B}_s=e_1e_2\boldsymbol{B}_s$, using Eq. (64) to expand $e_2\boldsymbol{B}_s$, and using
1205.5935v1_F00207	21	■ 1.000	$s-2$	$s-2$	Eq. (64) again to expand the product of e_1 with each term. The first term is of grade $s-2$, the middle two
1205.5935v1_F00208	21	■ 1.000	$\boldsymbol{B}_s$	$\boldsymbol{B}_s$	terms are of grade s (one grade lowering and one grade raising operation applied to $\boldsymbol{B}_s$), and the final term
1205.5935v1_F00209	21	■ 1.000	$s+2$	$s+2$	is of grade $s+2$.
1205.5935v1_F00210	21	■ 1.000	$r+s$	$r+s$	Theorem 9 tells us something important: while $\boldsymbol{A}_r\boldsymbol{B}_s$ is not an $r+s$ -vector, every term has grade $r+s-2j$
1205.5935v1_F00211	21	■ 1.000	$r+s-2j$	$r+s-2j$	Theorem 9 tells us something important: while $\boldsymbol{A}_r\boldsymbol{B}_s$ is not an $r+s$ -vector, every term has grade $r+s-2j$
1205.5935v1_F00212	21	■ 1.000	j	j	for some j , so $\boldsymbol{A}_r\boldsymbol{B}_s$ is even if $r+s$ is even and odd if $r+s$ is odd. That means that the product of two even
1205.5935v1_F00213	21	■ 1.000	$(-1)^{r+s-2j}=(-1)^{r+s}$	$(-1)^{r+s-2j}=(-1)^{r+s}$	because the product of two odd elements is also even.) Also, since $(-1)^{r+s-2j}=(-1)^{r+s}$, it follows that
1205.5935v1_F00214	22	■ 1.000	$\frac{1}{2}(-1)^r\boldsymbol{A}_ra\boldsymbol{B}_s$	$\frac{1}{2}(-1)^r\boldsymbol{A}_ra\boldsymbol{B}_s$	first of Eqs. (73) and then adding and subtracting $\frac{1}{2}(-1)^r\boldsymbol{A}_ra\boldsymbol{B}_s$,
1205.5935v1_F00215	22	■ 1.000	$j-1$	$j-1$	that a_j first has to be moved to the far left side of the outer product, which requires $j-1$ interchanges of
1205.5935v1_F00216	22	■ 1.000	$(-1)^{j-1}$	$(-1)^{j-1}$	adjacent vectors and introduces a factor of $(-1)^{j-1}$.
1205.5935v1_F00217	22	■ 1.000	$\boldsymbol{a}_1\boldsymbol{a}_2\cdots\boldsymbol{a}_r$	$a_1a_2\cdots a_r$	Theorem 12. An r -versor $a_1a_2\cdots a_r$ is a linear combination of terms, each of which is an outer product
1205.5935v1_F00218	22	■ 0.899	$r=0,1$	$r=0,1$	<i>Proof.</i> As usual, the proof is by induction. The result is true if $r=0,1$, or 2 , so assume it's
1205.5935v1_F00219	22	■ 1.000	$r-2$	$r-2$	of products of $r-2$ of the a_i , so by the $r-2$ result the first term is a linear combination of outer
1205.5935v1_F00220	23	■ 0.635	$\left.\boldsymbol{A}_r\right\rceil\left\lfloor\boldsymbol{B}_r\right\rceil=\boldsymbol{A}_r\left\lfloor\left\lfloor\boldsymbol{B}_r\right\rceil\right\rfloor=\left\lfloor\left\langle\boldsymbol{A}_r\right\rceil\boldsymbol{B}_r\right\rangle\right\rfloor$	$\boldsymbol{A}_r\rfloor\boldsymbol{B}_r=\boldsymbol{A}_r\lfloor\boldsymbol{B}_r=\langle\boldsymbol{A}_r\boldsymbol{B}_r\rangle$	1. $\boldsymbol{A}_r\rfloor\boldsymbol{B}_r=\boldsymbol{A}_r\lfloor\boldsymbol{B}_r=\langle\boldsymbol{A}_r\boldsymbol{B}_r\rangle$.
1205.5935v1_F00221	23	■ 0.995	$r>s$	$r>s$	2. If $r>s$ then $\boldsymbol{A}_r\rfloor\boldsymbol{B}_s=\boldsymbol{B}_s\lfloor\boldsymbol{A}_r=0$ because all negative-grade multivectors vanish.
1205.5935v1_F00222	23	■ 0.995	$\left.\boldsymbol{A}_r\right\rceil\left\lfloor\boldsymbol{B}_s\right\rceil=\boldsymbol{B}_s\left\lfloor\left\lfloor\boldsymbol{A}_r\right\rceil\right\rfloor=0$	$\boldsymbol{A}_r\rfloor\boldsymbol{B}_s=\boldsymbol{B}_s\lfloor\boldsymbol{A}_r=0$	2. If $r>s$ then $\boldsymbol{A}_r\rfloor\boldsymbol{B}_s=\boldsymbol{B}_s\lfloor\boldsymbol{A}_r=0$ because all negative-grade multivectors vanish.
1205.5935v1_F00223	23	■ 0.983	$\boldsymbol{A}_r\left\lfloor\left\lfloor\boldsymbol{B}_s\right\rceil\right\rfloor$	$\boldsymbol{A}_r\lfloor\boldsymbol{B}_s$	3. The lowest grade term in $\boldsymbol{A}_r\boldsymbol{B}_s$ is $\boldsymbol{A}_r\rfloor\boldsymbol{B}_s$ if $r\leq s$ and $\boldsymbol{A}_r\lfloor\boldsymbol{B}_s$ if $r\geq s$.
1205.5935v1_F00224	23	■ 0.983	$r\geq s$	$r\geq s$	3. The lowest grade term in $\boldsymbol{A}_r\boldsymbol{B}_s$ is $\boldsymbol{A}_r\rfloor\boldsymbol{B}_s$ if $r\leq s$ and $\boldsymbol{A}_r\lfloor\boldsymbol{B}_s$ if $r\geq s$.
1205.5935v1_F00225	23	■ 1.000	$\boldsymbol{A}_r\wedge\boldsymbol{B}_s$	$\boldsymbol{A}_r\wedge\boldsymbol{B}_s$	4. The highest grade term in $\boldsymbol{A}_r\boldsymbol{B}_s$ is $\boldsymbol{A}_r\wedge\boldsymbol{B}_s$.
1205.5935v1_F00226	23	■ 0.924	$\lambda\boldsymbol{A}=\lambda\left\lfloor\boldsymbol{A}=\lambda\wedge\boldsymbol{A}\right.$	$\lambda,\lambda\boldsymbol{A}=\lambda\rfloor\boldsymbol{A}=\lambda\wedge\boldsymbol{A}$	5. For any λ , $\lambda\boldsymbol{A}=\lambda\rfloor\boldsymbol{A}=\lambda\wedge\boldsymbol{A}$ and $\lambda\boldsymbol{A}=\lambda\lfloor\boldsymbol{A}=\lambda\wedge\boldsymbol{A}$. The product of a vector and any multivector
1205.5935v1_F00227	23	■ 0.924	$\boldsymbol{A}\left\lfloor\boldsymbol{A}=\boldsymbol{A}\left\lfloor\boldsymbol{A}=\boldsymbol{A}\wedge\boldsymbol{A}\right.$	$\lambda\boldsymbol{A}=\boldsymbol{A}\rfloor\boldsymbol{A}=\boldsymbol{A}\wedge\boldsymbol{A}$	5. For any λ , $\lambda\boldsymbol{A}=\lambda\rfloor\boldsymbol{A}=\lambda\wedge\boldsymbol{A}$ and $\lambda\boldsymbol{A}=\boldsymbol{A}\lfloor\boldsymbol{A}=\boldsymbol{A}\wedge\boldsymbol{A}$. The product of a vector and any multivector
1205.5935v1_F00228	23	■ 0.846	$\boldsymbol{B}_s\boldsymbol{A}_r$	$\boldsymbol{B}_s\boldsymbol{A}_r$	$\boldsymbol{B}_s\boldsymbol{A}_r$ are related, but not in a way that's easy to use, while going from $\boldsymbol{A}_r\rfloor\boldsymbol{B}_s$ to $\boldsymbol{B}_s\lfloor\boldsymbol{A}_r$ or from $\boldsymbol{A}_r\wedge\boldsymbol{B}_s$
1205.5935v1_F00229	23	■ 0.846	$\left.\boldsymbol{A}_r\right\rceil\left\lfloor\boldsymbol{B}_s\right\rceil$	$\boldsymbol{A}_r\rfloor\boldsymbol{B}_s$	$\boldsymbol{B}_s\boldsymbol{A}_r$ are related, but not in a way that's easy to use, while going from $\boldsymbol{A}_r\rfloor\boldsymbol{B}_s$ to $\boldsymbol{B}_s\lfloor\boldsymbol{A}_r$ or from $\boldsymbol{A}_r\wedge\boldsymbol{B}_s$
1205.5935v1_F00230	23	■ 0.846	$\boldsymbol{B}_s\left\lfloor\left\lfloor\boldsymbol{A}_r\right\rceil\right\rfloor$	$\boldsymbol{B}_s\lfloor\boldsymbol{A}_r$	$\boldsymbol{B}_s\boldsymbol{A}_r$ are related, but not in a way that's easy to use, while going from $\boldsymbol{A}_r\rfloor\boldsymbol{B}_s$ to $\boldsymbol{B}_s\lfloor\boldsymbol{A}_r$ or from $\boldsymbol{A}_r\wedge\boldsymbol{B}_s$
1205.5935v1_F00231	23	■ 1.000	$\boldsymbol{B}_s\wedge\boldsymbol{A}_r$	$\boldsymbol{B}_s\wedge\boldsymbol{A}_r$	to $\boldsymbol{B}_s\wedge\boldsymbol{A}_r$ is just a matter of a sign change (Eqs. (141) and (143)).
1205.5935v1_F00232	23	■ 1.000	$r+s-1$	$r+s-1$	The definitions also allow me to deduce three more identities from Eqs. (74); I take the $r+s-1$ -grade
1205.5935v1_F00233	23	■ 1.000	$r-s+1$	$r-s+1$	term of the first of Eqs. (74), the $r-s+1$ -grade term of the third, and the $s-r+1$ -grade term of the last,
1205.5935v1_F00234	23	■ 1.000	$s-r+1$	$s-r+1$	term of the first of Eqs. (74), the $r-s+1$ -grade term of the third, and the $s-r+1$ -grade term of the last,
1205.5935v1_F00235	24	■ 0.999	$\left\langle\boldsymbol{s}+\boldsymbol{t}\right\rangle$	$\rangle_{s+t}$	The crucial step is taken on the third line, where the $\langle\rangle_{s+t}$ is dropped. This can be done because
1205.5935v1_F00236	24	■ 1.000	$\boldsymbol{A}_r\boldsymbol{B}_s\boldsymbol{C}_t$	$\boldsymbol{A}_r\boldsymbol{B}_s\boldsymbol{C}_t$	the only term in $\boldsymbol{A}_r\boldsymbol{B}_s\boldsymbol{C}_t$ that has grade $r+s+t$ is the term that comes from multiplying $\boldsymbol{A}_r$ by
1205.5935v1_F00237	24	■ 1.000	$r+s+t$	$r+s+t$	the only term in $\boldsymbol{A}_r\boldsymbol{B}_s\boldsymbol{C}_t$ that has grade $r+s+t$ is the term that comes from multiplying $\boldsymbol{A}_r$ by
1205.5935v1_F00238	24	■ 1.000	$\boldsymbol{B}_s\boldsymbol{C}_t$	$\boldsymbol{B}_s\boldsymbol{C}_t$	the highest grade term in $\boldsymbol{B}_s\boldsymbol{C}_t$. From this it follows that $\boldsymbol{A}\wedge(\boldsymbol{B}\wedge\boldsymbol{C})=(\boldsymbol{A}\wedge\boldsymbol{B})\wedge\boldsymbol{C}$ for any $\boldsymbol{A}$,
1205.5935v1_F00239	24	■ 1.000	$\boldsymbol{A}\wedge(\boldsymbol{B}\wedge\boldsymbol{C})=(\boldsymbol{A}\wedge\boldsymbol{B})\wedge\boldsymbol{C}$	$\boldsymbol{A}\wedge(\boldsymbol{B}\wedge\boldsymbol{C})=(\boldsymbol{A}\wedge\boldsymbol{B})\wedge\boldsymbol{C}$	the highest grade term in $\boldsymbol{B}_s\boldsymbol{C}_t$. From this it follows that $\boldsymbol{A}\wedge(\boldsymbol{B}\wedge\boldsymbol{C})=(\boldsymbol{A}\wedge\boldsymbol{B})\wedge\boldsymbol{C}$ for any $\boldsymbol{A}$,
1205.5935v1_F00240	24	■ 1.000	$\boldsymbol{C}$	$\boldsymbol{C}$	$\boldsymbol{B}$, and $\boldsymbol{C}$. The third line also gives me Eq. (84). For the second relation, consider
1205.5935v1_F00241	24	■ 1.000	$r\leq s-t$	$r\leq s-t$	Now this vanishes automatically unless $r\leq s-t$, in which case
1205.5935v1_F00242	24	■ 1.000	$s-(r+t)$	$s-(r+t)$	because when the inequality is satisfied, the only term in $\boldsymbol{A}_r\boldsymbol{B}_s\boldsymbol{C}_t$ that has grade $s-(r+t)$ is
1205.5935v1_F00243	24	■ 1.000	$\boldsymbol{A}\left\lfloor(\boldsymbol{B}\left\lfloor\boldsymbol{C}\right\rceil)=(\boldsymbol{A}\left\rfloor\boldsymbol{B})\left\lfloor\boldsymbol{C}\right.\right.$	$\boldsymbol{A}\rfloor(\boldsymbol{B}\lfloor\boldsymbol{C})=(\boldsymbol{A}\rfloor\boldsymbol{B})\lfloor\boldsymbol{C}$	This expression also vanishes unless the inequality is satisfied, so $\boldsymbol{A}\rfloor(\boldsymbol{B}\lfloor\boldsymbol{C})=(\boldsymbol{A}\rfloor\boldsymbol{B})\lfloor\boldsymbol{C}$ in
1205.5935v1_F00244	25	■ 1.000	$r+s\leq t$	$r+s\leq t$	Now this vanishes automatically unless $r+s\leq t$, in which case logic similar to that used above
1205.5935v1_F00245	25	■ 0.930	$\boldsymbol{A}\left\rfloor(\boldsymbol{B}\left\rfloor\boldsymbol{C})=\right.$	$\boldsymbol{A}\rfloor(\boldsymbol{B}\rfloor\boldsymbol{C})=$	This final expression also vanishes unless the inequality is satisfied; therefore $\boldsymbol{A}\rfloor(\boldsymbol{B}\rfloor\boldsymbol{C})=$
1205.5935v1_F00246	25	■ 0.585	$(\boldsymbol{A}\wedge\boldsymbol{B})\left\rfloor\boldsymbol{C}\right.$	$(\boldsymbol{A}\wedge\boldsymbol{B})\rfloor\boldsymbol{C}$	$(\boldsymbol{A}\wedge\boldsymbol{B})\rfloor\boldsymbol{C}$ for all $\boldsymbol{A}$, $\boldsymbol{B}$, and $\boldsymbol{C}$. A similar proof shows that $(\boldsymbol{A}\lfloor\boldsymbol{B})\lfloor\boldsymbol{C}=\boldsymbol{A}\lfloor(\boldsymbol{B}\wedge\boldsymbol{C})$.
1205.5935v1_F00247	25	■ 0.585	$\boldsymbol{A}\left\lfloor\boldsymbol{B}\right\rfloor\boldsymbol{C}=\boldsymbol{A}\left\lfloor(\boldsymbol{B}\wedge\boldsymbol{C})\right.$	$\boldsymbol{A}\rfloor\boldsymbol{B}\rfloor\boldsymbol{C}=\boldsymbol{A}\rfloor(\boldsymbol{B}\wedge\boldsymbol{C})$	$(\boldsymbol{A}\wedge\boldsymbol{B})\rfloor\boldsymbol{C}$ for all $\boldsymbol{A}$, $\boldsymbol{B}$, and $\boldsymbol{C}$. A similar proof shows that $(\boldsymbol{A}\lfloor\boldsymbol{B})\lfloor\boldsymbol{C}=\boldsymbol{A}\lfloor(\boldsymbol{B}\wedge\boldsymbol{C})$.
1205.5935v1_F00248	25	■ 1.000	$\left.\boldsymbol{B}_s\right\rceil\left\lfloor\boldsymbol{A}_{\boldsymbol{r}}\right\rceil\left\lfloor\boldsymbol{B}_s\right\rceil$	$\boldsymbol{B}_s, \boldsymbol{A}_r\rfloor\boldsymbol{B}_s$	product, as I'll show in Section 4.3, is this: for blades $\boldsymbol{A}_r$ and $\boldsymbol{B}_s$, $\boldsymbol{A}_r\rfloor\boldsymbol{B}_s$ is also a blade, and it represents
1205.5935v1_F00249	25	■ 1.000	$\boldsymbol{B}_s$	$\boldsymbol{B}_s$	the subspace of vectors <i>orthogonal to $\boldsymbol{A}_r$ and contained in $\boldsymbol{B}_s$</i> . So the roles played by the factors in the
1205.5935v1_F00251	25	■ 0.887	$\left\{a_{i_j}\right\}_{j=1, \ldots, r}$	$\left\{a_{i_j}\right\}_{j=1, \ldots, r}$	where the sum is performed over all possible choices of $\{a_{i_j}\}_{j=1, \ldots, r}$ out of $\{a_i\}_{i=1, \ldots, s}$, and in each term i_1

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1285.5935v1_F08252	25	■ 0.887	$\left\{a_i\right\}_{i=1,\ldots,s}$	$\{a_i\}_{i=1,\dots,s}$	where the sum is performed over all possible choices of $\{a_{i_j}\}_{j=1,\dots,r}$ out of $\{a_i\}_{i=1,\dots,s}$, and in each term i_1
1285.5935v1_F08253	25	■ 0.887	$i_{\{1\}}$	i_1	where the sum is performed over all possible choices of $\{a_{i_j}\}_{j=1,\dots,r}$ out of $\{a_i\}_{i=1,\dots,s}$, and in each term i_1
1285.5935v1_F08254	25	■ 1.000	$i_{\{r\}}$	i_r	through i_r and i_{r+1} through i_s separately are in ascending order.
1285.5935v1_F08255	25	■ 1.000	$i_{\{r+1\}}$	i_{r+1}	through i_r and i_{r+1} through i_s separately are in ascending order.
1285.5935v1_F08256	25	■ 1.000	$i_{\{s\}}$	i_s	through i_r and i_{r+1} through i_s separately are in ascending order.
1285.5935v1_F08257	25	■ 1.000	$B_{\{r\}}$	B_r	them in their original order, in order to act on them with B_r . In both cases, permuting vectors to the far
1285.5935v1_F08258	25	■ 0.987	$\binom{s}{r}$	$\binom{s}{r}$	of a set s elements, the sum has $\binom{s}{r}$ terms.
1285.5935v1_F08259	25	■ 1.000	$\boldsymbol{B}_{\{\boldsymbol{B}_{\{r\}}=a\wedge\boldsymbol{B}_{r-1}\}}$	$\boldsymbol{B}_r = a \wedge \boldsymbol{B}_{r-1}$	true for $r-1$. Now $\boldsymbol{B}_r = a \wedge \boldsymbol{B}_{r-1}$ where a is a vector and $\boldsymbol{B}_{r-1}$ is an $r-1$ -blade, so using the
1285.5935v1_F08260	25	■ 1.000	$\boldsymbol{B}_{\{\boldsymbol{B}_{\{r\}}-\mathbf{1}\}}$	$\boldsymbol{B}_{r-1}$	true for $r-1$. Now $\boldsymbol{B}_r = a \wedge \boldsymbol{B}_{r-1}$ where a is a vector and $\boldsymbol{B}_{r-1}$ is an $r-1$ -blade, so using the
1285.5935v1_F08261	26	■ 0.991	$i_{\{r-1\}}$	i_{r-1}	where i_1 through i_{r-1} and i_r through i_s are in ascending order separately. Now I use Theorem
1285.5935v1_F08262	26	■ 0.880	$(s-r+1)\binom{s}{r-1}=r\binom{s}{r}$	$(s-r+1)\binom{s}{r-1}=r\binom{s}{r}$	This expression has $(s-r+1)\binom{s}{r-1}=r\binom{s}{r}$ terms, which is too many by a factor of r , so it's time
1285.5935v1_F08263	26	■ 1.000	$s-r$	$s-r$	outer product of the remaining $s-r$ vectors arranged in ascending order, and that every possible
1285.5935v1_F08264	26	■ 1.000	$C(\left\{a_{i_1},\ldots,a_{i_r}\right\})$	$C(a_{i_1},\dots,a_{i_r})$	choice of r vectors occurs. That means that for some choice of scalars $C(a_{i_1},\dots,a_{i_r})$,
1285.5935v1_F08265	26	■ 1.000	$a_{\{i_{\{1\}}\}}$	a_{i_1}	what the coefficients $C(a_{i_1},\dots,a_{i_r})$ are. Well, for a given choice of a_{i_1} through a_{i_r} , the coefficient
1285.5935v1_F08266	26	■ 1.000	$a_{\{i_{\{r\}}\}}$	a_{i_r}	what the coefficients $C(a_{i_1},\dots,a_{i_r})$ are. Well, for a given choice of a_{i_1} through a_{i_r} , the coefficient
1285.5935v1_F08267	26	■ 1.000	$a_{\{i_{\{j\}}\}}$	a_{i_j}	will include terms in which one of the a_{i_j} is in the inner product with a while the others are in
1285.5935v1_F08268	26	■ 1.000	$\epsilon_{\{j\}}$	ϵ_j	for some value of ϵ_j for each j . This sum has r terms, one for each of the a_{i_j} , which is exactly the
1285.5935v1_F08269	26	■ 0.982	$a_{\{i_{\{k\}}\}}$	a_{i_k}	out those positions. First consider every a_{i_k} in the inner product with $\boldsymbol{B}_{r-1}$ where $k < j$: each
1285.5935v1_F08270	26	■ 0.982	$k < j$	$k < j$	out those positions. First consider every a_{i_k} in the inner product with $\boldsymbol{B}_{r-1}$ where $k < j$: each
1285.5935v1_F08271	26	■ 1.000	$i_{\{k\}}$	i_k	one is moved from position i_k to position k , so it contributes $i_k - k$ to ϵ_j . Now consider the a_{i_k}
1285.5935v1_F08272	26	■ 1.000	k	k	one is moved from position i_k to position k , so it contributes $i_k - k$ to ϵ_j . Now consider the a_{i_k}
1285.5935v1_F08273	26	■ 1.000	$i_{\{k\}}-k$	$i_k - k$	one is moved from position i_k to position k , so it contributes $i_k - k$ to ϵ_j . Now consider the a_{i_k}
1285.5935v1_F08274	26	■ 1.000	$k > j$	$k > j$	where $k > j$: each one is moved from position i_k to position $k-1$ (because position j is empty),
1285.5935v1_F08275	26	■ 1.000	$k-1$	$k-1$	where $k > j$: each one is moved from position i_k to position $k-1$ (because position j is empty),
1285.5935v1_F08276	26	■ 1.000	$i_{\{k\}}-k+1$	$i_k - k + 1$	so it contributes $i_k - k + 1$.
1285.5935v1_F08277	26	■ 1.000	$a_{\{i_{\{j+1\}}\}}$	$a_{i_{j+1}}$	it is part of the second group of vectors. Therefore vectors $a_{i_{j+1}}$ through a_{i_r} had to be moved to
1285.5935v1_F08278	26	■ 1.000	$i_{\{j\}}$	i_j	its left, moving it from its original position i_j ahead to $i_j - j + r$. It is then moved to position r
1285.5935v1_F08279	26	■ 1.000	$i_{\{j\}}-j+r$	$i_j - j + r$	its left, moving it from its original position i_j ahead to $i_j - j + r$. It is then moved to position r
1285.5935v1_F08280	26	■ 1.000	$i_{\{j\}}-j$	$i_j - j$	for the inner product with a , so its contribution to ϵ_j is $i_j - j$. Therefore
1285.5935v1_F08281	27	■ 1.000	$v\wedge n\wedge n^{-1}$	$v \wedge nn^{-1}$	and $v \wedge nn^{-1}$ are without knowing how they were derived. (Notice I'm using order of operations to drop
1285.5935v1_F08282	27	■ 1.000	$n\vee n$	nnn	parentheses.) First let's do nnn^{-1} ; since this is proportional to nnn , I'll look at that instead. We know now
1285.5935v1_F08283	27	■ 1.000	$n\wedge v\wedge n$	$n \wedge v \wedge n$	is the outer product of the factors. In this case the outer product is $n \wedge v \wedge n$, which vanishes because n
1285.5935v1_F08284	27	■ 1.000	$v\wedge n\wedge n$	$v \wedge nn$	Next let's look at $v \wedge nn^{-1}$; again I'll consider $v \wedge nn$ because the answer will be the same. The easiest
1285.5935v1_F08285	27	■ 1.000	$v\wedge n\wedge n$	$v \wedge n \wedge n$	because the $v \wedge n \wedge n$ term vanishes. Even if you didn't know the remaining term had to be a vector, you
1285.5935v1_F08286	27	■ 0.996	$\left.\left(\left(v\wedge n\wedge n^{-1}\right)\right)\right\rfloor n$	$\left.\left(v \wedge nn^{-1}\right)\right\rfloor n$	I also asked how to calculate $\left.\left(v \wedge nn^{-1}\right)\right\rfloor n$ to verify that $v \wedge nn^{-1}$ really is perpendicular to n . (Re-
1285.5935v1_F08287	27	■ 1.000	$v\wedge n$	$v \wedge n$	since $v \wedge n$ has no scalar part. Easy, huh?
1285.5935v1_F08288	27	■ 1.000	$r,s\geq 1$	$r,s \geq 1$	Theorem 15. Let $\boldsymbol{A}_r$ and $\boldsymbol{B}_s$ be nonzero blades where $r,s \geq 1$.
1285.5935v1_F08289	27	■ 1.000	$\boldsymbol{A}_{\{\boldsymbol{B}_{\{r\}}\}}\wedge\boldsymbol{B}_{\{\boldsymbol{B}_{\{s\}}=0\}}$	$\boldsymbol{A}_r \wedge \boldsymbol{B}_s = 0$	(a) $\boldsymbol{A}_r \wedge \boldsymbol{B}_s = 0$ iff $\boldsymbol{A}_r$ and $\boldsymbol{B}_s$ share nonzero vectors.
1285.5935v1_F08290	27	■ 1.000	$\boldsymbol{A}_{\{\boldsymbol{B}_{\{r\}}\}}\wedge\boldsymbol{B}_{\{\boldsymbol{B}_{\{s\}}\}}$	$\boldsymbol{A}_r \wedge \boldsymbol{B}_s$	(b) $\boldsymbol{A}_r \wedge \boldsymbol{B}_s$, if nonzero, represents the direct sum of the corresponding subspaces.
1285.5935v1_F08291	28	■ 1.000	$a\wedge\boldsymbol{A}_{\{\boldsymbol{B}_{\{r\}}\}}\wedge\boldsymbol{B}_{\{\boldsymbol{B}_{\{s\}}=0\}}$	$a \wedge \boldsymbol{A}_r \wedge \boldsymbol{B}_s = 0$	immediately, since $a \wedge \boldsymbol{A}_r \wedge \boldsymbol{B}_s = 0$ iff a is a linear combination of the factors of $\boldsymbol{A}_r$ and $\boldsymbol{B}_s$.
1285.5935v1_F08292	28	■ 0.592	$a\wedge\boldsymbol{B}_{\{\boldsymbol{B}_{\{s\}}=0\}}$	$a \wedge \boldsymbol{B}_s = 0$	complement of $\boldsymbol{A}_r$ in $\boldsymbol{B}_s$ iff $a \rfloor \boldsymbol{A}_r = 0$ and $a \wedge \boldsymbol{B}_s = 0$. Now I'm ready to prove the result.
1285.5935v1_F08293	28	■ 0.977	$\left.\left(\left(\left(\boldsymbol{A}_{\{\boldsymbol{B}_{\{r\}}\}}\right)\right)\right)\right\rfloor\boldsymbol{B}_{\{\boldsymbol{B}_{\{s\}}=0\}}$	$\boldsymbol{A}_r \rfloor \boldsymbol{B}_s = 0$	(a) $\boldsymbol{A}_r \rfloor \boldsymbol{B}_s = 0$ iff $\boldsymbol{A}_r$ contains a nonzero vector orthogonal to $\boldsymbol{B}_s$.
1285.5935v1_F08294	28	■ 0.585	$r < s$	$r < s$	(b) If $r < s$ then $\boldsymbol{A}_r \rfloor \boldsymbol{B}_s$, if nonzero, is an $s-r$ -blade representing the orthogonal complement of $\boldsymbol{A}_r$ in
1285.5935v1_F08295	28	■ 0.585	$\left.\left(\left(\left(\boldsymbol{A}_{\{\boldsymbol{B}_{\{r\}}\}}\right)\right)\right)\right\rfloor\boldsymbol{B}_{\{\boldsymbol{B}_{\{s\}}\}}$	$\boldsymbol{A}_r \rfloor \boldsymbol{B}_s$	(b) If $r < s$ then $\boldsymbol{A}_r \rfloor \boldsymbol{B}_s$, if nonzero, is an $s-r$ -blade representing the orthogonal complement of $\boldsymbol{A}_r$ in
1285.5935v1_F08296	28	■ 1.000	$\boldsymbol{A}_{\{\boldsymbol{B}_{\{r\}}\}}=\boldsymbol{A}_{\{\boldsymbol{B}_{\{r\}}-\mathbf{1}\}}\wedge a_{\{1\}}$	$\boldsymbol{A}_r = \boldsymbol{A}_{r-1} \wedge a_1$	$\boldsymbol{A}_r = \boldsymbol{A}_{r-1} \wedge a_1$, which means
1285.5935v1_F08297	28	■ 0.749	$a_{\{1\}}\downarrow\boldsymbol{B}_{\{\boldsymbol{B}_{\{s\}}=0\}}$	$a_1 \rfloor \boldsymbol{B}_s = 0$	Suppose $\boldsymbol{A}_r$ contains a vector orthogonal to $\boldsymbol{B}_s$; then let that vector be a_1 , so $a_1 \rfloor \boldsymbol{B}_s = 0$, so

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1205.5935v1_F00298	28	■ 0.413	$\boldsymbol{A}_r \downharpoonleft \boldsymbol{B}_s = 0$	$\boldsymbol{A}_r \downharpoonleft \boldsymbol{B}_s = 0$	$\boldsymbol{A}_r \downharpoonleft \boldsymbol{B}_s = 0$. For the converse, assume $\boldsymbol{A}_r \downharpoonleft \boldsymbol{B}_s = 0$; then it follows that $\boldsymbol{A}_{r-1} \downharpoonleft (\boldsymbol{a}_1 \downharpoonleft \boldsymbol{B}_s) = 0$.
1205.5935v1_F00299	28	■ 0.413	$\left. \left. \boldsymbol{A}_r \right \boldsymbol{B}_s \right \boldsymbol{a}_1 = 0$	$\boldsymbol{A}_{r-1} \downharpoonleft (\boldsymbol{a}_1 \downharpoonleft \boldsymbol{B}_s) = 0$	$\boldsymbol{A}_r \downharpoonleft \boldsymbol{B}_s = 0$. For the converse, assume $\boldsymbol{A}_r \downharpoonleft \boldsymbol{B}_s = 0$; then it follows that $\boldsymbol{A}_{r-1} \downharpoonleft (\boldsymbol{a}_1 \downharpoonleft \boldsymbol{B}_s) = 0$.
1205.5935v1_F00300	28	■ 0.925	$\left. \boldsymbol{a}_1 \right \boldsymbol{B}_s = 0$	$\boldsymbol{a}_1 \downharpoonleft \boldsymbol{B}_s = 0$	There are now three possibilities. The first is $\boldsymbol{a}_1 \downharpoonleft \boldsymbol{B}_s = 0$, in which case $\boldsymbol{a}_1$ is orthogonal to $\boldsymbol{B}_s$
1205.5935v1_F00301	28	■ 0.994	$\boldsymbol{a}_2$	$\boldsymbol{a}_2$	and I'm done. If not, then by the $r-1$ result $\boldsymbol{A}_{r-1}$ contains a vector $\boldsymbol{a}_2$ orthogonal to $\boldsymbol{a}_1 \downharpoonleft \boldsymbol{B}_s$;
1205.5935v1_F00302	28	■ 0.994	$\left. \boldsymbol{a}_1 \right \boldsymbol{B}_s$	$\boldsymbol{a}_1 \downharpoonleft \boldsymbol{B}_s$	and I'm done. If not, then by the $r-1$ result $\boldsymbol{A}_{r-1}$ contains a vector $\boldsymbol{a}_2$ orthogonal to $\boldsymbol{a}_1 \downharpoonleft \boldsymbol{B}_s$;
1205.5935v1_F00303	28	■ 0.993	$\boldsymbol{b}_1 \wedge \cdots \wedge \boldsymbol{b}_s$	$\boldsymbol{b}_1 \wedge \cdots \wedge \boldsymbol{b}_s$	$\boldsymbol{b}_1 \wedge \cdots \wedge \boldsymbol{b}_s$ where $\boldsymbol{a}_1 \downharpoonleft \boldsymbol{b}_1 \neq 0$ while the other $\boldsymbol{b}_j$ are orthogonal to $\boldsymbol{a}_1$, so
1205.5935v1_F00304	28	■ 0.993	$\boldsymbol{a}_1 \downharpoonleft \boldsymbol{b}_1 \neq 0$	$\boldsymbol{a}_1 \downharpoonleft \boldsymbol{b}_1 \neq 0$	$\boldsymbol{b}_1 \wedge \cdots \wedge \boldsymbol{b}_s$ where $\boldsymbol{a}_1 \downharpoonleft \boldsymbol{b}_1 \neq 0$ while the other $\boldsymbol{b}_j$ are orthogonal to $\boldsymbol{a}_1$, so
1205.5935v1_F00305	28	■ 0.470	$\boldsymbol{a}_2 \downharpoonleft \boldsymbol{b}_1 \neq 0$	$\boldsymbol{a}_2 \downharpoonleft \boldsymbol{b}_1 \neq 0$	The only way $\boldsymbol{a}_2$ can be orthogonal to $\boldsymbol{a}_1 \downharpoonleft \boldsymbol{B}_s$ but not $\boldsymbol{B}_s$ is if $\boldsymbol{a}_2 \downharpoonleft \boldsymbol{b}_1 \neq 0$ while $\boldsymbol{a}_2$ is orthogonal
1205.5935v1_F00306	28	■ 0.999	$\boldsymbol{b}_1$	$\boldsymbol{b}_1$	to all the $\boldsymbol{b}_j$, including $\boldsymbol{b}_1$, by construction. Thus it's orthogonal to $\boldsymbol{B}_s$. So in all three cases
1205.5935v1_F00307	28	■ 0.997	$\boldsymbol{A}_r \downharpoonleft \boldsymbol{B}_s =$	$\boldsymbol{A}_r \downharpoonleft \boldsymbol{B}_s =$	$\boldsymbol{A}_r$ contains a vector orthogonal to $\boldsymbol{B}_s$. To prove part (b), assume $r < s$ and $\boldsymbol{A}_r \downharpoonleft \boldsymbol{B}_s =$
1205.5935v1_F00308	28	■ 0.301	$\boldsymbol{A}_{r-1} \downharpoonleft (\boldsymbol{a}_1 \downharpoonleft \boldsymbol{B}_s) \neq 0$	$\boldsymbol{A}_{r-1} \downharpoonleft (\boldsymbol{a}_1 \downharpoonleft \boldsymbol{B}_s) \neq 0$	$\boldsymbol{A}_{r-1} \downharpoonleft (\boldsymbol{a}_1 \downharpoonleft \boldsymbol{B}_s) \neq 0$. Then by the $r-1$ result $\boldsymbol{A}_r \downharpoonleft \boldsymbol{B}_s$ is the space of all vectors in $\boldsymbol{a}_1 \downharpoonleft \boldsymbol{B}_s$ that
1205.5935v1_F00309	29	■ 0.975	$r > s; \boldsymbol{A}_r \downharpoonleft \boldsymbol{B}_s = 0$	$r > s; \boldsymbol{A}_r \downharpoonleft \boldsymbol{B}_s = 0$	Now for the case $r > s$; $\boldsymbol{A}_r \downharpoonleft \boldsymbol{B}_s = 0$ automatically, so I can forget about part (b) and
1205.5935v1_F00310	29	■ 0.958	$\boldsymbol{B}_s \downharpoonleft \boldsymbol{A}_r$	$\boldsymbol{B}_s \downharpoonleft \boldsymbol{A}_r$	I only need to show that $\boldsymbol{A}_r$ always contains a vector orthogonal to $\boldsymbol{B}_s$. Consider $\boldsymbol{B}_s \downharpoonleft \boldsymbol{A}_r$;
1205.5935v1_F00311	29	■ 1.000	$\boldsymbol{B}_p$	$\boldsymbol{B}_p$	and I'm done. If it does vanish, then $\boldsymbol{B}_s$ contains vectors orthogonal to $\boldsymbol{A}_r$; let $\boldsymbol{B}_p$ represent
1205.5935v1_F00312	29	■ 0.997	$p = s$	$p = s$	the subspace of all such vectors. If $p = s$, then $\boldsymbol{B}_s$ is orthogonal to $\boldsymbol{A}_r$ and I'm also done. If
1205.5935v1_F00313	29	■ 0.998	$p < s$	$p < s$	$p < s$, then $\boldsymbol{B}_s = \boldsymbol{B}_p \wedge \boldsymbol{B}_{s-p}$ where $\boldsymbol{B}_{s-p}$ contains no vectors orthogonal to $\boldsymbol{A}_r$, which implies
1205.5935v1_F00314	29	■ 0.998	$\boldsymbol{B}_s = \boldsymbol{B}_p \wedge \boldsymbol{B}_{s-p}$	$\boldsymbol{B}_s = \boldsymbol{B}_p \wedge \boldsymbol{B}_{s-p}$	$p < s$, then $\boldsymbol{B}_s = \boldsymbol{B}_p \wedge \boldsymbol{B}_{s-p}$ where $\boldsymbol{B}_{s-p}$ contains no vectors orthogonal to $\boldsymbol{A}_r$, which implies
1205.5935v1_F00315	29	■ 0.998	$\boldsymbol{B}_{s-p}$	$\boldsymbol{B}_{s-p}$	$p < s$, then $\boldsymbol{B}_s = \boldsymbol{B}_p \wedge \boldsymbol{B}_{s-p}$ where $\boldsymbol{B}_{s-p}$ contains no vectors orthogonal to $\boldsymbol{A}_r$, which implies
1205.5935v1_F00316	29	■ 0.680	$\boldsymbol{B}_{s-p} \downharpoonleft \boldsymbol{A}_r \neq 0$	$\boldsymbol{B}_{s-p} \downharpoonleft \boldsymbol{A}_r \neq 0$	$\boldsymbol{B}_{s-p} \downharpoonleft \boldsymbol{A}_r \neq 0$. Then any vector in $\boldsymbol{B}_{s-p} \downharpoonleft \boldsymbol{A}_r$ lies in $\boldsymbol{A}_r$ and is orthogonal to $\boldsymbol{B}_{s-p}$; but by
1205.5935v1_F00317	29	■ 0.680	$\boldsymbol{B}_{s-p} \downharpoonleft \boldsymbol{A}_r$	$\boldsymbol{B}_{s-p} \downharpoonleft \boldsymbol{A}_r$	$\boldsymbol{B}_{s-p} \downharpoonleft \boldsymbol{A}_r \neq 0$. Then any vector in $\boldsymbol{B}_{s-p} \downharpoonleft \boldsymbol{A}_r$ lies in $\boldsymbol{A}_r$ and is orthogonal to $\boldsymbol{B}_{s-p}$; but by
1205.5935v1_F00318	29	■ 1.000	$\boldsymbol{A}_r \left[\boldsymbol{B}_s = (-1)^{r(s-1)} \boldsymbol{B}_s \right] \boldsymbol{A}_r$	$\boldsymbol{A}_r \left[\boldsymbol{B}_s = (-1)^{r(s-1)} \boldsymbol{B}_s \right] \boldsymbol{A}_r$	Incidentally, I'll show in Section 5.2 that $\boldsymbol{A}_r \downharpoonleft \boldsymbol{B}_s = (-1)^{r(s-1)} \boldsymbol{B}_s \downharpoonleft \boldsymbol{A}_r$ for any r - and s -vectors, so the
1205.5935v1_F00319	29	■ 1.000	$\boldsymbol{A}$	$\boldsymbol{A}$	To explain the next few theorems, I need some facts about blade $\boldsymbol{A}$ to be proved in Section 5.4.
1205.5935v1_F00320	29	■ 1.000	$\boldsymbol{A}^2$	$\boldsymbol{A}^2$	1. $\boldsymbol{A}^2$ is a scalar.
1205.5935v1_F00321	29	■ 1.000	$\boldsymbol{A}^2 \neq 0$	$\boldsymbol{A}^2 \neq 0$	2. $\boldsymbol{A}$ is invertible iff $\boldsymbol{A}^2 \neq 0$, and $\boldsymbol{A}^{-1} = \boldsymbol{A}/\boldsymbol{A}^2$. Therefore $\boldsymbol{A}$ and $\boldsymbol{A}^{-1}$ represent the same subspace.
1205.5935v1_F00322	29	■ 1.000	$\boldsymbol{A}^{-1} = \boldsymbol{A}/\boldsymbol{A}^2$	$\boldsymbol{A}^{-1} = \boldsymbol{A}/\boldsymbol{A}^2$	2. $\boldsymbol{A}$ is invertible iff $\boldsymbol{A}^2 \neq 0$, and $\boldsymbol{A}^{-1} = \boldsymbol{A}/\boldsymbol{A}^2$. Therefore $\boldsymbol{A}$ and $\boldsymbol{A}^{-1}$ represent the same subspace.
1205.5935v1_F00323	29	■ 1.000	$\boldsymbol{A}^{-1}$	$\boldsymbol{A}^{-1}$	2. $\boldsymbol{A}$ is invertible iff $\boldsymbol{A}^2 \neq 0$, and $\boldsymbol{A}^{-1} = \boldsymbol{A}/\boldsymbol{A}^2$. Therefore $\boldsymbol{A}$ and $\boldsymbol{A}^{-1}$ represent the same subspace.
1205.5935v1_F00324	29	■ 1.000	$1 \leq r \leq s$	$1 \leq r \leq s$	Theorem 17. Let $\boldsymbol{A}_r$ and $\boldsymbol{B}_s$ be nonzero blades where $1 \leq r \leq s$.
1205.5935v1_F00325	29	■ 0.978	$\boldsymbol{A}_r \boldsymbol{B}_s = \boldsymbol{A}_r \downharpoonleft \boldsymbol{B}_s$	$\boldsymbol{A}_r \boldsymbol{B}_s = \boldsymbol{A}_r \downharpoonleft \boldsymbol{B}_s$	(a) If $\boldsymbol{A}_r$ is a subspace of $\boldsymbol{B}_s$, then $\boldsymbol{A}_r \boldsymbol{B}_s = \boldsymbol{A}_r \downharpoonleft \boldsymbol{B}_s$.
1205.5935v1_F00326	29	■ 1.000	$s = 1$	$s = 1$	(b) The converse is true if either (1) $r = 1$ or $s = 1$ or (2) $\boldsymbol{A}_r$ or $\boldsymbol{B}_s$ is invertible.
1205.5935v1_F00327	29	■ 0.986	$\boldsymbol{a} \downharpoonleft \boldsymbol{B}_s$	$\boldsymbol{a} \downharpoonleft \boldsymbol{B}_s$	<i>Proof.</i> First let $r = 1$; then $\boldsymbol{a} \downharpoonleft \boldsymbol{B}_s = \boldsymbol{a} \downharpoonleft \boldsymbol{B}_s$ iff $\boldsymbol{a} \wedge \boldsymbol{B}_s = 0$, which is true iff $\boldsymbol{a}$ belongs to $\boldsymbol{B}_s$.
1205.5935v1_F00328	29	■ 0.960	$\boldsymbol{B}_s; \boldsymbol{I}$	$\boldsymbol{B}_s; \boldsymbol{I}$	Now assume the result is true for $r-1$ and let $\boldsymbol{A}_r$ be a subspace of $\boldsymbol{B}_s$; I can write $\boldsymbol{A}_r = \boldsymbol{A}_{r-1} \boldsymbol{a}$
1205.5935v1_F00329	29	■ 0.960	$\boldsymbol{A}_r = \boldsymbol{A}_{r-1} \boldsymbol{a}$	$\boldsymbol{A}_r = \boldsymbol{A}_{r-1} \boldsymbol{a}$	Now assume the result is true for $r-1$ and let $\boldsymbol{A}_r$ be a subspace of $\boldsymbol{B}_s$; I can write $\boldsymbol{A}_r = \boldsymbol{A}_{r-1} \boldsymbol{a}$
1205.5935v1_F00330	29	■ 0.659	$\boldsymbol{a} \downharpoonleft \boldsymbol{B}_s$	$\boldsymbol{a} \downharpoonleft \boldsymbol{B}_s$	Now $\boldsymbol{a} \downharpoonleft \boldsymbol{B}_s$ is the orthogonal complement of $\boldsymbol{a}$ in $\boldsymbol{B}_s$, which means $\boldsymbol{a} \downharpoonleft \boldsymbol{B}_s$ contains $\boldsymbol{A}_{r-1}$, so by
1205.5935v1_F00331	29	■ 0.659	$\boldsymbol{a} \downharpoonleft \boldsymbol{B}_s$	$\boldsymbol{a} \downharpoonleft \boldsymbol{B}_s$	Now $\boldsymbol{a} \downharpoonleft \boldsymbol{B}_s$ is the orthogonal complement of $\boldsymbol{a}$ in $\boldsymbol{B}_s$, which means $\boldsymbol{a} \downharpoonleft \boldsymbol{B}_s$ contains $\boldsymbol{A}_{r-1}$, so by
1205.5935v1_F00332	29	■ 0.389	$\boldsymbol{a} \downharpoonleft \boldsymbol{B}_s = 0$	$\boldsymbol{a} \downharpoonleft \boldsymbol{B}_s = 0$	which is the desired result. (If $\boldsymbol{a} \downharpoonleft \boldsymbol{B}_s = 0$ then $\boldsymbol{A}_r \downharpoonleft \boldsymbol{B}_s = 0$ also by Theorem 16, so if one side
1205.5935v1_F00333	29	■ 0.969	$\boldsymbol{B}_s = \boldsymbol{A}_r^{-1} \boldsymbol{A}_r \downharpoonleft \boldsymbol{B}_s$	$\boldsymbol{B}_s = \boldsymbol{A}_r^{-1} \boldsymbol{A}_r \downharpoonleft \boldsymbol{B}_s$	implies $r = 1$, so assume $\boldsymbol{A}_r$ is invertible; then $\boldsymbol{A}_r \boldsymbol{B}_s = \boldsymbol{A}_r \downharpoonleft \boldsymbol{B}_s$ implies $\boldsymbol{B}_s = \boldsymbol{A}_r^{-1} \boldsymbol{A}_r \downharpoonleft \boldsymbol{B}_s$. By
1205.5935v1_F00334	29	■ 0.826	$\boldsymbol{B}_s \neq 0$	$\boldsymbol{B}_s \neq 0$	assumption $\boldsymbol{B}_s \neq 0$, so $\boldsymbol{A}_r \downharpoonleft \boldsymbol{B}_s \neq 0$ also. Therefore if $r = s$, then $\boldsymbol{B}_s$ is just a nonzero multiple

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1205.5935v1_F00335	29	■ 0.826	$\left.\boldsymbol{A}_r\right\rfloor\boldsymbol{B}_s\neq 0$	$\boldsymbol{A}_r\rfloor\boldsymbol{B}_s\neq 0$	assumption $\boldsymbol{B}_s\neq 0$, so $\boldsymbol{A}_r\rfloor\boldsymbol{B}_s\neq 0$ also. Therefore if $r=s$, then $\boldsymbol{B}_s$ is just a nonzero multiple
1205.5935v1_F00336	29	■ 0.826	$r=s$	$r=s$	assumption $\boldsymbol{B}_s\neq 0$, so $\boldsymbol{A}_r\rfloor\boldsymbol{B}_s\neq 0$ also. Therefore if $r=s$, then $\boldsymbol{B}_s$ is just a nonzero multiple
1205.5935v1_F00337	29	■ 1.000	$\boldsymbol{A}_r^{-1}$	$\boldsymbol{A}_r^{-1}$	of $\boldsymbol{A}_r^{-1}$. Since $\boldsymbol{A}_r$ and $\boldsymbol{A}_r^{-1}$ represent the same subspace, $\boldsymbol{A}_r$ and $\boldsymbol{B}_s$ also represent the same
1205.5935v1_F00338	29	■ 0.763	$\left.\boldsymbol{A}_r\right\rfloor\boldsymbol{B}_s$	$\boldsymbol{A}_r^{-1}\boldsymbol{A}_r\rfloor\boldsymbol{B}_s$	subspace. If $r<s$, then since $\boldsymbol{B}_s$ is an s -vector, $\boldsymbol{A}_r^{-1}\boldsymbol{A}_r\rfloor\boldsymbol{B}_s$ must be too; but only its highest
1205.5935v1_F00339	29	■ 0.386	$\boldsymbol{B}_s=\boldsymbol{A}_r^{-1}\wedge(\boldsymbol{A}_r\rfloor\boldsymbol{B}_s)$	$\boldsymbol{B}_s=\boldsymbol{A}_r^{-1}\wedge(\boldsymbol{A}_r\rfloor\boldsymbol{B}_s)$	outer product, so $\boldsymbol{B}_s=\boldsymbol{A}_r^{-1}\wedge(\boldsymbol{A}_r\rfloor\boldsymbol{B}_s)$. Therefore $\boldsymbol{B}_s$ is the direct sum of $\boldsymbol{A}_r\rfloor\boldsymbol{B}_s$ and $\boldsymbol{A}_r^{-1}$;
1205.5935v1_F00340	29	■ 0.386	$\boldsymbol{A}_r\rfloor\boldsymbol{B}_s$	$\boldsymbol{A}_r\rfloor\boldsymbol{B}_s$	outer product, so $\boldsymbol{B}_s=\boldsymbol{A}_r^{-1}\wedge(\boldsymbol{A}_r\rfloor\boldsymbol{B}_s)$. Therefore $\boldsymbol{B}_s$ is the direct sum of $\boldsymbol{A}_r\rfloor\boldsymbol{B}_s$ and $\boldsymbol{A}_r^{-1}$;
1205.5935v1_F00341	30	■ 0.951	$\boldsymbol{A}_r=\boldsymbol{A}_r\rfloor\boldsymbol{B}_s\boldsymbol{B}_s^{-1}$	$\boldsymbol{A}_r=\boldsymbol{A}_r\rfloor\boldsymbol{B}_s\boldsymbol{B}_s^{-1}$	when $\boldsymbol{B}_s$ is invertible is similar; $\boldsymbol{A}_r\boldsymbol{B}_s=\boldsymbol{A}_r\rfloor\boldsymbol{B}_s$ implies $\boldsymbol{A}_r=\boldsymbol{A}_r\rfloor\boldsymbol{B}_s\boldsymbol{B}_s^{-1}$, so if $r=s$, $\boldsymbol{A}_r$ is
1205.5935v1_F00342	30	■ 0.951	$r=s,\boldsymbol{A}_r$	$r=s,\boldsymbol{A}_r$	when $\boldsymbol{B}_s$ is invertible is similar; $\boldsymbol{A}_r\boldsymbol{B}_s=\boldsymbol{A}_r\rfloor\boldsymbol{B}_s$ implies $\boldsymbol{A}_r=\boldsymbol{A}_r\rfloor\boldsymbol{B}_s\boldsymbol{B}_s^{-1}$, so if $r=s$, $\boldsymbol{A}_r$ is
1205.5935v1_F00343	30	■ 0.996	$\boldsymbol{A}_r\rfloor\boldsymbol{B}_s\boldsymbol{B}_s^{-1}$	$\boldsymbol{A}_r\rfloor\boldsymbol{B}_s\boldsymbol{B}_s^{-1}$	a nonzero multiple of $\boldsymbol{B}_s$. If $r<s$, then since $\boldsymbol{A}_r$ is an r -vector, $\boldsymbol{A}_r\rfloor\boldsymbol{B}_s\boldsymbol{B}_s^{-1}$ must be too; but
1205.5935v1_F00344	30	■ 0.986	$\boldsymbol{A}_r=(\boldsymbol{A}_r\rfloor\boldsymbol{B}_s)\rfloor\boldsymbol{B}_s^{-1}$	$\boldsymbol{A}_r=(\boldsymbol{A}_r\rfloor\boldsymbol{B}_s)\rfloor\boldsymbol{B}_s^{-1}$	product, so $\boldsymbol{A}_r=(\boldsymbol{A}_r\rfloor\boldsymbol{B}_s)\rfloor\boldsymbol{B}_s^{-1}$. Therefore, $\boldsymbol{A}_r$ is a subspace of $\boldsymbol{B}_s^{-1}$, and thus of $\boldsymbol{B}_s$.
1205.5935v1_F00345	30	■ 0.986	$\boldsymbol{B}_s^{-1}$	$\boldsymbol{B}_s^{-1}$	product, so $\boldsymbol{A}_r=(\boldsymbol{A}_r\rfloor\boldsymbol{B}_s)\rfloor\boldsymbol{B}_s^{-1}$. Therefore, $\boldsymbol{A}_r$ is a subspace of $\boldsymbol{B}_s^{-1}$, and thus of $\boldsymbol{B}_s$.
1205.5935v1_F00346	30	■ 1.000	$\boldsymbol{A}_r\boldsymbol{B}_s=\boldsymbol{A}_r\wedge\boldsymbol{B}_s$	$\boldsymbol{A}_r\boldsymbol{B}_s=\boldsymbol{A}_r\wedge\boldsymbol{B}_s$	(a) If $\boldsymbol{A}_r$ and $\boldsymbol{B}_s$ are orthogonal, then $\boldsymbol{A}_r\boldsymbol{B}_s=\boldsymbol{A}_r\wedge\boldsymbol{B}_s$.
1205.5935v1_F00347	30	■ 1.000	$b_1b_2\cdots b_s$	$b_1b_2\cdots b_s$	each other; similarly, $\boldsymbol{B}_s$ can be expressed $b_1b_2\cdots b_s$ where the b_j are also orthogonal to each
1205.5935v1_F00348	30	■ 0.893	$a\boldsymbol{B}_s=a\wedge\boldsymbol{B}_s$	$a\boldsymbol{B}_s=a\wedge\boldsymbol{B}_s$	To prove the converse, first let $r=1$; then $a\boldsymbol{B}_s=a\wedge\boldsymbol{B}_s$ iff $a\rfloor\boldsymbol{B}_s=0$, which is true iff
1205.5935v1_F00349	30	■ 0.893	$a\rfloor\boldsymbol{B}_s=0$	$a\rfloor\boldsymbol{B}_s=0$	To prove the converse, first let $r=1$; then $a\boldsymbol{B}_s=a\wedge\boldsymbol{B}_s$ iff $a\rfloor\boldsymbol{B}_s=0$, which is true iff
1205.5935v1_F00350	30	■ 0.995	$\boldsymbol{B}_s=$	$\boldsymbol{B}_s=$	a is orthogonal to $\boldsymbol{B}_s$. Now assume $\boldsymbol{A}_r$ is invertible and let $\boldsymbol{A}_r\boldsymbol{B}_s=\boldsymbol{A}_r\wedge\boldsymbol{B}_s$; then $\boldsymbol{B}_s=$
1205.5935v1_F00351	30	■ 0.987	$\boldsymbol{A}_r^{-1}\boldsymbol{A}_r\wedge\boldsymbol{B}_s$	$\boldsymbol{A}_r^{-1}\boldsymbol{A}_r\wedge\boldsymbol{B}_s$	$\boldsymbol{A}_r^{-1}\boldsymbol{A}_r\wedge\boldsymbol{B}_s$. Since $\boldsymbol{A}_r^{-1}$ represents the same subspace as $\boldsymbol{A}_r$, which is a subspace of $\boldsymbol{A}_r\wedge\boldsymbol{B}_s$,
1205.5935v1_F00352	30	■ 0.998	$\boldsymbol{B}_s=\boldsymbol{A}_r^{-1}\rfloor(\boldsymbol{A}_r\wedge\boldsymbol{B}_s)$	$\boldsymbol{B}_s=\boldsymbol{A}_r^{-1}\rfloor(\boldsymbol{A}_r\wedge\boldsymbol{B}_s)$	it follows that $\boldsymbol{B}_s=\boldsymbol{A}_r^{-1}\rfloor(\boldsymbol{A}_r\wedge\boldsymbol{B}_s)$, so $\boldsymbol{B}_s$ is orthogonal to $\boldsymbol{A}_r$. The proof when $s=1$ or $\boldsymbol{B}_s$
1205.5935v1_F00353	30	■ 0.999	e_1,e_2,e_3	e_1,e_2,e_3	example to show why not. Consider an algebra with orthogonal vectors e_1,e_2,e_3 , and e_4 such that e_2 is null
1205.5935v1_F00354	30	■ 0.999	e_4	e_4	example to show why not. Consider an algebra with orthogonal vectors e_1,e_2,e_3 , and e_4 such that e_2 is null
1205.5935v1_F00355	30	■ 0.979	$\boldsymbol{A}_3=e_1e_2e_3$	$\boldsymbol{A}_3=e_1e_2e_3$	be violated.) Let $\boldsymbol{A}_3=e_1e_2e_3$ and $\boldsymbol{B}_3=e_2e_3e_4$; then $e_2^2=0$ implies $\boldsymbol{A}_3\boldsymbol{B}_3=0$, so $\boldsymbol{A}_3\rfloor\boldsymbol{B}_3=\boldsymbol{A}_3\wedge\boldsymbol{B}_3=0$
1205.5935v1_F00356	30	■ 0.979	$\boldsymbol{B}_3=e_2e_3e_4$	$\boldsymbol{B}_3=e_2e_3e_4$	be violated.) Let $\boldsymbol{A}_3=e_1e_2e_3$ and $\boldsymbol{B}_3=e_2e_3e_4$; then $e_2^2=0$ implies $\boldsymbol{A}_3\boldsymbol{B}_3=0$, so $\boldsymbol{A}_3\rfloor\boldsymbol{B}_3=\boldsymbol{A}_3\wedge\boldsymbol{B}_3=0$
1205.5935v1_F00357	30	■ 0.979	$e_2^2=0$	$e_2^2=0$	be violated.) Let $\boldsymbol{A}_3=e_1e_2e_3$ and $\boldsymbol{B}_3=e_2e_3e_4$; then $e_2^2=0$ implies $\boldsymbol{A}_3\boldsymbol{B}_3=0$, so $\boldsymbol{A}_3\rfloor\boldsymbol{B}_3=\boldsymbol{A}_3\wedge\boldsymbol{B}_3=0$
1205.5935v1_F00358	30	■ 0.979	$\boldsymbol{A}_3\boldsymbol{B}_3=0$	$\boldsymbol{A}_3\boldsymbol{B}_3=0$	be violated.) Let $\boldsymbol{A}_3=e_1e_2e_3$ and $\boldsymbol{B}_3=e_2e_3e_4$; then $e_2^2=0$ implies $\boldsymbol{A}_3\boldsymbol{B}_3=0$, so $\boldsymbol{A}_3\rfloor\boldsymbol{B}_3=\boldsymbol{A}_3\wedge\boldsymbol{B}_3=0$
1205.5935v1_F00359	30	■ 0.979	$\boldsymbol{A}_3\rfloor\boldsymbol{B}_3=\boldsymbol{A}_3\wedge\boldsymbol{B}_3=0$	$\boldsymbol{A}_3\rfloor\boldsymbol{B}_3=\boldsymbol{A}_3\wedge\boldsymbol{B}_3=0$	be violated.) Let $\boldsymbol{A}_3=e_1e_2e_3$ and $\boldsymbol{B}_3=e_2e_3e_4$; then $e_2^2=0$ implies $\boldsymbol{A}_3\boldsymbol{B}_3=0$, so $\boldsymbol{A}_3\rfloor\boldsymbol{B}_3=\boldsymbol{A}_3\wedge\boldsymbol{B}_3=0$
1205.5935v1_F00360	30	■ 1.000	$\boldsymbol{A}_3$	$\boldsymbol{A}_3$	also. However, neither $\boldsymbol{A}_3$ nor $\boldsymbol{B}_3$ is a subspace of the other, so the converse part of Theorem 17 doesn't
1205.5935v1_F00361	30	■ 1.000	$\boldsymbol{B}_3$	$\boldsymbol{B}_3$	also. However, neither $\boldsymbol{A}_3$ nor $\boldsymbol{B}_3$ is a subspace of the other, so the converse part of Theorem 17 doesn't
1205.5935v1_F00362	30	■ 1.000	e_3	e_3	hold, and the subspaces are not orthogonal to each other (they both contain non-null e_3), so the converse
1205.5935v1_F00363	30	■ 1.000	$\boldsymbol{A}_r\boldsymbol{B}_s=\boldsymbol{A}_r\rfloor\boldsymbol{B}_s$	$\boldsymbol{A}_r\boldsymbol{B}_s=\boldsymbol{A}_r\rfloor\boldsymbol{B}_s$	Incidentally, $\boldsymbol{A}_r\boldsymbol{B}_s=\boldsymbol{A}_r\rfloor\boldsymbol{B}_s$ implies $\boldsymbol{B}_s\boldsymbol{A}_r=\boldsymbol{B}_s\rfloor\boldsymbol{A}_r$ for any r - and s -vectors, and the same is true if
1205.5935v1_F00364	30	■ 1.000	$\boldsymbol{B}_s\boldsymbol{A}_r=\boldsymbol{B}_s\rfloor\boldsymbol{A}_r$	$\boldsymbol{B}_s\boldsymbol{A}_r=\boldsymbol{B}_s\rfloor\boldsymbol{A}_r$	Incidentally, $\boldsymbol{A}_r\boldsymbol{B}_s=\boldsymbol{A}_r\rfloor\boldsymbol{B}_s$ implies $\boldsymbol{B}_s\boldsymbol{A}_r=\boldsymbol{B}_s\rfloor\boldsymbol{A}_r$ for any r - and s -vectors, and the same is true if
1205.5935v1_F00365	30	■ 1.000	z	z	in $\boldsymbol{B}_s$. (For example, three-dimensional Euclidean space is the direct sum of the z axis and its orthogonal
1205.5935v1_F00366	30	■ 0.838	xy	xy	complement, the xy plane.) Using our theorems, that suggests something like $\boldsymbol{B}_s=\boldsymbol{A}_r\wedge(\boldsymbol{A}_r\rfloor\boldsymbol{B}_s)$. Now

Identifier	Page	Conf.	LaTeX source	Rendered	Image
1285.5935v1_F08367	30	■ 0.838	$\boldsymbol{B}_s = \boldsymbol{A}_r \wedge (\boldsymbol{A}_r \rfloor \boldsymbol{B}_s)$	$\boldsymbol{B}_s = \boldsymbol{A}_r \wedge (\boldsymbol{A}_r \rfloor \boldsymbol{B}_s)$	complement, the xy plane.) Using our theorems, that suggests something like $\boldsymbol{B}_s = \boldsymbol{A}_r \wedge (\boldsymbol{A}_r \rfloor \boldsymbol{B}_s)$. Now
1285.5935v1_F08368	30	■ 0.637	$\boldsymbol{B}_s = \boldsymbol{A}_r \wedge (\boldsymbol{A}_r^{-1} \rfloor \boldsymbol{B}_s)$	$\boldsymbol{B}_s = \boldsymbol{A}_r \wedge (\boldsymbol{A}_r^{-1} \rfloor \boldsymbol{B}_s)$	That's easy to fix, though: maybe $\boldsymbol{B}_s = \boldsymbol{A}_r \wedge (\boldsymbol{A}_r^{-1} \rfloor \boldsymbol{B}_s)$ instead. That turns out to be right, but with some
1285.5935v1_F08369	30	■ 0.657	$\boldsymbol{A}_r^2 \boldsymbol{B}_s =$	$\boldsymbol{A}_r^2 \boldsymbol{B}_s =$	Theorem 19. If $1 \leq r \leq s$ and nonzero blades $\boldsymbol{A}_r$ and $\boldsymbol{B}_s$ satisfy $\boldsymbol{A}_r \boldsymbol{B}_s = \boldsymbol{A}_r \rfloor \boldsymbol{B}_s$, then $\boldsymbol{A}_r^2 \boldsymbol{B}_s =$
1285.5935v1_F08370	30	■ 0.804	$\boldsymbol{A}_r \wedge (\boldsymbol{A}_r \rfloor \boldsymbol{B}_s)$	$\boldsymbol{A}_r \wedge (\boldsymbol{A}_r \rfloor \boldsymbol{B}_s)$	$\boldsymbol{A}_r \wedge (\boldsymbol{A}_r \rfloor \boldsymbol{B}_s)$.
1285.5935v1_F08371	31	■ 1.000	$\boldsymbol{A}_r^2$	$\boldsymbol{A}_r^2$	Since $\boldsymbol{A}_r^2$ is a number, the left hand side is an s -vector; so the right hand side must be also. Using
1285.5935v1_F08372	31	■ 1.000	$\boldsymbol{A}_r^2 \neq 0$	$\boldsymbol{A}_r^2 \neq 0$	either $\boldsymbol{A}_r$ is invertible or it's not; first suppose it is. Then $\boldsymbol{A}_r^2 \neq 0$, so taking the result of Theorem 19 and
1285.5935v1_F08373	31	■ 0.804	$\boldsymbol{B}_s = \boldsymbol{A}_r \wedge (\boldsymbol{A}_r^{-1} \rfloor \boldsymbol{B}_s)$	$\boldsymbol{B}_s = \boldsymbol{A}_r \wedge (\boldsymbol{A}_r^{-1} \rfloor \boldsymbol{B}_s)$	dividing by $\boldsymbol{A}_r^2$ yields $\boldsymbol{B}_s = \boldsymbol{A}_r \wedge (\boldsymbol{A}_r^{-1} \rfloor \boldsymbol{B}_s)$, which is the result I was after. Notice I had to assume $\boldsymbol{A}_r$ was
1285.5935v1_F08374	31	■ 0.991	$\boldsymbol{A}_r \wedge (\boldsymbol{A}_r \rfloor \boldsymbol{B}_s) = 0$	$\boldsymbol{A}_r \wedge (\boldsymbol{A}_r \rfloor \boldsymbol{B}_s) = 0$	or it isn't, in which case that vector also lies in $\boldsymbol{A}_r \rfloor \boldsymbol{B}_s$, so $\boldsymbol{A}_r \wedge (\boldsymbol{A}_r \rfloor \boldsymbol{B}_s) = 0$. Since $\boldsymbol{A}_r^2 = 0$, the theorem
1285.5935v1_F08375	31	■ 0.991	$\boldsymbol{A}_r^2 = 0$	$\boldsymbol{A}_r^2 = 0$	or it isn't, in which case that vector also lies in $\boldsymbol{A}_r \rfloor \boldsymbol{B}_s$, so $\boldsymbol{A}_r \wedge (\boldsymbol{A}_r \rfloor \boldsymbol{B}_s) = 0$. Since $\boldsymbol{A}_r^2 = 0$, the theorem
1285.5935v1_F08376	31	■ 1.000	$(A^{-1})^* A^* =$	$(A^{-1})^* A^* =$	for any multivector A . (That's what makes it an involution.) Suppose A is invertible; then since $(A^{-1})^* A^* =$
1285.5935v1_F08377	31	■ 1.000	$(A^{-1} A)^* = 1^* = 1$	$(A^{-1} A)^* = 1^* = 1$	$(A^{-1} A)^* = 1^* = 1$,
1285.5935v1_F08378	32	■ 1.000	$A \rfloor a$	$A \rfloor a$	the definitions of $A \rfloor a$ and $A \wedge a$ from Eqs. (43) and (52) become
1285.5935v1_F08379	32	■ 1.000	$A \wedge a$	$A \wedge a$	the definitions of $A \rfloor a$ and $A \wedge a$ from Eqs. (43) and (52) become
1285.5935v1_F08380	32	■ 0.999	A^{**r}	A^{**r}	Every once in a while I'll use **r to indicate grade involution taken r times. A^{**r} equals A if r is even and A^*
1285.5935v1_F08381	32	■ 0.999	A^{*r}	A^{*r}	Every once in a while I'll use *r to indicate grade involution taken r times. A^{*r} equals A if r is even and A^*
1285.5935v1_F08382	32	■ 0.999	A^*	A^*	Every once in a while I'll use *r to indicate grade involution taken r times. A^{*r} equals A if r is even and A^*
1285.5935v1_F08383	32	■ 0.995	$\dagger$	$\dagger$	The second operation is called <i>reversion</i> or <i>taking the reverse</i> and is represented by a $\dagger$. It's a little more
1285.5935v1_F08384	33	■ 1.000	$r(r-1)/2$	$r(r-1)/2$	because $r(r-1)/2$ interchanges are needed to return the vectors to their original order; therefore
1285.5935v1_F08385	33	■ 1.000	$(A^{-1})^\dagger A^\dagger = (AA^{-1})^\dagger = 1^\dagger = 1$	$(A^{-1})^\dagger A^\dagger = (AA^{-1})^\dagger = 1^\dagger = 1$	invertible; then since $(A^{-1})^\dagger A^\dagger = (AA^{-1})^\dagger = 1^\dagger = 1$,
1285.5935v1_F08386	33	■ 1.000	$\langle A_r B_s \rangle = \langle B_s A_r \rangle = 0$	$\langle A_r B_s \rangle = \langle B_s A_r \rangle = 0$	multivectors of different grades doesn't have a scalar part, so $\langle A_r B_s \rangle = \langle B_s A_r \rangle = 0$ if $r \neq s$, and when $r = s$
1285.5935v1_F08387	33	■ 1.000	$r \neq s$	$r \neq s$	multivectors of different grades doesn't have a scalar part, so $\langle A_r B_s \rangle = \langle B_s A_r \rangle = 0$ if $r \neq s$, and when $r = s$
1285.5935v1_F08388	33	■ 1.000	$\langle A_r B_s \rangle = \langle B_s A_r \rangle$	$\langle A_r B_s \rangle = \langle B_s A_r \rangle$	Therefore $\langle A_r B_s \rangle = \langle B_s A_r \rangle$ in general. Since the geometric product is distributive, I can go all the way to
1285.5935v1_F08389	34	■ 1.000	$j = r$	$j = r$	special cases. First, suppose $r \leq s$; then Eq. (140) with $j = r$ refers to the lowest grade term, so
1285.5935v1_F08390	34	■ 1.000	$j = 0$	$j = 0$	but in all other cases it commutes. Without any restrictions on r and s , Eq. (140) when $j = 0$ gives for the
1285.5935v1_F08391	34	■ 0.951	$(nvn)^\dagger = nvn$	$(nvn)^\dagger = nvn$	example, let me reconsider the product nvn of three vectors from Section 1.3. Notice that $(nvn)^\dagger = nvn$.
1285.5935v1_F08392	34	■ 1.000	$\ddagger$	$\ddagger$	It's represented by a $\ddagger$ and defined as follows:
1285.5935v1_F08393	35	■ 1.000	$A * B = \langle AB \rangle$	$A * B = \langle AB \rangle$	(Some authors define $A * B = \langle AB \rangle$. I'll tell you why I don't shortly.) First consider the scalar product of
1285.5935v1_F08394	36	■ 1.000	$\langle AB^\dagger \rangle$	$\langle AB^\dagger \rangle$	The next to last equality shows that an equivalent definition of the scalar product is $\langle AB^\dagger \rangle$. The scalar
1285.5935v1_F08395	36	■ 1.000	$A * (B \wedge C)$	$A * (B \wedge C)$	able to factor the term B out of expressions like $A * (B \wedge C)$. You do this by defining an inner product
1285.5935v1_F08396	36	■ 1.000	$ A ^2 = 0$	$ A ^2 = 0$	A is said to be <i>null</i> if $ A ^2 = 0$ and a <i>unit</i> multivector if $ A ^2 = \pm 1$. (Despite the notation, $ A ^2$ can be
1285.5935v1_F08397	36	■ 1.000	$ A ^2 = \pm 1$	$ A ^2 = \pm 1$	A is said to be <i>null</i> if $ A ^2 = 0$ and a <i>unit</i> multivector if $ A ^2 = \pm 1$. (Despite the notation, $ A ^2$ can be
1285.5935v1_F08398	36	■ 1.000	$ A ^2$	$ A ^2$	A is said to be <i>null</i> if $ A ^2 = 0$ and a <i>unit</i> multivector if $ A ^2 = \pm 1$. (Despite the notation, $ A ^2$ can be
1285.5935v1_F08399	36	■ 1.000	$ A ^n$	$ A ^n$	I define $ A ^n$ for other powers n in the obvious way as a power of $ A ^2$, but due care should be taken that
1285.5935v1_F08400	37	■ 1.000	$A^\dagger A$	$A^\dagger A$	Notice also that if A is a versor then $A^\dagger A$ also equals $ A ^2$. This gives me a couple of useful results.
1285.5935v1_F08401	37	■ 1.000	$B = C$	$B = C$	follows by setting $B = C$.
1285.5935v1_F08402	37	■ 1.000	$ A \neq 0$	$ A \neq 0$	<i>Proof.</i> If $ A \neq 0$, then clearly Eq. (168) gives an inverse of A , so A must be invertible.
1285.5935v1_F08403	37	■ 0.998	$AB = 1$	$AB = 1$	Conversely, suppose A is invertible; then there exists a B such that $AB = 1$. Then it follows that
1285.5935v1_F08404	37	■ 1.000	$B^\dagger A^\dagger = 1$	$B^\dagger A^\dagger = 1$	$B^\dagger A^\dagger = 1$ also, so
1285.5935v1_F08405	37	■ 1.000	$ A^{-1} ^2$	$ A^{-1} ^2$	above expression. For the squared norm, just calculate $ A^{-1} ^2$ using Eq. (168).

Identifier	Page	Conf.	LaTeX source	Rendered	Image
1205.5935v1_F00406	38	■1.000	$\backslash\boldsymbol{\mathrm{A}}_{\backslash\boldsymbol{\mathrm{r}}}\wedge\{\dagger\}=(-1)^{r(r-1)/2}\boldsymbol{\mathrm{A}}_{\boldsymbol{\mathrm{r}}}, \boldsymbol{\mathrm{A}} ^2=\boldsymbol{\mathrm{A}}^\dagger\boldsymbol{\mathrm{A}}$	$\boldsymbol{A}_r^\dagger = (-1)^{r(r-1)/2}\boldsymbol{A}_r, \boldsymbol{A} ^2 = \boldsymbol{A}^\dagger \boldsymbol{A}$	few more results are also true. Since $\boldsymbol{A}_r^\dagger = (-1)^{r(r-1)/2}\boldsymbol{A}_r$, $ \boldsymbol{A} ^2 = \boldsymbol{A}^\dagger \boldsymbol{A}$ becomes $ \boldsymbol{A}_r ^2 = (-1)^{r(r-1)/2}\boldsymbol{A}_r^2$.
1205.5935v1_F00407	38	■1.000	$\backslash\left \boldsymbol{\mathrm{A}}_{\backslash\boldsymbol{\mathrm{r}}}\right ^2=(-1)^{r(r-1)/2}\boldsymbol{\mathrm{A}}_{\boldsymbol{\mathrm{r}}}^2$	$ \boldsymbol{A}_r ^2 = (-1)^{r(r-1)/2}\boldsymbol{A}_r^2$	few more results are also true. Since $\boldsymbol{A}_r^\dagger = (-1)^{r(r-1)/2}\boldsymbol{A}_r$, $ \boldsymbol{A} ^2 = \boldsymbol{A}^\dagger \boldsymbol{A}$ becomes $ \boldsymbol{A}_r ^2 = (-1)^{r(r-1)/2}\boldsymbol{A}_r^2$.
1205.5935v1_F00408	38	■0.760	$\backslash\boldsymbol{\mathrm{A}}_{\backslash\boldsymbol{\mathrm{r}}}\wedge\{2\}=\pm 1$	$\boldsymbol{A}_r^2 = \pm 1$	satisfy $\boldsymbol{A}_r^2 = \pm 1$, although that ± 1 may not be the blade’s squared norm. It also follows that the inverse of
1205.5935v1_F00409	38	■0.999	$(\boldsymbol{\mathrm{a}}\wedge\boldsymbol{\mathrm{b}})^2$	$(\boldsymbol{a}\wedge\boldsymbol{b})^2$	the original blade by its square. I actually calculated $(\boldsymbol{a}\wedge\boldsymbol{b})^2$ in Section 1.1, and the result was $-\boldsymbol{a}^2\boldsymbol{b}^2\sin^2\theta$.
1205.5935v1_F00410	38	■0.999	$-\boldsymbol{\mathrm{a}}^2\boldsymbol{\mathrm{b}}^2\sin^2\theta$	$-\boldsymbol{a}^2\boldsymbol{b}^2\sin^2\theta$	the original blade by its square. I actually calculated $(\boldsymbol{a}\wedge\boldsymbol{b})^2$ in Section 1.1, and the result was $-\boldsymbol{a}^2\boldsymbol{b}^2\sin^2\theta$.
1205.5935v1_F00411	38	■1.000	$\boldsymbol{\mathrm{A}}_{\backslash\boldsymbol{\mathrm{r}}}=\boldsymbol{\mathrm{e}}_1\boldsymbol{\mathrm{e}}_2\cdots\boldsymbol{\mathrm{e}}_r$	$\boldsymbol{A}_r = \boldsymbol{e}_1\boldsymbol{e}_2\cdots\boldsymbol{e}_r$	<i>Proof.</i> $\boldsymbol{A}_r = \boldsymbol{e}_1\boldsymbol{e}_2\cdots\boldsymbol{e}_r$ is null iff $\boldsymbol{e}_i^2 = 0$ for at least one i , in which case $\boldsymbol{e}_i$ is orthogonal to
1205.5935v1_F00412	38	■1.000	$\boldsymbol{\mathrm{e}}_{\boldsymbol{\mathrm{i}}}\wedge\{2\}=0$	$\boldsymbol{e}_i^2 = 0$	<i>Proof.</i> $\boldsymbol{A}_r = \boldsymbol{e}_1\boldsymbol{e}_2\cdots\boldsymbol{e}_r$ is null iff $\boldsymbol{e}_i^2 = 0$ for at least one i , in which case $\boldsymbol{e}_i$ is orthogonal to
1205.5935v1_F00413	38	■1.000	$\backslash\left\{\boldsymbol{\mathrm{e}}_{\boldsymbol{\mathrm{j}}}\right\}_{\boldsymbol{\mathrm{i}}=1,\ldots,r}$	$\{\boldsymbol{e}_j\}_{i=1,\dots,r}$	every vector in the span of $\{\boldsymbol{e}_j\}_{i=1,\dots,r}$, which is just $\boldsymbol{A}_r$. That means that either (a) $\boldsymbol{e}_i = 0$ or
1205.5935v1_F00414	38	■1.000	$\boldsymbol{\mathrm{e}}_{\boldsymbol{\mathrm{i}}}=0$	$\boldsymbol{e}_i = 0$	every vector in the span of $\{\boldsymbol{e}_j\}_{i=1,\dots,r}$, which is just $\boldsymbol{A}_r$. That means that either (a) $\boldsymbol{e}_i = 0$ or
1205.5935v1_F00415	38	■1.000	$\boldsymbol{\mathrm{e}}_{\boldsymbol{\mathrm{i}}}\neq 0$	$\boldsymbol{e}_i \neq 0$	(b) $\boldsymbol{e}_i \neq 0$ but the inner product is degenerate on $\boldsymbol{A}_r$. Since $\boldsymbol{A}_r \neq 0$, none of the $\boldsymbol{e}_i$ vanish, so
1205.5935v1_F00416	38	■1.000	$\boldsymbol{\mathrm{A}}_{\backslash\boldsymbol{\mathrm{r}}}\neq 0$	$\boldsymbol{A}_r \neq 0$	(b) $\boldsymbol{e}_i \neq 0$ but the inner product is degenerate on $\boldsymbol{A}_r$. Since $\boldsymbol{A}_r \neq 0$, none of the $\boldsymbol{e}_i$ vanish, so
1205.5935v1_F00417	38	■1.000	$\boldsymbol{\mathrm{A}}_{\boldsymbol{\mathrm{r}}}\boldsymbol{\mathrm{B}}_{\boldsymbol{\mathrm{s}}}=0$	$\boldsymbol{A}_r\boldsymbol{B}_s = 0$	Theorem 24. If $r, s \geq 1$ and nonzero versors $\boldsymbol{A}_r$ and $\boldsymbol{B}_s$ satisfy $\boldsymbol{A}_r\boldsymbol{B}_s = 0$, then both versors are null.
1205.5935v1_F00418	38	■1.000	$\backslash\left \boldsymbol{\mathrm{A}}_{\boldsymbol{\mathrm{r}}}\right ^2\boldsymbol{\mathrm{B}}_{\boldsymbol{\mathrm{s}}}=\boldsymbol{\mathrm{A}}_{\boldsymbol{\mathrm{r}}}^\dagger\boldsymbol{\mathrm{A}}_{\boldsymbol{\mathrm{r}}}\boldsymbol{\mathrm{B}}_{\boldsymbol{\mathrm{s}}} = 0$	$ \boldsymbol{A}_r ^2\boldsymbol{B}_s = \boldsymbol{A}_r^\dagger\boldsymbol{A}_r\boldsymbol{B}_s = 0$	<i>Proof.</i> This one is easy: if $\boldsymbol{A}_r\boldsymbol{B}_s = 0$, then $ \boldsymbol{A}_r ^2\boldsymbol{B}_s = \boldsymbol{A}_r^\dagger\boldsymbol{A}_r\boldsymbol{B}_s = 0$ also. Now since $\boldsymbol{B}_s$
1205.5935v1_F00419	38	■1.000	$\backslash\left \boldsymbol{\mathrm{A}}_{\boldsymbol{\mathrm{r}}}\right ^2=0$	$ \boldsymbol{A}_r ^2 = 0$	is assumed nonzero, it follows that $ \boldsymbol{A}_r ^2 = 0$, or $\boldsymbol{A}_r$ is null. Going back to $\boldsymbol{A}_r\boldsymbol{B}_s = 0$ and
1205.5935v1_F00420	38	■1.000	$\boldsymbol{\mathrm{B}}_{\boldsymbol{\mathrm{s}}}^\dagger$	$\boldsymbol{B}_s^\dagger$	multiplying from the right by $\boldsymbol{B}_s^\dagger$ establishes that $\boldsymbol{B}_s$ is also null.
1205.5935v1_F00421	38	■1.000	$\boldsymbol{\mathrm{A}}_{\backslash\boldsymbol{\mathrm{r}}}\wedge\boldsymbol{\mathrm{A}}_{\boldsymbol{\mathrm{r}}}=\boldsymbol{\mathrm{0}}\right)$	$\boldsymbol{A}_r(\boldsymbol{a}\wedge\boldsymbol{A}_r = 0)$	A special case of this theorem arises if a vector $\boldsymbol{a}$ both lies in blade $\boldsymbol{A}_r$ ($\boldsymbol{a}\wedge\boldsymbol{A}_r = 0$) and is orthogonal
1205.5935v1_F00422 [refined: source]	38	■0.491	$\backslash\left(\boldsymbol{\mathrm{a}}\rfloor\boldsymbol{\mathrm{A}}_{\backslash\boldsymbol{\mathrm{r}}}\right)=0\right)$	$(\boldsymbol{a}\rfloor\boldsymbol{A}_r = 0)$	to it ($\boldsymbol{a}\rfloor\boldsymbol{A}_r = 0$). In that case $\boldsymbol{a}\boldsymbol{A}_r = 0$, and we say $\boldsymbol{a}$ annihilates $\boldsymbol{A}_r$. The theorem tells us that $\boldsymbol{a}$ must
1205.5935v1_F00423	38	■0.491	$\boldsymbol{\mathrm{a}}\wedge\boldsymbol{\mathrm{A}}_{\boldsymbol{\mathrm{r}}}=\boldsymbol{\mathrm{0}}$	$\boldsymbol{a}\boldsymbol{A}_r = 0$	to it ($\boldsymbol{a}\rfloor\boldsymbol{A}_r = 0$). In that case $\boldsymbol{a}\boldsymbol{A}_r = 0$, and we say $\boldsymbol{a}$ annihilates $\boldsymbol{A}_r$. The theorem tells us that $\boldsymbol{a}$ must
1205.5935v1_F00424	39	■1.000	$\backslash\left.\boldsymbol{\mathrm{B}}\right\rfloor\boldsymbol{\mathrm{A}}_{\backslash\boldsymbol{\mathrm{r}}}^{-1}$	$\boldsymbol{B}\rfloor\boldsymbol{A}_r^{-1}$	then the dual of any multivector $\boldsymbol{B}$ by $\boldsymbol{A}_r$ is $\boldsymbol{B}\rfloor\boldsymbol{A}_r^{-1}$. (Duality gets its own symbol only in a special case,
1205.5935v1_F00425	39	■1.000	$\boldsymbol{\mathrm{s}}>\boldsymbol{\mathrm{r}}$	$\boldsymbol{s} > \boldsymbol{r}$	1. If $\boldsymbol{s} > \boldsymbol{r}$, the dual of $\boldsymbol{B}_s$ vanishes.
1205.5935v1_F00426	39	■1.000	$\boldsymbol{\mathrm{s}}=\boldsymbol{\mathrm{r}}$	$\boldsymbol{s} = \boldsymbol{r}$	2. If $\boldsymbol{s} = \boldsymbol{r}$, the dual of $\boldsymbol{B}_s$ is a scalar which is zero iff $\boldsymbol{B}_s$ contains a vector orthogonal to $\boldsymbol{A}_r$ (Theorem
1205.5935v1_F00427	39	■0.996	$\boldsymbol{\mathrm{s}}<\boldsymbol{\mathrm{r}}$	$\boldsymbol{s} < \boldsymbol{r}$	3. If $\boldsymbol{s} < \boldsymbol{r}$, the dual of $\boldsymbol{B}_s$ is either zero or an $\boldsymbol{r} - \boldsymbol{s}$ -blade representing the orthogonal complement of $\boldsymbol{B}_s$
1205.5935v1_F00428	39	■0.996	$\boldsymbol{\mathrm{r}}-\boldsymbol{\mathrm{s}}$	$\boldsymbol{r} - \boldsymbol{s}$	3. If $\boldsymbol{s} < \boldsymbol{r}$, the dual of $\boldsymbol{B}_s$ is either zero or an $\boldsymbol{r} - \boldsymbol{s}$ -blade representing the orthogonal complement of $\boldsymbol{B}_s$
1205.5935v1_F00429	39	■0.998	$\boldsymbol{\mathrm{B}}_{\backslash\boldsymbol{\mathrm{s}}}\boldsymbol{\mathrm{A}}_{\backslash\boldsymbol{\mathrm{r}}}^{-1}$	$\boldsymbol{B}_s\boldsymbol{A}_r^{-1}$	in $\boldsymbol{A}_r$ (Theorem 16 again). If $\boldsymbol{B}_s$ was inside $\boldsymbol{A}_r$ to begin with, the dual of $\boldsymbol{B}_s$ is just $\boldsymbol{B}_s\boldsymbol{A}_r^{-1}$ (Theorem
1205.5935v1_F00430	39	■1.000	$\boldsymbol{\mathrm{I}}$	$\boldsymbol{I}$	element $\boldsymbol{I}$ to be a unit $\boldsymbol{n}$ -blade. This determines $\boldsymbol{I}$ to within a sign. (Some people call a volume element a
1205.5935v1_F00431	39	■1.000	$ \boldsymbol{\mathrm{I}} ^2$	$ \boldsymbol{I} ^2$	<i>pseudoscalar</i> , but I won’t.) In fact, I can calculate $ \boldsymbol{I} ^2$ explicitly. Let $\{\boldsymbol{e}_i\}_{i=1,\dots,r}$ be an orthonormal basis,
1205.5935v1_F00432	39	■0.635	$\boldsymbol{\mathrm{I}}=\boldsymbol{\mathrm{e}}_1\boldsymbol{\mathrm{e}}_2\cdots\boldsymbol{\mathrm{e}}_{\boldsymbol{\mathrm{n}}}$	$\boldsymbol{I} = \boldsymbol{e}_1\boldsymbol{e}_2\cdots\boldsymbol{e}_n$	and suppose $\boldsymbol{p}$ of the $\boldsymbol{e}_i$ square to -1 while the rest square to 1. Let $\boldsymbol{I} = \boldsymbol{e}_1\boldsymbol{e}_2\cdots\boldsymbol{e}_n$; then using Eq. (164),
1205.5935v1_F00433	39	■0.999	$ \boldsymbol{\mathrm{I}} ^2=1$	$ \boldsymbol{I} ^2 = 1$	Therefore $ \boldsymbol{I} ^2 = 1$ in any Euclidean space, while $ \boldsymbol{I} ^2 = -1$ in Minkowski space. Since $ \boldsymbol{I} ^2 = \boldsymbol{I}^\dagger\boldsymbol{I}$, this
1205.5935v1_F00434	39	■0.999	$ \boldsymbol{\mathrm{I}} ^2=-1$	$ \boldsymbol{I} ^2 = -1$	Therefore $ \boldsymbol{I} ^2 = 1$ in any Euclidean space, while $ \boldsymbol{I} ^2 = -1$ in Minkowski space. Since $ \boldsymbol{I} ^2 = \boldsymbol{I}^\dagger\boldsymbol{I}$, this
1205.5935v1_F00435	39	■0.999	$ \boldsymbol{\mathrm{I}} ^2=\boldsymbol{\mathrm{I}}^\dagger\boldsymbol{\mathrm{I}}$	$ \boldsymbol{I} ^2 = \boldsymbol{I}^\dagger\boldsymbol{I}$	Therefore $ \boldsymbol{I} ^2 = 1$ in any Euclidean space, while $ \boldsymbol{I} ^2 = -1$ in Minkowski space. Since $ \boldsymbol{I} ^2 = \boldsymbol{I}^\dagger\boldsymbol{I}$, this
1205.5935v1_F00436	39	■1.000	$\boldsymbol{\mathrm{A}}_{\backslash\boldsymbol{\mathrm{r}}}\wedge\boldsymbol{\mathrm{A}}_{\boldsymbol{\mathrm{s}}-\boldsymbol{\mathrm{r}}}=\boldsymbol{\mathrm{A}}_{\boldsymbol{\mathrm{s}}}$	$\boldsymbol{A}_r\wedge\boldsymbol{A}_{s-r} = \boldsymbol{A}_s$	there exists a blade $\boldsymbol{A}_{s-r}$ such that $\boldsymbol{A}_r\wedge\boldsymbol{A}_{s-r} = \boldsymbol{A}_s$. Thus $\boldsymbol{A}_r$ is a factor of a nonzero blade,
1205.5935v1_F00437	39	■0.835	$\boldsymbol{\mathrm{A}}^\perp$	$\boldsymbol{A}^\perp$	Unless otherwise specified, “the dual of $\boldsymbol{A}$ ” means “the dual of $\boldsymbol{A}$ by $\boldsymbol{I}$ ” and is denoted $\boldsymbol{A}^\perp$. Let’s
1205.5935v1_F00438	39	■1.000	$\boldsymbol{\mathrm{s}}>\boldsymbol{\mathrm{n}}$	$\boldsymbol{s} > \boldsymbol{n}$	1. There are no $\boldsymbol{s}$ -blades for $\boldsymbol{s} > \boldsymbol{n}$, so the first option can’t happen.
1205.5935v1_F00439	39	■1.000	$\boldsymbol{\mathrm{B}}_{\boldsymbol{\mathrm{n}}}=\lambda\boldsymbol{\mathrm{I}}$	$\boldsymbol{B}_n = \lambda\boldsymbol{I}$	2. Any $\boldsymbol{n}$ -blade $\boldsymbol{B}_n = \lambda\boldsymbol{I}$ for some λ , in which case the dual of $\boldsymbol{B}_n$ is just λ .
1205.5935v1_F00440	39	■1.000	$\boldsymbol{\mathrm{B}}_{\backslash\boldsymbol{\mathrm{n}}}$	$\boldsymbol{B}_n$	2. Any $\boldsymbol{n}$ -blade $\boldsymbol{B}_n = \lambda\boldsymbol{I}$ for some λ , in which case the dual of $\boldsymbol{B}_n$ is just λ .
1205.5935v1_F00441	40	■1.000	$\boldsymbol{\mathrm{s}}<\boldsymbol{\mathrm{n}}$	$\boldsymbol{s} < \boldsymbol{n}$	3. If $\boldsymbol{s} < \boldsymbol{n}$, then $\boldsymbol{B}_s$ represents a subspace of the full space, so the dual of $\boldsymbol{B}_s$ is just $\boldsymbol{B}_s\boldsymbol{I}^{-1}$. It cannot
1205.5935v1_F00442	40	■1.000	$\boldsymbol{\mathrm{B}}_{\backslash\boldsymbol{\mathrm{s}}}\boldsymbol{\mathrm{I}}^{-1}$	$\boldsymbol{B}_s\boldsymbol{I}^{-1}$	3. If $\boldsymbol{s} < \boldsymbol{n}$, then $\boldsymbol{B}_s$ represents a subspace of the full space, so the dual of $\boldsymbol{B}_s$ is just $\boldsymbol{B}_s\boldsymbol{I}^{-1}$. It cannot
1205.5935v1_F00443	40	■0.731	$\boldsymbol{\mathrm{I}}^{-1}$	$\boldsymbol{I}^{-1}$	dual by $\boldsymbol{I}^{-1}$ instead of $\boldsymbol{I}$ is the inverse operation; it’s denoted by $\boldsymbol{A}^\perp$. Since $\boldsymbol{I}^{-1} = \boldsymbol{I}/\boldsymbol{I}^2$, $\boldsymbol{A}^\perp$ and $\boldsymbol{A}^{-\perp}$
1205.5935v1_F00444	40	■0.731	$\boldsymbol{\mathrm{A}}^{-\perp}$	$\boldsymbol{A}^{-\perp}$	dual by $\boldsymbol{I}^{-1}$ instead of $\boldsymbol{I}$ is the inverse operation; it’s denoted by $\boldsymbol{A}^\perp$. Since $\boldsymbol{I}^{-1} = \boldsymbol{I}/\boldsymbol{I}^2$, $\boldsymbol{A}^\perp$ and $\boldsymbol{A}^{-\perp}$
1205.5935v1_F00445	40	■0.731	$\boldsymbol{\mathrm{I}}^{-1}=\boldsymbol{\mathrm{I}}/\boldsymbol{\mathrm{I}}^2,\boldsymbol{\mathrm{A}}^\perp$	$\boldsymbol{I}^{-1} = \boldsymbol{I}/\boldsymbol{I}^2, \boldsymbol{A}^\perp$	dual by $\boldsymbol{I}^{-1}$ instead of $\boldsymbol{I}$ is the inverse operation; it’s denoted by $\boldsymbol{A}^\perp$. Since $\boldsymbol{I}^{-1} = \boldsymbol{I}/\boldsymbol{I}^2$, $\boldsymbol{A}^\perp$ and $\boldsymbol{A}^{-\perp}$
1205.5935v1_F00446	40	■1.000	$\boldsymbol{\mathrm{I}}^2$	$\boldsymbol{I}^2$	differ only by a factor of $\boldsymbol{I}^2$, so the duality transformation is its own inverse up to at most a sign.
1205.5935v1_F00447	40	■0.999	$\boldsymbol{\mathrm{n}}-1$	$\boldsymbol{n} - 1$	Theorem 26. If the vector space is $\boldsymbol{n}$ -dimensional, every $\boldsymbol{n} - 1$ -vector is an $\boldsymbol{n} - 1$ -blade.
1205.5935v1_F00448	40	■0.999	$\boldsymbol{\mathrm{A}}_{\boldsymbol{\mathrm{n}}-1}$	$\boldsymbol{A}_{n-1}$	<i>Proof.</i> Let $\boldsymbol{A}_{n-1}$ be an $\boldsymbol{n} - 1$ -vector. The dual of $\boldsymbol{A}_{n-1}$ is a vector, so $\boldsymbol{A}_{n-1}$ is the dual
1205.5935v1_F00449	41	■1.000	$\boldsymbol{\mathrm{C}}=\boldsymbol{\mathrm{I}}^{-1}$	$\boldsymbol{C} = \boldsymbol{I}^{-1}$	the third of Eqs. (83) and set $\boldsymbol{C} = \boldsymbol{I}^{-1}$. That gets you the first equation. Then replace $\boldsymbol{B}$ with $\boldsymbol{B}^\perp$ and take
1205.5935v1_F00450	41	■1.000	$\boldsymbol{\mathrm{B}}^\perp$	$\boldsymbol{B}^\perp$	the third of Eqs. (83) and set $\boldsymbol{C} = \boldsymbol{I}^{-1}$. That gets you the first equation. Then replace $\boldsymbol{B}$ with $\boldsymbol{B}^\perp$ and take

Identifier	Page	Conf.	LaTeX source	Rendered	Image
1285.5935v1_F08451	41	■ 0.993	$\left(\mathbf{a} \downharpoonleft \mathbf{A}_r\right)^{\perp}=\mathbf{a} \wedge \mathbf{A}_r^{\perp}$	$(a \downharpoonleft \mathbf{A}_r)^{\perp}=a \wedge \mathbf{A}_r^{\perp}$	the inverse dual of both sides; that gets you the other equation.) A special case of this is $(a \downharpoonleft \mathbf{A}_r)^{\perp}=a \wedge \mathbf{A}_r^{\perp}$,
1285.5935v1_F08452	41	■ 1.000	$\mathbf{A}_r^{\perp}$	$\mathbf{A}_r^{\perp}$	which means vector a is orthogonal to subspace $\mathbf{A}_r$ iff a lies in $\mathbf{A}_r^{\perp}$. That’s further confirmation that duals
1285.5935v1_F08453	41	■ 0.996	$\mathbf{a} \downharpoonleft \mathbf{A}_r^{\perp}=0$	$a \downharpoonleft \mathbf{A}_r^{\perp}=0$	that $a \wedge \mathbf{A}_r=0$) and a <i>dual</i> representation (all a such that $a \downharpoonleft \mathbf{A}_r^{\perp}=0$). These two representations are both
1285.5935v1_F08454	42	■ 1.000	$\frac{1}{2}$	$\frac{1}{2}$	Notice the factor of $\frac{1}{2}$, which is not present in the usual definition of the commutator, used for example in
1285.5935v1_F08455	42	■ 1.000	$\mathbf{A} \times(\mathbf{B} \mathbf{C})$	$\mathbf{A} \times(\mathbf{B} \mathbf{C})$	on the algebra (it obeys the Leibnitz rule). Use this identity to expand $\mathbf{A} \times(\mathbf{B} \mathbf{C})$ and $\mathbf{A} \times(\mathbf{C} \mathbf{B})$ and take
1285.5935v1_F08456	42	■ 1.000	$\mathbf{A} \times(\mathbf{C} \mathbf{B})$	$\mathbf{A} \times(\mathbf{C} \mathbf{B})$	on the algebra (it obeys the Leibnitz rule). Use this identity to expand $\mathbf{A} \times(\mathbf{B} \mathbf{C})$ and $\mathbf{A} \times(\mathbf{C} \mathbf{B})$ and take
1285.5935v1_F08457	42	■ 1.000	$\mathbf{A}=\langle \mathbf{A}\rangle_{+}+\langle \mathbf{A}\rangle_{-}$	$\mathbf{A}=\langle \mathbf{A}\rangle_{+}+\langle \mathbf{A}\rangle_{-}$	be expressed nicely by decomposing a general multivector as $\mathbf{A}=\langle \mathbf{A}\rangle_{+}+\langle \mathbf{A}\rangle_{-}$ and recalling Eqs. (42) and
1285.5935v1_F08458	42	■ 1.000	$\mathbf{A}=\lambda+\mu\langle \mathbf{I}\rangle_{-}$	$\mathbf{A}=\lambda+\mu\langle \mathbf{I}\rangle_{-}$	3. $\mathbf{A}=\lambda+\mu\langle \mathbf{I}\rangle_{-}$.
1285.5935v1_F08459	42	■ 1.000	$\lambda+\mu \mathbf{I}$	$\lambda+\mu \mathbf{I}$	Item 3 is my sneaky way of saying $\mathbf{A}$ equals λ in even-dimensional vector spaces and $\lambda+\mu \mathbf{I}$ in odd-
1285.5935v1_F08460	42	■ 1.000	$\mathbf{a} \times \mathbf{I}=\mathbf{a} \wedge \mathbf{I}$	$a \times \mathbf{I}=a \wedge \mathbf{I}$	Now for item 3. The first of Eqs. (193) tells me that $\mathbf{a} \times \mathbf{I}=\mathbf{a} \wedge \mathbf{I}$ if the vector space is
1285.5935v1_F08461	42	■ 0.958	$\mathbf{a} \downharpoonleft \mathbf{I}$	$a \downharpoonleft \mathbf{I}$	odd-dimensional and $a \downharpoonleft \mathbf{I}$ if the space is even-dimensional. Now $a \wedge \mathbf{I}=0$ and $a \downharpoonleft \mathbf{I} \neq 0$ for all
1285.5935v1_F08462	42	■ 0.958	$\mathbf{a} \wedge \mathbf{I}=0$	$a \wedge \mathbf{I}=0$	odd-dimensional and $a \downharpoonleft \mathbf{I}$ if the space is even-dimensional. Now $a \wedge \mathbf{I}=0$ and $a \downharpoonleft \mathbf{I} \neq 0$ for all
1285.5935v1_F08463	42	■ 0.958	$\mathbf{a} \downharpoonleft \mathbf{I} \neq 0$	$a \downharpoonleft \mathbf{I} \neq 0$	odd-dimensional and $a \downharpoonleft \mathbf{I}$ if the space is even-dimensional. Now $a \wedge \mathbf{I}=0$ and $a \downharpoonleft \mathbf{I} \neq 0$ for all
1285.5935v1_F08464	42	■ 0.614	$0< r < n$	$0< r < n$	To finish off, let $\mathbf{A}_r$ be an r -blade where $0< r < n$. If r is even, then $\mathbf{a} \times \mathbf{A}_r=\mathbf{a} \downharpoonleft \mathbf{A}_r$, and if
1285.5935v1_F08465	42	■ 0.614	$\left(\mathbf{a} \times \mathbf{A}_r\right)^{\perp}=\mathbf{a} \downharpoonleft \mathbf{A}_r$	$a \times \mathbf{A}_r=a \downharpoonleft \mathbf{A}_r$	To finish off, let $\mathbf{A}_r$ be an r -blade where $0< r < n$. If r is even, then $\mathbf{a} \times \mathbf{A}_r=\mathbf{a} \downharpoonleft \mathbf{A}_r$, and if
1285.5935v1_F08466	42	■ 1.000	$\mathbf{a} \times \mathbf{A}_r=\mathbf{a} \wedge \mathbf{A}_r$	$a \times \mathbf{A}_r=a \wedge \mathbf{A}_r$	If r is odd, then $\mathbf{a} \times \mathbf{A}_r=\mathbf{a} \wedge \mathbf{A}_r$. Since $r < n$ there certainly exists a vector a outside $\mathbf{A}_r$, in
1285.5935v1_F08467	42	■ 1.000	$r < n$	$r < n$	If r is odd, then $\mathbf{a} \times \mathbf{A}_r=\mathbf{a} \wedge \mathbf{A}_r$. Since $r < n$ there certainly exists a vector a outside $\mathbf{A}_r$, in
1285.5935v1_F08468	43	■ 0.623	$r \geq 2$	$r \geq 2$	The first equation is obvious when $r \geq 2$; for $r < 2$, recall that in such cases $\mathbf{A}_2 \downharpoonleft \mathbf{A}_r=0$. The
1285.5935v1_F08469	43	■ 0.623	$r < 2$	$r < 2$	The first equation is obvious when $r \geq 2$; for $r < 2$, recall that in such cases $\mathbf{A}_2 \downharpoonleft \mathbf{A}_r=0$. The
1285.5935v1_F08470	43	■ 0.623	$\mathbf{A}_2 \downharpoonleft \mathbf{A}_r=0$	$\mathbf{A}_2 \downharpoonleft \mathbf{A}_r=0$	The first equation is obvious when $r \geq 2$; for $r < 2$, recall that in such cases $\mathbf{A}_2 \downharpoonleft \mathbf{A}_r=0$. The
1285.5935v1_F08471	43	■ 1.000	$j=1$	$j=1$	under interchange and Eq. (140) when $j=1$. Subtracting these equations yields
1285.5935v1_F08472	43	■ 1.000	$\mathbf{A}=\mathbf{A}_2$	$\mathbf{A}=\mathbf{A}_2$	Since commutation with a bivector is grade preserving, the identity in Eq. (189) still holds if $\mathbf{A}=\mathbf{A}_2$
1285.5935v1_F08473	43	■ 1.000	$\mathbf{B}_{r-1}=\mathbf{a}_1 \wedge \mathbf{a}_2 \wedge \cdots \wedge \mathbf{a}_{r-1}$	$\mathbf{B}_{r-1}=\mathbf{a}_1 \wedge \mathbf{a}_2 \wedge \cdots \wedge \mathbf{a}_{r-1}$	where $\mathbf{B}_{r-1}=\mathbf{a}_1 \wedge \mathbf{a}_2 \wedge \cdots \wedge \mathbf{a}_{r-1}$. Applying the last of Eqs. (197) and the $r-1$ result yields
1285.5935v1_F08474	44	■ 1.000	$\mathbb{R}^2$	$\mathbb{R}^2$	First I’ll consider the real plane $\mathbb{R}^2$ with the Euclidean scalar product; this is often denoted $\mathbb{E}^2$. It has an
1285.5935v1_F08475	44	■ 1.000	$\mathbb{E}^2$	$\mathbb{E}^2$	First I’ll consider the real plane $\mathbb{R}^2$ with the Euclidean scalar product; this is often denoted $\mathbb{E}^2$. It has an
1285.5935v1_F08476	44	■ 1.000	$\left\{\mathbf{e}_1, \mathbf{e}_2\right\}$	$\left\{e_1, e_2\right\}$	orthonormal basis $\left\{e_1, e_2\right\}$, which produces a geometric algebra spanned by the elements 1, e_1 , e_2 , and $e_1 e_2$.
1285.5935v1_F08477	44	■ 1.000	$1, \mathbf{e}_1, \mathbf{e}_2$	$1, e_1, e_2$	orthonormal basis $\left\{e_1, e_2\right\}$, which produces a geometric algebra spanned by the elements 1, e_1 , e_2 , and $e_1 e_2$.
1285.5935v1_F08478	44	■ 1.000	$\mathbf{e}_1 \mathbf{e}_2$	$e_1 e_2$	orthonormal basis $\left\{e_1, e_2\right\}$, which produces a geometric algebra spanned by the elements 1, e_1 , e_2 , and $e_1 e_2$.
1285.5935v1_F08479	44	■ 0.999	$\mathbf{I}^2=-1$	$\mathbf{I}^2=-1$	$\mathbf{I}^2=-1$. It defines a right-handed orientation, and a few examples show that all vectors anticommute with
1285.5935v1_F08480	44	■ 1.000	$\pi / 2$	$\pi / 2$	a vector clockwise through $\pi / 2$. Similarly, right multiplication rotates a vector counterclockwise through the
1285.5935v1_F08481	44	■ 0.978	$\mathbf{Z}=\mathbf{x}+\mathbf{I} \mathbf{y}$	$\mathbf{Z}=\mathbf{x}+\mathbf{I} \mathbf{y}$	A generic even multivector can be written $\mathbf{Z}=\mathbf{x}+\mathbf{I} \mathbf{y}$ where x and y are real numbers and $\mathbf{I}^2=-1$; in
1285.5935v1_F08482	44	■ 0.978	$\mathbf{y}$	$\mathbf{y}$	A generic even multivector can be written $\mathbf{Z}=\mathbf{x}+\mathbf{I} \mathbf{y}$ where x and y are real numbers and $\mathbf{I}^2=-1$; in
1285.5935v1_F08483	44	■ 1.000	$\mathbf{z}=r e^{i \theta}$	$\mathbf{z}=r e^{i \theta}$	the polar form $\mathbf{z}=r e^{i \theta}$ reminds us that $\mathbf{z}$ also represents a rotation through angle θ followed by a dilatation
1285.5935v1_F08484	44	■ 1.000	$\mathbf{z}=\mathbf{x} \mathbf{e}_1+\mathbf{y} \mathbf{e}_2$	$\mathbf{z}=x e_1+y e_2$	the space of vectors; a generic vector in $\mathbb{E}^2$ takes the form $\mathbf{z}=\mathbf{x} \mathbf{e}_1+\mathbf{y} \mathbf{e}_2$ where x and y are real numbers,
1285.5935v1_F08485	44	■ 1.000	$\mathbf{z}^{\prime}=-\mathbf{e}_2 \mathbf{z} \mathbf{e}_2$	$\mathbf{z}^{\prime}=-e_2 z e_2$	part. Therefore complex conjugation is a reflection along the e_2 axis, which takes $\mathbf{z}$ to $\mathbf{z}^{\prime}=-e_2 \mathbf{z} e_2$. What
1285.5935v1_F08486	45	■ 1.000	$\mathbf{w}$	$\mathbf{w}$	taking the reverse. Now let $\mathbf{w}$ and $\mathbf{z}$ be vectors with corresponding even elements W and Z ; it follows that
1285.5935v1_F08487	45	■ 1.000	$\mathbf{W}$	$\mathbf{W}$	taking the reverse. Now let $\mathbf{w}$ and $\mathbf{z}$ be vectors with corresponding even elements W and Z ; it follows that
1285.5935v1_F08488	45	■ 1.000	$\mathbf{Z}$	$\mathbf{Z}$	taking the reverse. Now let $\mathbf{w}$ and $\mathbf{z}$ be vectors with corresponding even elements W and Z ; it follows that
1285.5935v1_F08489	45	■ 1.000	$\mathbf{r} \exp (-\mathbf{I} \theta)$	$\mathbf{r} \exp (-\mathbf{I} \theta)$	Now for rotations. An element $\mathbf{W}$ of the even subalgebra has a polar form $\mathbf{r} \exp (-\mathbf{I} \theta)$ for some r and θ .
1285.5935v1_F08490	45	■ 0.911	$\mathbf{R}=\exp (-\mathbf{I} \theta / 2)$	$\mathbf{R}=\exp (-\mathbf{I} \theta / 2)$	where I defined $\mathbf{R}=\exp (-\mathbf{I} \theta / 2)$. Clearly $\mathbf{R}$ is a rotor, so multiplication by $\mathbf{W}$ performs a counterclockwise
1285.5935v1_F08491	45	■ 1.000	$\exp (\mathbf{I} \theta)$	$\exp (\mathbf{I} \theta)$	by $\exp (\mathbf{I} \theta)$, exactly as you’d expect.
1285.5935v1_F08492	45	■ 1.000	$\mathbb{E}^3$	$\mathbb{E}^3$	Much that is true in two dimensions carries over to three: $\mathbb{E}^3$ has an orthonormal basis $\left\{e_1, e_2, e_3\right\}$, so its
1285.5935v1_F08493	45	■ 1.000	$\left\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\right\}$	$\left\{e_1, e_2, e_3\right\}$	Much that is true in two dimensions carries over to three: $\mathbb{E}^3$ has an orthonormal basis $\left\{e_1, e_2, e_3\right\}$, so its
1285.5935v1_F08494	45	■ 1.000	$1, \mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3, \mathbf{e}_1 \mathbf{e}_2, \mathbf{e}_1 \mathbf{e}_3, \mathbf{e}_2 \mathbf{e}_3$	$1, e_1, e_2, e_3, e_1 e_2, e_1 e_3, e_2 e_3$	geometric algebra is spanned by 1, e_1 , e_2 , e_3 , $e_1 e_2$, $e_1 e_3$, $e_2 e_3$, and the volume element $e_1 e_2 e_3$. This volume
1285.5935v1_F08495	45	■ 1.000	$\mathbf{e}_1 \mathbf{e}_2 \mathbf{e}_3$	$e_1 e_2 e_3$	geometric algebra is spanned by 1, e_1 , e_2 , e_3 , $e_1 e_2$, $e_1 e_3$, $e_2 e_3$, and the volume element $e_1 e_2 e_3$. This volume
1285.5935v1_F08496	45	■ 0.998	$\left \mathbf{I}\right ^2=\mathbf{I}^{\dagger} \mathbf{I}=1$	$\left \mathbf{I}\right ^2=\mathbf{I}^{\dagger} \mathbf{I}=1$	element is also denoted $\mathbf{I}$, defines a right-handed orientation, satisfies $\left \mathbf{I}\right ^2=\mathbf{I}^{\dagger} \mathbf{I}=1$, and squares to -1 .
1285.5935v1_F08497	46	■ 0.998	$-\mathbf{I}$	$-\mathbf{I}$	multiplication by $-\mathbf{I}$ and vectors and bivectors commute with $\mathbf{I}$, I can use Eq. (208) to write
1285.5935v1_F08498	46	■ 1.000	$\mathbf{W}^{\dagger} \mathbf{Z}$	$\mathbf{W}^{\dagger} \mathbf{Z}$	which is the three-dimensional analog of the product $\mathbf{W}^{\dagger} \mathbf{Z}$ of complex numbers $\mathbf{W}$ and $\mathbf{Z}$. Another popular
1285.5935v1_F08499	46	■ 1.000	$\mathbf{a} \cdot \mathbf{b} \times \mathbf{c}$	$\mathbf{a} \cdot \mathbf{b} \times \mathbf{c}$	product in traditional vector algebra is the triple product $\mathbf{a} \cdot \mathbf{b} \times \mathbf{c}$, which relates to geometric algebra by

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1205.5935v1_F00500	46	■ 1.000	<code>a \times (b \times c)</code>	$a \times (b \times c)$	This form makes the cyclic property of the triple product obvious. The triple cross product $a \times (b \times c)$ is
1205.5935v1_F00501	46	■ 0.832	<code>(a \wedge b)^\perp = a \downharpoonleft b^\perp</code>	$(a \wedge b)^\perp = a \downharpoonleft b^\perp$	I'd like to say a little more about cross products. Since $(a \wedge b)^\perp = a \downharpoonleft b^\perp$ and in three dimensions $b^\perp$ is a
1205.5935v1_F00502	46	■ 0.832	<code>b^\perp</code>	$b^\perp$	I'd like to say a little more about cross products. Since $(a \wedge b)^\perp = a \downharpoonleft b^\perp$ and in three dimensions $b^\perp$ is a
1205.5935v1_F00503	46	■ 1.000	<code>a \times b</code>	$a \times b$	bivector, it follows that $a \times b$ is also the inner product of a and a bivector. You may recall that in classical
1205.5935v1_F00504	46	■ 1.000	<code>\boldsymbol{v}=\boldsymbol{\omega} \times \boldsymbol{r}</code>	$\boldsymbol{v} = \boldsymbol{\omega} \times \boldsymbol{r}$	mechanics, linear and angular velocity are related by a cross product: $\boldsymbol{v} = \boldsymbol{\omega} \times \boldsymbol{r}$. The geometric algebra
1205.5935v1_F00505	46	■ 1.000	<code>v=\Omega\left\lfloor r\right.</code>	$v = \Omega \lfloor r$	equivalent is $v = \Omega \lfloor r$, where $\Omega = \boldsymbol{\omega}^\perp$ is an angular velocity bivector in the instantaneous plane of rotation.
1205.5935v1_F00506	46	■ 1.000	<code>\Omega=\boldsymbol{\omega}^\perp</code>	$\Omega = \boldsymbol{\omega}^\perp$	equivalent is $v = \Omega \lfloor r$, where $\Omega = \boldsymbol{\omega}^\perp$ is an angular velocity bivector in the instantaneous plane of rotation.
1205.5935v1_F00507	46	■ 1.000	<code>\boldsymbol{F}=q \boldsymbol{v} \times \boldsymbol{B}</code>	$\boldsymbol{F} = q\boldsymbol{v} \times \boldsymbol{B}$	the magnetic part of the Lorentz force on a point charge is $\boldsymbol{F} = q\boldsymbol{v} \times \boldsymbol{B}$, where $\boldsymbol{B}$ is the magnetic field
1205.5935v1_F00508	46	■ 1.000	<code>\boldsymbol{B}</code>	$\boldsymbol{B}$	the magnetic part of the Lorentz force on a point charge is $\boldsymbol{F} = q\boldsymbol{v} \times \boldsymbol{B}$, where $\boldsymbol{B}$ is the magnetic field
1205.5935v1_F00509	46	■ 0.777	<code>F=q v \downharpoonleft B</code>	$F = qv \downharpoonleft B$	vector. In geometric algebra this becomes $F = qv \downharpoonleft B$, where $B = \boldsymbol{B}^\perp$ is the magnetic field bivector. I'll
1205.5935v1_F00510	46	■ 0.777	<code>B=B^\perp</code>	$B = B^\perp$	vector. In geometric algebra this becomes $F = qv \downharpoonleft B$, where $B = \boldsymbol{B}^\perp$ is the magnetic field bivector. I'll
1205.5935v1_F00511	46	■ 0.884	<code>\boldsymbol{A}_{\mathbf{2}}</code>	$\boldsymbol{A}_2$	are multiples of $\boldsymbol{I}$), and I can show using geometry that all bivectors are 2-blades. Consider two 2-blades $\boldsymbol{A}_2$
1205.5935v1_F00512	46	■ 1.000	<code>\boldsymbol{B}_{\mathbf{2}}</code>	$\boldsymbol{B}_2$	and $\boldsymbol{B}_2$; each represents a plane passing through the origin, and any two such planes in three dimensions
1205.5935v1_F00513	46	■ 1.000	<code>\boldsymbol{A}_{\mathbf{2}}=a \wedge b</code>	$\boldsymbol{A}_2 = a \wedge b$	share a common line. Therefore $\boldsymbol{A}_2 = a \wedge b$ and $\boldsymbol{B}_2 = a \wedge c$ where a is a vector along the line shared by the
1205.5935v1_F00514	46	■ 1.000	<code>\boldsymbol{B}_{\mathbf{2}}=a \wedge c</code>	$\boldsymbol{B}_2 = a \wedge c$	share a common line. Therefore $\boldsymbol{A}_2 = a \wedge b$ and $\boldsymbol{B}_2 = a \wedge c$ where a is a vector along the line shared by the
1205.5935v1_F00515	46	■ 1.000	<code>a=a_{\parallel}+a_{\perp}</code>	$a = a_\parallel + a_\perp$	like; next I'll do a vector times a bivector. Let a be a vector and $\boldsymbol{B}$ be a bivector; then $a = a_\parallel + a_\perp$ where
1205.5935v1_F00516	46	■ 1.000	<code>a_{\parallel}</code>	$a_\parallel$	$a_\parallel$ lies in the plane determined by $\boldsymbol{B}$ and $a_\perp$ is perpendicular to it. In that case there exists a vector b
1205.5935v1_F00517	46	■ 1.000	<code>a_{\perp}</code>	$a_\perp$	$a_\parallel$ lies in the plane determined by $\boldsymbol{B}$ and $a_\perp$ is perpendicular to it. In that case there exists a vector b
1205.5935v1_F00518	46	■ 1.000	<code>\boldsymbol{B}=a_{\parallel} b</code>	$\boldsymbol{B} = a_\parallel b$	perpendicular to $a_\parallel$ such that $\boldsymbol{B} = a_\parallel b$, so
1205.5935v1_F00519	46	■ 0.973	<code>a \downharpoonleft \boldsymbol{B}</code>	$a \downharpoonleft \boldsymbol{B}$	the trivector defined by $\boldsymbol{B}$ and the component of a perpendicular to it. Clearly the vector is $a \downharpoonleft \boldsymbol{B}$ and the
1205.5935v1_F00520	46	■ 1.000	<code>a \wedge \boldsymbol{B}</code>	$a \wedge \boldsymbol{B}$	trivector is $a \wedge \boldsymbol{B}$. The trivector can also be written
1205.5935v1_F00521	47	■ 1.000	<code>\boldsymbol{T}</code>	$\boldsymbol{T}$	Vector times trivector is even easier. If a is a vector and $\boldsymbol{T}$ is a trivector, then $\boldsymbol{T} = abc$ where b and c
1205.5935v1_F00522	47	■ 1.000	<code>\boldsymbol{T}=a b c</code>	$\boldsymbol{T} = abc$	Vector times trivector is even easier. If a is a vector and $\boldsymbol{T}$ is a trivector, then $\boldsymbol{T} = abc$ where b and c
1205.5935v1_F00523	47	■ 0.922	<code>a \boldsymbol{T}</code>	$a\boldsymbol{T}$	So $a\boldsymbol{T}$ is a bivector representing the plane to which a is perpendicular. This is clearly $a \downharpoonleft \boldsymbol{T}$.
1205.5935v1_F00524	47	■ 0.922	<code>a \downharpoonleft \boldsymbol{T}</code>	$a \downharpoonleft \boldsymbol{T}$	So $a\boldsymbol{T}$ is a bivector representing the plane to which a is perpendicular. This is clearly $a \downharpoonleft \boldsymbol{T}$.
1205.5935v1_F00525	47	■ 0.999	<code>\boldsymbol{A}_{\mathbf{2}}=b a</code>	$\boldsymbol{A}_2 = ba$	vector a lie along the direction shared by bivectors $\boldsymbol{A}_2$ and $\boldsymbol{B}_2$, so $\boldsymbol{A}_2 = ba$ and $\boldsymbol{B}_2 = ac$ where b and c are
1205.5935v1_F00526	47	■ 0.999	<code>\boldsymbol{B}_{\mathbf{2}}=a c</code>	$\boldsymbol{B}_2 = ac$	vector a lie along the direction shared by bivectors $\boldsymbol{A}_2$ and $\boldsymbol{B}_2$, so $\boldsymbol{A}_2 = ba$ and $\boldsymbol{B}_2 = ac$ where b and c are
1205.5935v1_F00527	47	■ 0.976	<code>\left.\boldsymbol{A}_{\mathbf{2}}\right\rfloor \boldsymbol{B}_2=\boldsymbol{A}_2 \lfloor \boldsymbol{B}_2</code> <code>\boldsymbol{B}_{\mathbf{2}}=\boldsymbol{A}_{\mathbf{2}} \lfloor \boldsymbol{B}_2</code> <code>\left.\boldsymbol{B}_2\right\rfloor \boldsymbol{A}_2=\boldsymbol{B}_2 \lfloor \boldsymbol{A}_2</code>	$\boldsymbol{A}_2 \rfloor \boldsymbol{B}_2 = \boldsymbol{A}_2 \lfloor \boldsymbol{B}_2$	line. The first term is $\boldsymbol{A}_2 \rfloor \boldsymbol{B}_2 = \boldsymbol{A}_2 \lfloor \boldsymbol{B}_2$ and the second term is $\boldsymbol{A}_2 \times \boldsymbol{B}_2$.
1205.5935v1_F00528	47	■ 0.976	<code>\boldsymbol{A}_{\mathbf{2}} \times \boldsymbol{B}_2</code>	$\boldsymbol{A}_2 \times \boldsymbol{B}_2$	line. The first term is $\boldsymbol{A}_2 \rfloor \boldsymbol{B}_2 = \boldsymbol{A}_2 \lfloor \boldsymbol{B}_2$ and the second term is $\boldsymbol{A}_2 \times \boldsymbol{B}_2$.
1205.5935v1_F00529	47	■ 1.000	<code>\boldsymbol{B}=a b</code>	$\boldsymbol{B} = ab$	Next, a bivector times a trivector: if bivector $\boldsymbol{B} = ab$ where a and b are perpendicular, then there exists
1205.5935v1_F00530	47	■ 1.000	<code>\boldsymbol{T}=b a c</code>	$\boldsymbol{T} = bac$	vector c perpendicular to a and b such that trivector $\boldsymbol{T} = bac$, in which case
1205.5935v1_F00531	47	■ 0.984	<code>\boldsymbol{B} \rfloor \boldsymbol{T}</code>	$\boldsymbol{B} \rfloor \boldsymbol{T}$	also $\boldsymbol{B} \rfloor \boldsymbol{T}$.
1205.5935v1_F00532	47	■ 1.000	<code>\hat{\boldsymbol{x}}, \hat{\boldsymbol{y}}</code>	$\hat{\boldsymbol{x}}, \hat{\boldsymbol{y}}$	a Euclidean Clifford algebra, but the idea that they are literally matrix representations of $\hat{\boldsymbol{x}}, \hat{\boldsymbol{y}}$, and $\hat{\boldsymbol{z}}$ is not
1205.5935v1_F00533	47	■ 1.000	<code>\hat{\boldsymbol{z}}</code>	$\hat{\boldsymbol{z}}$	a Euclidean Clifford algebra, but the idea that they are literally matrix representations of $\hat{\boldsymbol{x}}, \hat{\boldsymbol{y}}$, and $\hat{\boldsymbol{z}}$ is not
1205.5935v1_F00534	47	■ 1.000	<code>\boldsymbol{B}_i</code>	$\boldsymbol{B}_i$	so the even subalgebra of the algebra on $\mathbb{E}^3$, which is spanned by 1 and the $\boldsymbol{B}_i$, is isomorphic to the
1205.5935v1_F00535	48	■ 1.000	<code>P_u(v)</code>	$P_u(v)$	(The second parts of each equation are easy to verify.) $P_u(v)$ is parallel to u , $R_u(v)$ is orthogonal to u , and
1205.5935v1_F00536	48	■ 1.000	<code>u, R_u(v)</code>	$u, R_u(v)$	(The second parts of each equation are easy to verify.) $P_u(v)$ is parallel to u , $R_u(v)$ is orthogonal to u , and
1205.5935v1_F00537	48	■ 1.000	<code>P_u(v)+R_u(v)=v</code>	$P_u(v) + R_u(v) = v$	$P_u(v) + R_u(v) = v$. These operations require u to be invertible, so it can't be a null vector. I promised in
1205.5935v1_F00538	48	■ 1.000	<code>S</code>	S	Let's take a moment to consider the general notion of projection into a subspace. Let S be a subspace (S is
1205.5935v1_F00539	48	■ 1.000	<code>v \in S</code>	$v \in S$	not a blade this time; it really is the subspace itself) and let a be a vector not in S . Then for any $v \in S$ I can
1205.5935v1_F00540	48	■ 1.000	<code>a=v+(a-v)</code>	$a = v + (a - v)$	write $a = v + (a - v)$, which is the sum of a vector in S and a vector not in S . So which v is the “projection”
1205.5935v1_F00541	48	■ 1.000	<code>S_1</code>	S_1	of a into S ? We can't say without further information. For example, consider two subspaces S_1 and S_2 that
1205.5935v1_F00542	48	■ 1.000	<code>S_2</code>	S_2	of a into S ? We can't say without further information. For example, consider two subspaces S_1 and S_2 that
1205.5935v1_F00543	48	■ 1.000	<code>S^\perp</code>	$S^\perp$	two terms, one in subspace S and one in $S^\perp$, the orthogonal complement of S . This works only if S and
1205.5935v1_F00544	48	■ 0.603	<code>P_{\boldsymbol{A_r}}(a)+R_{\boldsymbol{A_r}}(a)=a</code>	$P_{\boldsymbol{A_r}}(a) + R_{\boldsymbol{A_r}}(a) = a$	<i>Proof.</i> First, it's clear that $P_{\boldsymbol{A_r}}(a) + R_{\boldsymbol{A_r}}(a) = a$. Now, $a \downharpoonleft \boldsymbol{A_r}$ is the dual of a by $\boldsymbol{A_r}^{-1}$; it is
1205.5935v1_F00545	48	■ 0.603	<code>\left.a\right\rfloor \boldsymbol{A_r}</code>	$a \downharpoonleft \boldsymbol{A_r}$	<i>Proof.</i> First, it's clear that $P_{\boldsymbol{A_r}}(a) + R_{\boldsymbol{A_r}}(a) = a$. Now, $a \downharpoonleft \boldsymbol{A_r}$ is the dual of a by $\boldsymbol{A_r}^{-1}$; it is
1205.5935v1_F00546	48	■ 1.000	<code>\left(a \wedge \boldsymbol{A_r}\right)^{-1}</code>	$(a \wedge \boldsymbol{A_r})^{-1}$	with $\boldsymbol{A_r}^{-1}$ equals the right inner product, and is the dual of $\boldsymbol{A_r}^{-1}$ by $(a \wedge \boldsymbol{A_r})^{-1}$, so the result is
1205.5935v1_F00547	49	■ 0.998	<code>P_{\boldsymbol{A_r}}(a)</code>	$P_{\boldsymbol{A_r}}(a)$	I can demonstrate directly that $P_{\boldsymbol{A_r}}(a)$ lies in $\boldsymbol{A_r}$:
1205.5935v1_F00548	49	■ 1.000	<code>R_{\boldsymbol{A_r}}(a)</code>	$R_{\boldsymbol{A_r}}(a)$	Similarly, I can show that $R_{\boldsymbol{A_r}}(a)$ is orthogonal to $\boldsymbol{A_r}$ as follows:
1205.5935v1_F00549	49	■ 1.000	<code>b_1=a_1</code>	$b_1 = a_1$	set $\{b_j\}_{j=1,\dots,r}$ as follows. Let $b_1 = a_1$ to start with. Then b_2 equals a_2 minus its projection onto b_1 , or
1205.5935v1_F00550	49	■ 1.000	<code>b_2</code>	b_2	set $\{b_j\}_{j=1,\dots,r}$ as follows. Let $b_1 = a_1$ to start with. Then b_2 equals a_2 minus its projection onto b_1 , or
1205.5935v1_F00551	49	■ 1.000	<code>b_3</code>	b_3	equivalently the orthogonal rejection of a_2 from b_1 . Next, b_3 equals the orthogonal rejection of a_3 from the

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1205.5935v1_F00552	49	■ 1.000	$\mathbf{a}_{\{3\}}$	a_3	equivalently the orthogonal rejection of a_2 from b_1 . Next, b_3 equals the orthogonal rejection of a_3 from the
1205.5935v1_F00553	49	■ 0.984	$\boldsymbol{B}_{\mathbf{j}} = \mathbf{b}_1 \wedge \cdots \wedge \mathbf{b}_j$	$\boldsymbol{B}_j = b_1 \wedge \cdots \wedge b_j$	2. For each j starting with 1, let $\boldsymbol{B}_j = b_1 \wedge \cdots \wedge b_j$.
1205.5935v1_F00554	49	■ 0.992	$\mathbf{b}_{\{j+1\}} = \mathbf{a}_{\{j+1\}} \wedge \boldsymbol{B}_{\mathbf{j}}$	$b_{j+1} = a_{j+1} \wedge \boldsymbol{B}_j \boldsymbol{B}_j^{-1}$	3. Then let $b_{j+1} = a_{j+1} \wedge \boldsymbol{B}_j \boldsymbol{B}_j^{-1}$.
1205.5935v1_F00555	49	■ 1.000	$\boldsymbol{B}_{\mathbf{j}}$	$\boldsymbol{B}_j$	This procedure will work only if each $\boldsymbol{B}_j$ is invertible, which is why it is normally used only in Euclidean
1205.5935v1_F00556	49	■ 0.999	$\boldsymbol{A}_{\mathbf{r}} \wedge \boldsymbol{B}_{\mathbf{s}} = \boldsymbol{A}_{\mathbf{r}} \boldsymbol{B}_{\mathbf{s}}$	$\boldsymbol{A}_r \wedge \boldsymbol{B}_s = \boldsymbol{A}_r \boldsymbol{B}_s$	directly. By Theorem 18, $\boldsymbol{A}_r \wedge \boldsymbol{B}_s = \boldsymbol{A}_r \boldsymbol{B}_s$, so using the first of Eqs. (74) I find
1205.5935v1_F00557	49	■ 0.798	$a \rfloor \boldsymbol{B}_{\mathbf{s}}$	$a \rfloor \boldsymbol{B}_s$	Now let's work on that last term. If $\boldsymbol{A}_r$ and $\boldsymbol{B}_s$ are orthogonal, $\boldsymbol{A}_r$ and $a \rfloor \boldsymbol{B}_s$ are too, so their product is
1205.5935v1_F00558	49	■ 0.998	$(-1)^{r(s-1)}$	$(-1)^{r(s-1)}$	an outer product, so I can interchange them and pick up a factor of $(-1)^{r(s-1)}$. And since $\boldsymbol{A}_r^{-1}$ and $\boldsymbol{B}_s^{-1}$
1205.5935v1_F00559	49	■ 1.000	$(-1)^{rs}$	$(-1)^{rs}$	a factor of $(-1)^{rs}$. Putting all this in Eq. (228),
1205.5935v1_F00560	50	■ 1.000	$\mathbf{b}_{\{j\}}$	$\{b_j\}_{j=1,\dots,s}$	Now suppose the set $\{b_j\}_{j=1,\dots,s}$ is linearly independent but their projections are not. That would happen
1205.5935v1_F00561	50	■ 0.977	$\mathbf{b}_{\{j\}}$	$\{b_j\}_{j=1,\dots,s}$	Theorem 32. For any invertible blade $\boldsymbol{A}_r$ and vectors $\{b_j\}_{j=1,\dots,s}$,
1205.5935v1_F00562	50	■ 0.999	$P_{\boldsymbol{A}_r}(\mathbf{b}_{\mathbf{j}}) + R_{\boldsymbol{A}_r}(\mathbf{b}_{\mathbf{j}})$	$b_j = P_{\boldsymbol{A}_r}(b_j) + R_{\boldsymbol{A}_r}(b_j)$	<i>Proof.</i> I start with the first equation. Since each $b_j = P_{\boldsymbol{A}_r}(b_j) + R_{\boldsymbol{A}_r}(b_j)$, the outer product
1205.5935v1_F00563	50	■ 0.830	$P_{\boldsymbol{A}_r}(\mathbf{b}_1) \wedge \cdots \wedge P_{\boldsymbol{A}_r}(\mathbf{b}_s)$	$P_{\boldsymbol{A}_r}(b_1) \wedge \cdots \wedge P_{\boldsymbol{A}_r}(b_s)$	$b_1 \wedge \cdots \wedge b_s$ can be written as a sum of terms, one of which equals $P_{\boldsymbol{A}_r}(b_1) \wedge \cdots \wedge P_{\boldsymbol{A}_r}(b_s)$ while
1205.5935v1_F00564	50	■ 1.000	$R_{\boldsymbol{A}_r}(\mathbf{b}_{\mathbf{j}})$	$R_{\boldsymbol{A}_r}(b_j)$	each of the others contains at least one $R_{\boldsymbol{A}_r}(b_j)$. Consider what happens to each term when
1205.5935v1_F00565	50	■ 1.000	$P_{\boldsymbol{A}_r}(\mathbf{b}_1) \wedge \cdots \wedge P_{\boldsymbol{A}_r}(\mathbf{b}_s)$	$P_{\boldsymbol{A}_r}(b_1) \wedge \cdots \wedge P_{\boldsymbol{A}_r}(b_s)$	you take the inner product with $\boldsymbol{A}_r$ and multiply by $\boldsymbol{A}_r^{-1}$. The term $P_{\boldsymbol{A}_r}(b_1) \wedge \cdots \wedge P_{\boldsymbol{A}_r}(b_s)$ lies
1205.5935v1_F00566	50	■ 0.952	$R_{\boldsymbol{A}_r}(\mathbf{b}_1) \wedge \cdots \wedge R_{\boldsymbol{A}_r}(\mathbf{b}_s)$	$R_{\boldsymbol{A}_r}(b_1) \wedge \cdots \wedge R_{\boldsymbol{A}_r}(b_s)$	one of them equals $R_{\boldsymbol{A}_r}(b_1) \wedge \cdots \wedge R_{\boldsymbol{A}_r}(b_s)$ while each of the others contains at least one $P_{\boldsymbol{A}_r}(b_j)$.
1205.5935v1_F00567	50	■ 0.952	$P_{\boldsymbol{A}_r}(\mathbf{b}_{\mathbf{j}})$	$P_{\boldsymbol{A}_r}(b_j)$	one of them equals $R_{\boldsymbol{A}_r}(b_1) \wedge \cdots \wedge R_{\boldsymbol{A}_r}(b_s)$ while each of the others contains at least one $P_{\boldsymbol{A}_r}(b_j)$.
1205.5935v1_F00568	50	■ 0.919	$\boldsymbol{P}_{\boldsymbol{A}_r}(\boldsymbol{B}_{\mathbf{s}}) = \boldsymbol{B}_{\mathbf{s}} \rfloor \boldsymbol{A}_r \boldsymbol{A}_r^{-1}$	$P_{\boldsymbol{A}_r}(\boldsymbol{B}_s) = \boldsymbol{B}_s \rfloor \boldsymbol{A}_r \boldsymbol{A}_r^{-1}$	Therefore $P_{\boldsymbol{A}_r}(\boldsymbol{B}_s) = \boldsymbol{B}_s \rfloor \boldsymbol{A}_r \boldsymbol{A}_r^{-1}$ and $R_{\boldsymbol{A}_r}(\boldsymbol{B}_s) = \boldsymbol{B}_s \wedge \boldsymbol{A}_r \boldsymbol{A}_r^{-1}$ for any blade $\boldsymbol{B}_s$. Taking the obvious
1205.5935v1_F00569	50	■ 0.919	$R_{\boldsymbol{A}_r}(\boldsymbol{B}_{\mathbf{s}}) = \boldsymbol{B}_{\mathbf{s}} \wedge \boldsymbol{A}_r \boldsymbol{A}_r^{-1}$	$R_{\boldsymbol{A}_r}(\boldsymbol{B}_s) = \boldsymbol{B}_s \wedge \boldsymbol{A}_r \boldsymbol{A}_r^{-1}$	Therefore $P_{\boldsymbol{A}_r}(\boldsymbol{B}_s) = \boldsymbol{B}_s \rfloor \boldsymbol{A}_r \boldsymbol{A}_r^{-1}$ and $R_{\boldsymbol{A}_r}(\boldsymbol{B}_s) = \boldsymbol{B}_s \wedge \boldsymbol{A}_r \boldsymbol{A}_r^{-1}$ for any blade $\boldsymbol{B}_s$. Taking the obvious
1205.5935v1_F00570	51	■ 0.529	$A \rfloor \boldsymbol{B}_{\mathbf{s}}$	$A \rfloor \boldsymbol{B}_s$	the norm squared of $A \rfloor \boldsymbol{B}_s$ equals the norm squared of $P_{\boldsymbol{B}_s}(A)$ times the norm squared of $\boldsymbol{B}_s$, and a similar
1205.5935v1_F00571	51	■ 0.529	$P_{\boldsymbol{B}_s}(A)$	$P_{\boldsymbol{B}_s}(A)$	the norm squared of $A \rfloor \boldsymbol{B}_s$ equals the norm squared of $P_{\boldsymbol{B}_s}(A)$ times the norm squared of $\boldsymbol{B}_s$, and a similar
1205.5935v1_F00572	51	■ 0.628	$P_{\boldsymbol{A}_r}(B) + R_{\boldsymbol{A}_r}(B) = B$	$P_{\boldsymbol{A}_r}(B) + R_{\boldsymbol{A}_r}(B) = B$	one property: we no longer have $P_{\boldsymbol{A}_r}(B) + R_{\boldsymbol{A}_r}(B) = B$ in general. This makes geometric sense, however,
1205.5935v1_F00573	52	■ 0.730	a^{*r}	a^{*r}	In the last line, a^{*r} means a is grade involuted r times; I introduced the notation back in Section 5.1.
1205.5935v1_F00574	52	■ 1.000	$\boldsymbol{A}_{\mathbf{r}} = a_1 a_2 \cdots a_r$	$\boldsymbol{A}_r = a_1 a_2 \cdots a_r$	Another way to arrive at this formula is to write $\boldsymbol{A}_r = a_1 a_2 \cdots a_r$ and reflect a along each of the a_j in
1205.5935v1_F00575	52	■ 1.000	$a B_{\mathbf{s}} a$	$a B_s a$	<i>Proof.</i> The theorem is true for versors if it's true for vectors, so I'll look at $a B_s a$. Using
1205.5935v1_F00576	52	■ 0.993	$(-1)^{s-1} (B_{\mathbf{s}} \rfloor a) \rfloor a = (-1)^{s-1} B_{\mathbf{s}} \rfloor (a \wedge a) = 0$	$(-1)^{s-1} (B_s \rfloor a) \rfloor a = (-1)^{s-1} B_s \rfloor (a \wedge a) = 0$	as $(-1)^{s-1} (B_s \rfloor a) \rfloor a = (-1)^{s-1} B_s \rfloor (a \wedge a) = 0$. Since Theorem 22 tells me that the inverse of
1205.5935v1_F00577	52	■ 1.000	$(\boldsymbol{A} \boldsymbol{B} \boldsymbol{A}^{-1}) \wedge (\boldsymbol{A} \boldsymbol{C} \boldsymbol{A}^{-1}) = \boldsymbol{A} (\boldsymbol{B} \wedge \boldsymbol{C}) \boldsymbol{A}^{-1}$	$(\boldsymbol{A} \boldsymbol{B} \boldsymbol{A}^{-1}) \wedge (\boldsymbol{A} \boldsymbol{C} \boldsymbol{A}^{-1}) = \boldsymbol{A} (\boldsymbol{B} \wedge \boldsymbol{C}) \boldsymbol{A}^{-1}$	Therefore if A is invertible, $(\boldsymbol{A} \boldsymbol{B} \boldsymbol{A}^{-1}) \wedge (\boldsymbol{A} \boldsymbol{C} \boldsymbol{A}^{-1}) = \boldsymbol{A} (\boldsymbol{B} \wedge \boldsymbol{C}) \boldsymbol{A}^{-1}$.
1205.5935v1_F00578	52	■ 1.000	C_t	C_t	<i>Proof.</i> The result is true for general B and C if it's true for B_s and C_t , and I've already
1205.5935v1_F00579	52	■ 1.000	$ A ^2 \neq 0$	$ A ^2 \neq 0$	If A is invertible, then $ A ^2 \neq 0$, so dividing both sides by $ A ^4$ yields the desired result.
1205.5935v1_F00580	52	■ 1.000	$ A ^4$	$ A ^4$	If A is invertible, then $ A ^2 \neq 0$, so dividing both sides by $ A ^4$ yields the desired result.
1205.5935v1_F00581	53	■ 1.000	$\boldsymbol{B}_{\mathbf{s}} = b_1 \wedge \cdots \wedge b_s$	$\boldsymbol{B}_s = b_1 \wedge \cdots \wedge b_s$	Now for reflections. If $\boldsymbol{B}_s = b_1 \wedge \cdots \wedge b_s$, then
1205.5935v1_F00582	53	■ 1.000	$\boldsymbol{I}^* = (-1)^n \boldsymbol{I}$	$\boldsymbol{I}^* = (-1)^n \boldsymbol{I}$	To show that this really is true, start with Eq. (178), grade involute both sides, and use $\boldsymbol{I}^* = (-1)^n \boldsymbol{I}$. Then
1205.5935v1_F00583	54	■ 0.999	$\boldsymbol{A}'_{\mathbf{r}} = a'_1 \wedge \cdots \wedge a'_r$	$\boldsymbol{A}'_r = a'_1 \wedge \cdots \wedge a'_r$	the argument for reflections: if $\boldsymbol{A}_r = a_1 \wedge \cdots \wedge a_r$, then $\boldsymbol{A}'_r = a'_1 \wedge \cdots \wedge a'_r$. Therefore the rotation by R is
1205.5935v1_F00584	54	■ 1.000	$S R S^{-1}$	$S R S^{-1}$	if R and S are rotors in the same plane, $S R S^{-1}$ should represent the same rotation as R . You can show
1205.5935v1_F00585	54	■ 0.993	$S R S^{-1} = R S S^{-1} = R$	$S R S^{-1} = R S S^{-1} = R$	$S R S^{-1} = R S S^{-1} = R$.

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1285.5935v1_F08586	54	■ 1.000	$R_{\{1\}}$	R_1	geometric algebra: the product of two rotations is another rotation. If R_1 represents the first rotation and
1285.5935v1_F08587	54	■ 1.000	$R_{\{2\}}$	R_2	R_2 the second, then their product is $R = R_2 R_1$. We lose no generality by demanding that both R_1 and
1285.5935v1_F08588	54	■ 1.000	$R=R_{\{2\}}\ R_{\{1\}}$	$R = R_2 R_1$	R_2 the second, then their product is $R = R_2 R_1$. We lose no generality by demanding that both R_1 and
1285.5935v1_F08589	54	■ 1.000	$R=\langle R\rangle+\langle R\rangle_2$	$R = \langle R \rangle + \langle R \rangle_2$	zero and two, so R is actually a scalar plus a bivector: $R = \langle R \rangle + \langle R \rangle_2$. Therefore $ R ^2 = 1$ becomes
1285.5935v1_F08590	54	■ 1.000	$ R ^{\wedge 2}=1$	$ R ^2 = 1$	zero and two, so R is actually a scalar plus a bivector: $R = \langle R \rangle + \langle R \rangle_2$. Therefore $ R ^2 = 1$ becomes
1285.5935v1_F08591	54	■ 1.000	$\langle R\rangle^{\wedge 2}+\left \langle R\rangle_2\right ^2=1$	$\langle R \rangle^2 + \langle R \rangle_2 ^2 = 1$	$\langle R \rangle^2 + \langle R \rangle_2 ^2 = 1$. That tells me that $R = \cos(\theta/2) - B \sin(\theta/2)$ for some θ and unit bivector B . And in
1285.5935v1_F08592	54	■ 1.000	$R=\cos (\theta / 2)-B \sin (\theta / 2)$	$R = \cos(\theta/2) - B \sin(\theta/2)$	$\langle R \rangle^2 + \langle R \rangle_2 ^2 = 1$. That tells me that $R = \cos(\theta/2) - B \sin(\theta/2)$ for some θ and unit bivector B . And in
1285.5935v1_F08593	54	■ 0.910	$R=\exp (-\boldsymbol{B} \theta / 2)$	$R = \exp(-\boldsymbol{B}\theta/2)$	three dimensions every bivector is a 2-blade, so $\boldsymbol{B}$ represents some plane, and thus $R = \exp(-\boldsymbol{B}\theta/2)$, which
1285.5935v1_F08594	54	■ 0.977	$R_{\{2\}}\ R_{\{1\}}$	$R_2 R_1$	result, because $R_2 R_1$ could have a 4-vector part.
1285.5935v1_F08595	54	■ 1.000	$\mathcal{G}^n$	$\mathcal{G}^n$	Now I'll consider a geometric algebra $\mathcal{G}^n$ in which the space of vectors has finite dimension n . Let $\{a_i\}_{i=1,\dots,n}$
1285.5935v1_F08596	54	■ 1.000	$\left\{a_i\right\}_{i=1, \ldots, n}$	$\{a_i\}_{i=1,\dots,n}$	Now I'll consider a geometric algebra $\mathcal{G}^n$ in which the space of vectors has finite dimension n . Let $\{a_i\}_{i=1,\dots,n}$
1285.5935v1_F08597	54	■ 1.000	$a_{\{1\}}\ a_{\{2\}}\ \cdots\ a_{\{r\}}$	$a_{i_1} a_{i_2} \cdots a_{i_r}$	a generic element of the algebra will be the sum of a scalar and terms of the form $a_{i_1} a_{i_2} \cdots a_{i_r}$ for $r \leq n$.
1285.5935v1_F08598	54	■ 1.000	$r \leq n$	$r \leq n$	a generic element of the algebra will be the sum of a scalar and terms of the form $a_{i_1} a_{i_2} \cdots a_{i_r}$ for $r \leq n$.
1285.5935v1_F08599	54	■ 1.000	$\left\{a_i\right\}$	$\{a_{i_i}\}$	Theorem 12 tells me that any such element is a linear combination of blades made up of the $\{a_{i_j}\}$; therefore
1285.5935v1_F08600	54	■ 1.000	$a_{\{1\}} \wedge a_{\{2\}} \wedge \cdots \wedge a_{\{r\}}$	$a_{i_1} \wedge a_{i_2} \wedge \cdots \wedge a_{i_r}$	the scalar 1 and the blades $a_{i_1} \wedge a_{i_2} \wedge \cdots \wedge a_{i_r}$ generate the whole geometric algebra. I'll now show that they
1285.5935v1_F08601	55	■ 1.000	$\left\{a^i\right\}_{i=1, \ldots, n}$	$\{a^i\}_{i=1,\dots,n}$	Given a frame $\{a_i\}_{i=1,\dots,n}$, another frame $\{a^i\}_{i=1,\dots,n}$ is called a <i>reciprocal frame</i> to the first if it satisfies
1285.5935v1_F08602	55	■ 1.000	$a^{\{i\}}$	a^i	If such a set of vectors exists, it is a frame because Eq. (256) guarantees that the a^i are linearly independent,
1285.5935v1_F08603	55	■ 1.000	$a^{\{j\}}$	a^j	so they form a basis. To construct such vectors, consider their definition: a^j should be orthogonal to all of
1285.5935v1_F08604	55	■ 1.000	$a_{\{N\}}=a_{\{1\}} \wedge a_{\{2\}} \wedge \cdots \wedge a_{\{n\}}$	$a_N = a_1 \wedge a_2 \wedge \cdots \wedge a_n$	Let $a_N = a_1 \wedge a_2 \wedge \cdots \wedge a_n$; then a_N is a (possibly unnormalized) volume element. (Even though a_N is a
1285.5935v1_F08605	55	■ 1.000	$a_{\{N\}}$	a_N	Let $a_N = a_1 \wedge a_2 \wedge \cdots \wedge a_n$; then a_N is a (possibly unnormalized) volume element. (Even though a_N is a
1285.5935v1_F08606	55	■ 1.000	$\left\{a^i\right\}$	$\{a^i\}$	$\{a^i\}$ is a reciprocal frame because, using the first of Eqs. (181),
1285.5935v1_F08607	55	■ 1.000	$i \neq j$	$i \neq j$	Now if $i \neq j$ then a_i equals one of the other vectors in the outer product, so the whole thing vanishes. If
1285.5935v1_F08608	55	■ 0.997	$i=j$	$i = j$	$i = j$, I move a_i past the first $i - 1$ vectors to its original spot, which cancels out the $(-1)^{j-1}$ prefactor.
1285.5935v1_F08609	55	■ 0.997	$i-1$	$i - 1$	$i = j$, I move a_i past the first $i - 1$ vectors to its original spot, which cancels out the $(-1)^{j-1}$ prefactor.
1285.5935v1_F08610	55	■ 1.000	$\left\{a_i\right\}$	$\{a_i\}$	Since both $\{a_i\}$ and $\{a^j\}$ are bases for the vectors, any vector v can be written $v = \sum v^i a_i$ or $v = \sum v_j a^j$.
1285.5935v1_F08611	55	■ 1.000	$\left\{a^j\right\}$	$\{a^j\}$	Since both $\{a_i\}$ and $\{a^j\}$ are bases for the vectors, any vector v can be written $v = \sum v^i a_i$ or $v = \sum v_j a^j$.
1285.5935v1_F08612	55	■ 1.000	$v=\sum v^{\{i\}}\ a_{\{i\}}$	$v = \sum v^i a_i$	Since both $\{a_i\}$ and $\{a^j\}$ are bases for the vectors, any vector v can be written $v = \sum v^i a_i$ or $v = \sum v_j a^j$.
1285.5935v1_F08613	55	■ 1.000	$v=\sum v_{\{j\}}\ a^{\{j\}}$	$v = \sum v_j a^j$	Since both $\{a_i\}$ and $\{a^j\}$ are bases for the vectors, any vector v can be written $v = \sum v^i a_i$ or $v = \sum v_j a^j$.
1285.5935v1_F08614	55	■ 0.424	$\left.v^{\{i\}}=v\right\rangle\lrcorner a^{\{i\}}$	$v^i = v \lrcorner a^i$	In fact, it's obvious that $v^i = v \lrcorner a^i$ and $v_j = v \lrcorner a_j$, so the components of v on either basis are easily calculated
1285.5935v1_F08615	55	■ 0.424	$\left.v_{\{j\}}=v\right\rangle\lrcorner a_{\{j\}}$	$v_j = v \lrcorner a_j$	In fact, it's obvious that $v^i = v \lrcorner a^i$ and $v_j = v \lrcorner a_j$, so the components of v on either basis are easily calculated
1285.5935v1_F08616	55	■ 1.000	$v^{\{i\}}=v * a^{\{i\}}$	$v^i = v * a^i$	Since v is a vector, the expressions for its components can be written $v^i = v * a^i$ and $v_j = v * a_j$, where
1285.5935v1_F08617	55	■ 1.000	$v_{\{j\}}=v * a_{\{j\}}$	$v_j = v * a_j$	Since v is a vector, the expressions for its components can be written $v^i = v * a^i$ and $v_j = v * a_j$, where
1285.5935v1_F08618	55	■ 1.000	$b^{\{i\}}$	b^i	components b^i and c_i .
1285.5935v1_F08619	55	■ 1.000	$c_{\{i\}}$	c_i	components b^i and c_i .
1285.5935v1_F08620	56	■ 0.968	$a^{\{i\}}=\sum_j\left(a^{\{i\}} \downharpoonleft a^{\{j\}}\right) a_{\{j\}}$	$a^i = \sum_j (a^i \lrcorner a^j) a_j$	To evaluate the second term, expand a^i on the original frame to get $a^i = \sum_j (a^i \lrcorner a^j) a_j$, so
1285.5935v1_F08621	56	■ 0.956	$a^{\{i\}} \downharpoonleft a^{\{j\}}$	$a^i \lrcorner a^j$	because $a^i \lrcorner a^j$ is symmetric in i and j while $a_i \wedge a_j$ is antisymmetric. The proof that $\sum_i a^i a_i = n$
1285.5935v1_F08622	56	■ 0.956	$a_{\{i\}} \wedge a_{\{j\}}$	$a_i \wedge a_j$	because $a^i \lrcorner a^j$ is symmetric in i and j while $a_i \wedge a_j$ is antisymmetric. The proof that $\sum_i a^i a_i = n$
1285.5935v1_F08623	56	■ 0.956	$\sum_i a^{\{i\}}\ a_{\{i\}}=n$	$\sum_i a^i a_i = n$	because $a^i \lrcorner a^j$ is symmetric in i and j while $a_i \wedge a_j$ is antisymmetric. The proof that $\sum_i a^i a_i = n$
1285.5935v1_F08624	56	■ 1.000	I	I	Before I continue, I need some fancy new notation. Let I be a string of indices $i_1, i_2, \dots, i_r$, and given a
1285.5935v1_F08625	56	■ 1.000	$i_{\{1\}}, i_{\{2\}}, \ldots, i_{\{r\}}$	$i_1, i_2, \dots, i_r$	Before I continue, I need some fancy new notation. Let I be a string of indices $i_1, i_2, \dots, i_r$, and given a
1285.5935v1_F08626	56	■ 1.000	$a_{\{I\}}$	a_I	string I let a_I be defined by
1285.5935v1_F08627	56	■ 1.000	$a^{\{I\}}$	a^I	and similarly for a^I . I will use the symbol N only to refer to the string $1, 2, \dots, n$, to be consistent with a_N
1285.5935v1_F08628	56	■ 1.000	N	N	and similarly for a^I . I will use the symbol N only to refer to the string $1, 2, \dots, n$, to be consistent with a_N
1285.5935v1_F08629	56	■ 1.000	$1, 2, \ldots, n$	$1, 2, \dots, n$	and similarly for a^I . I will use the symbol N only to refer to the string $1, 2, \dots, n$, to be consistent with a_N
1285.5935v1_F08630	56	■ 1.000	$a_{\{I\}}=a^{\{I\}}=1$	$a_I = a^I = 1$	in the previous section. I also allow I to be the “empty” sequence, in which case I define $a_I = a^I = 1$. Then
1285.5935v1_F08631	56	■ 1.000	$a_{\{I\}}=a^{\{I\}}=0$	$a_I = a^I = 0$	1. $a_I = a^I = 0$ iff the string I contains at least one index twice.
1285.5935v1_F08632	56	■ 1.000	J	J	2. If I and J contain the same elements but in a different order, then $a_I = (\operatorname{sgn} \sigma) a_J$ and $a^I = (\operatorname{sgn} \sigma) a^J$,
1285.5935v1_F08633	56	■ 1.000	$a_{\{I\}}=(\operatorname{operatorname{sgn}} \sigma) a_{\{J\}}$	$a_I = (\operatorname{sgn} \sigma) a_J$	2. If I and J contain the same elements but in a different order, then $a_I = (\operatorname{sgn} \sigma) a_J$ and $a^I = (\operatorname{sgn} \sigma) a^J$,
1285.5935v1_F08634	56	■ 1.000	$a^{\{I\}}=(\operatorname{operatorname{sgn}} \sigma) a^{\{J\}}$	$a^I = (\operatorname{sgn} \sigma) a^J$	2. If I and J contain the same elements but in a different order, then $a_I = (\operatorname{sgn} \sigma) a_J$ and $a^I = (\operatorname{sgn} \sigma) a^J$,
1285.5935v1_F08635	56	■ 1.000	$\left\{a_I\right\}$	$\{a_I\}$	3. Theorem 12 tells me that given a frame $\{a_i\}$ for the vectors, the set $\{a_I\}$ (or $\{a^I\}$) where I ranges
1285.5935v1_F08636	56	■ 1.000	$\left\{a^I\right\}$	$\{a^I\}$	3. Theorem 12 tells me that given a frame $\{a_i\}$ for the vectors, the set $\{a_I\}$ (or $\{a^I\}$) where I ranges
1285.5935v1_F08637	56	■ 1.000	$i_{\{1\}}<i_{\{2\}}<\cdots<i_{\{r\}}$	$i_1 < i_2 < \cdots < i_r$	over all <i>increasing</i> sequences generates $\mathcal{G}^n$. (I is an increasing sequence if $i_1 < i_2 < \cdots < i_r$.)

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1285.5935v1_F08638	56	■1.000	$\backslash\delta_{J}^I$	δ_J^I	where δ_J^I vanishes if either I or J repeats indices or if I is not a permutation of J (including having a
1285.5935v1_F08639	56	■1.000	$a_{\{J\}}$	a_J	a^I and a_J have different grades); to get the other results, I let $I = i_1 < i_2 < \cdots < i_r$ and
1285.5935v1_F08640	56	■1.000	$I=i_{\{1\}<i_{\{2\}<\cdots<i_{\{r\}}$	$I = i_1 < i_2 < \cdots < i_r$	a^I and a_J have different grades); to get the other results, I let $I = i_1 < i_2 < \cdots < i_r$ and
1285.5935v1_F08641	56	■1.000	$J=j_{\{1\}<j_{\{2\}<\cdots<j_{\{r\}}$	$J = j_1 < j_2 < \cdots < j_r$	$J = j_1 < j_2 < \cdots < j_r$ and use Eq. (156) to find
1285.5935v1_F08642	56	■1.000	$j_{\{k\}}$	j_k	Now consider the case where the i_k equal the j_k ; using the third of Eqs. (83) and Eq. (77),
1285.5935v1_F08643	57	■1.000	$i_{\{2\}}$	i_2	which can be repeated for the i_2 term and for each successive term until the final result
1285.5935v1_F08644	57	■1.000	$j_{\{l\}}$	j_l	is reached. Now suppose that i_k equals none of the j_l ; then when the evaluation of the scalar
1285.5935v1_F08645	57	■0.983	$\backslash\mathrm{left.a^{\{i_{\{k\}}\}}\backslash\mathrm{right}\rfloor\mathrm{floor}~a_{\{j_{\{l\}}\}}$	$a^{i_k} \rfloor a_{j_l}$	product as shown above reaches the k th iteration, all of the $a^{i_k} \rfloor a_{j_l}$ terms will vanish, and so will
1285.5935v1_F08646	57	■1.000	$\backslash\mathrm{binom}\{n\}\{r\}$	$\binom{n}{r}$	$\binom{n}{r}$, it follows that
1285.5935v1_F08647	57	■1.000	$B^{\wedge\{I\}}$	B^I	components B^I and C_I .
1285.5935v1_F08648	57	■1.000	$C_{\{I\}}$	C_I	components B^I and C_I .
1285.5935v1_F08649	57	■1.000	$A * B$	$A * B$	1. $A * B$ for every multivector B .
1285.5935v1_F08650	57	■0.477	$\backslash\mathrm{angle}~A\backslash\mathrm{rangle}$	$\langle A \rangle$	2. $\langle A \rangle$ and $a \rfloor A$ for every vector a .
1285.5935v1_F08651	57	■0.477	$a\backslash\mathrm{rfloor}~A$	$a \rfloor A$	2. $\langle A \rangle$ and $a \rfloor A$ for every vector a .
1285.5935v1_F08652	57	■0.950	$\backslash\mathrm{angle}~A\backslash\mathrm{rangle}=\emptyset$	$\langle A \rangle = 0$	By the distributive property, part 2 is equivalent to this statement: if $\langle A \rangle = 0$ and all $a \rfloor A = 0$,
1285.5935v1_F08653	57	■0.950	$a~\backslash\mathrm{downharpoonleft}~A=\emptyset$	$a \rfloor A = 0$	By the distributive property, part 2 is equivalent to this statement: if $\langle A \rangle = 0$ and all $a \rfloor A = 0$,
1285.5935v1_F08654	57	■1.000	$A=\emptyset$	$A = 0$	then $A = 0$. So that's what I'll prove. To do this, I assume A is an r -vector A_r ; if the result is
1285.5935v1_F08655	57	■1.000	$\backslash\mathrm{left}.\backslash\mathrm{left}(a^{\{i_{\{r\}}\}}~\backslash\mathrm{wedge}~\cdots~\backslash\mathrm{wedge}~a^{\{i_{\{2\}}\}}~\backslash\mathrm{wedge}~a^{\{i_{\{1\}}\}}\backslash\mathrm{right})\backslash\mathrm{right}\rfloor\mathrm{floor}~A_{\{r\}}$	$\left(a^{i_r} \wedge \cdots \wedge a^{i_2} \wedge a^{i_1}\right) \rfloor A_r$	defined by $\{a_i\}$ is $(a^{i_r} \wedge \cdots \wedge a^{i_2} \wedge a^{i_1}) \rfloor A_r$ for some strictly ascending choice of i_1 through i_r .
1285.5935v1_F08656	57	■0.998	$a^{\{i_{\{1\}}\}}~\backslash\mathrm{downharpoonleft}~A_{\{r\}}=\emptyset$	$a^{i_1} \rfloor A_r = 0$	since $a^{i_1} \rfloor A_r = 0$. So all the components of A_r vanish, so $A_r = 0$.
1285.5935v1_F08657	57	■0.998	$A_{\{r\}}=\emptyset$	$A_r = 0$	since $a^{i_1} \rfloor A_r = 0$. So all the components of A_r vanish, so $A_r = 0$.
1285.5935v1_F08658	58	■0.930	$\backslash\mathrm{angle}~A\backslash\mathrm{rangle},\backslash\mathrm{angle}~A\backslash\mathrm{rangle}_{\{1\}}$	$\langle A \rangle, \langle A \rangle_1$	This result can be extended <i>ad nauseum</i> : A is uniquely determined by $\langle A \rangle$, $\langle A \rangle_1$, and $A_2 \rfloor A$ for every
1285.5935v1_F08659	58	■0.930	$A_{\{2\}}~\backslash\mathrm{downharpoonleft}~A$	$A_2 \rfloor A$	This result can be extended <i>ad nauseum</i> : A is uniquely determined by $\langle A \rangle$, $\langle A \rangle_1$, and $A_2 \rfloor A$ for every
1285.5935v1_F08660	58	■1.000	$A_{\{2\}}$	A_2	bivector A_2 , and so on.
1285.5935v1_F08661	58	■0.931	$\backslash\mathrm{sum}_{\{i\}}~a^{\{i\}}~\backslash\mathrm{wedge}~a_{\{i\}}=\emptyset$	$\sum_i a^i \wedge a_i = 0$	because I showed in the proof of Theorem 35 that $\sum_i a^i \wedge a_i = 0$.
1285.5935v1_F08662	58	■1.000	$A_{\{r\}}=a^{\{j_{\{1\}}\}}~\backslash\mathrm{wedge}~\cdots~\backslash\mathrm{wedge}~a^{\{j_{\{r\}}\}}$	$A_r = a^{j_1} \wedge \cdots \wedge a^{j_r}$	so I have to prove it only on a basis; and I know just the basis to use. Let $A_r = a^{j_1} \wedge \cdots \wedge a^{j_r}$
1285.5935v1_F08663	58	■1.000	$a^{\{j_{\{k\}}\}}$	a^{j_k}	In each term of the sum, I move a^{j_k} past $k - 1$ other vectors to return it to its original spot,
1285.5935v1_F08664	58	■1.000	$(-1)^{\wedge\{k-1\}}$	$(-1)^{k-1}$	which cancels the $(-1)^{k-1}$ factor, so
1285.5935v1_F08665	58	■1.000	$a^{\{N\}}$	a^N	Just as the original frame has a volume element a_N , the reciprocal frame has a volume element a^N defined
1285.5935v1_F08666	58	■1.000	$a^{\{N\}}=a^{\{1\}}~\backslash\mathrm{wedge}~\cdots~\backslash\mathrm{wedge}~a^{\{n\}}$	$a^N = a^1 \wedge \cdots \wedge a^n$	in an analogous way: $a^N = a^1 \wedge \cdots \wedge a^n$. Now a_N and a^N have to be scalar multiples of each other, and
1285.5935v1_F08667	58	■0.997	$a_{\{N\}} * a^{\{N\}}=1$	$a_N * a^N = 1$	since Theorem 36 shows that $a_N * a^N = 1$, I conclude that
1285.5935v1_F08668	59	■1.000	$a^{\{i\}}~\backslash\mathrm{wedge}~a^{\{j\}}$	$a^i \wedge a^j$	vectors except a_i , and similarly for a^j ; therefore $a^i \wedge a^j$ is orthogonal to the outer product of all the frame
1285.5935v1_F08669	59	■1.000	$a_{\{1\}}~\backslash\mathrm{wedge}~\cdots~\backslash\mathrm{wedge}~\check{a}_{\{i\}}~\backslash\mathrm{wedge}~\cdots~\backslash\mathrm{wedge}~\check{a}_{\{j\}}~\backslash\mathrm{wedge}~\cdots~\backslash\mathrm{wedge}~a_{\{n\}}$	$a_1 \wedge \cdots \wedge \check{a}_i \wedge \cdots \wedge \check{a}_j \wedge \cdots \wedge a_n$	vectors except a_i and a_j . Therefore $a^i \wedge a^j$ is dual to $a_1 \wedge \cdots \wedge \check{a}_i \wedge \cdots \wedge \check{a}_j \wedge \cdots \wedge a_n$. To make this easier to
1285.5935v1_F08670	59	■1.000	$I^{\wedge\{c\}}$	I^c	write out, for any string of indices I let me define I^c to be its ascending complement, so I^c includes exactly
1285.5935v1_F08671	59	■1.000	$a_{\{I^{\wedge\{c\}}\}}$	a_{I^c}	the indices not in I in ascending order. In these terms, a^I is dual to a_{I^c} . To be more precise, I have
1285.5935v1_F08672	59	■1.000	$a^{\{I\}}~a_{\{N\}}$	$a^I a_N$	<i>Proof.</i> To prove this, I'll calculate $a^I a_N$.
1285.5935v1_F08673	59	■0.696	$\backslash\mathrm{left.a^{\{i_{\{r\}}\}}\backslash\mathrm{right}\rfloor\mathrm{floor}~a_{\{N\}}$	$a^{i_r} \rfloor a_N$	To calculate $a^{i_r} \rfloor a_N$ I use Eq. (77):
1285.5935v1_F08674	59	■0.886	$a^{\{i_{\{r-1\}}\}}~\backslash\mathrm{downharpoonleft}~a_{\{i_{\{r\}}\}^{\{c\}}}$	$a^{i_{r-1}} \rfloor a_{i_r^c}$	When I evaluate $a^{i_{r-1}} \rfloor a_{i_r^c}$ using Eq. (77) again, I remove $a_{i_{r-1}}$ and multiply by $(-1)^{i_{r-1}-1}$.
1285.5935v1_F08675	59	■0.886	$a_{\{i_{\{r-1\}}\}}$	$a_{i_{r-1}}$	When I evaluate $a^{i_{r-1}} \rfloor a_{i_r^c}$ using Eq. (77) again, I remove $a_{i_{r-1}}$ and multiply by $(-1)^{i_{r-1}-1}$.
1285.5935v1_F08676	59	■0.886	$(-1)^{\wedge\{i_{\{r-1\}}-1\}}$	$(-1)^{i_{r-1}-1}$	When I evaluate $a^{i_{r-1}} \rfloor a_{i_r^c}$ using Eq. (77) again, I remove $a_{i_{r-1}}$ and multiply by $(-1)^{i_{r-1}-1}$.
1285.5935v1_F08677	59	■1.000	$a_{\{i_{\{r\}}\}^{\{c\}}}$	$a_{i_r^c}$	position i_{r-1} in $a_{i_r^c}$.) Thus each step removes a factor a_{i_j} from a_N and multiplies by $(-1)^{i_j-1}$,
1285.5935v1_F08678	59	■1.000	$(-1)^{\wedge\{i_{\{j\}}-1\}}$	$(-1)^{i_j-1}$	position i_{r-1} in $a_{i_r^c}$.) Thus each step removes a factor a_{i_j} from a_N and multiplies by $(-1)^{i_j-1}$,
1285.5935v1_F08679	59	■1.000	$I=N$	$I = N$	Now I'll take care of the special cases on each extreme: I is empty and $I = N$. When I is empty,
1285.5935v1_F08680	59	■1.000	$a^{\{I\}}=a_{\{N\}}~a_{\{N\}}^{-1}=1$	$a^I = a_N a_N^{-1} = 1$	Eq. (280) reduces to $a^I = a_N a_N^{-1} = 1$, which is correct, and when $I = N$, I^c is empty so $a_{I^c} = 1$,
1285.5935v1_F08681	59	■1.000	$I=N,~I^{\wedge\{c\}}$	$I = N, I^c$	Eq. (280) reduces to $a^I = a_N a_N^{-1} = 1$, which is correct, and when $I = N$, I^c is empty so $a_{I^c} = 1$,
1285.5935v1_F08682	59	■1.000	$a_{\{I^{\wedge\{c\}}\}}=1$	$a_{I^c} = 1$	Eq. (280) reduces to $a^I = a_N a_N^{-1} = 1$, which is correct, and when $I = N$, I^c is empty so $a_{I^c} = 1$,
1285.5935v1_F08683	60	■1.000	$\backslash\mathrm{left}\{e_{\{i\}}\}\backslash\mathrm{right}\backslash$	$\{e_i\}$	denote $\{e_i\}$. Then the reciprocal frame is a lot easier to find: it's clear on inspection that $e^i = e_i^{-1} = e_i^{-2} e_i$
1285.5935v1_F08684	60	■1.000	$e^{\{i\}}=e_{\{i\}}^{\{-1\}}=e_{\{i\}}^{\{-2\}}~e_{\{i\}}$	$e^i = e_i^{-1} = e_i^{-2} e_i$	denote $\{e_i\}$. Then the reciprocal frame is a lot easier to find: it's clear on inspection that $e^i = e_i^{-1} = e_i^{-2} e_i$
1285.5935v1_F08685	60	■1.000	$e_{\{i\}}^{\{2\}}= \pm 1$	$e_i^2 = \pm 1$	fits the bill. Since the frame vectors are normalized, $e_i^2 = \pm 1$, so $e^i = \pm e_i$. For any I , let I_m be the number
1285.5935v1_F08686	60	■1.000	$e^{\{i\}}= \pm e_{\{i\}}$	$e^i = \pm e_i$	fits the bill. Since the frame vectors are normalized, $e_i^2 = \pm 1$, so $e^i = \pm e_i$. For any I , let I_m be the number
1285.5935v1_F08687	60	■1.000	$I_{\{m\}}$	I_m	fits the bill. Since the frame vectors are normalized, $e_i^2 = \pm 1$, so $e^i = \pm e_i$. For any I , let I_m be the number

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1285.5935v1_F08688	60	■ 1.000	$\mathbf{e}_{\{I\}}$	e_I	of elements in the product for e_I that have negative square; then $e^I = (-1)^{I_m} e_I$, so the multivector basis
1285.5935v1_F08689	60	■ 1.000	$\mathbf{e}^{\{I\}} = (-1)^{\{I_m\}} \mathbf{e}_{\{I\}}$	$e^I = (-1)^{I_m} e_I$	of elements in the product for e_I that have negative square; then $e^I = (-1)^{I_m} e_I$, so the multivector basis
1285.5935v1_F08690	60	■ 1.000	$\mathbf{I}_{\{m\}} = 0$	$I_m = 0$	If the space of vectors is Euclidean, then $I_m = 0$ for any I , so the magnitude is positive definite, so the entire
1285.5935v1_F08691	60	■ 1.000	$\left\{ \mathbf{a}_{\{j\}} \right\}$	$\{a_j\}$	sum above are the reciprocal frame to $\{a_j\}$, or
1285.5935v1_F08692	60	■ 1.000	$\boldsymbol{A}^{\mathbf{r}} = \mathbf{a}^{\{1\}} \wedge \cdots \wedge \mathbf{a}^{\mathbf{r}}$	$\boldsymbol{A}^{\mathbf{r}} = a^1 \wedge \cdots \wedge a^r$	Since the reciprocal frame volume element $\boldsymbol{A}^{\mathbf{r}} = a^1 \wedge \cdots \wedge a^r$ equals $\boldsymbol{A}_{\mathbf{r}}/ \boldsymbol{A}_{\mathbf{r}} ^2$ (cf. Eq. (285)), projection
1285.5935v1_F08693	60	■ 1.000	$\boldsymbol{A}_{\mathbf{r}}/ \boldsymbol{A}_{\mathbf{r}} ^2$	$\boldsymbol{A}_{\mathbf{r}}/ \boldsymbol{A}_{\mathbf{r}} ^2$	Since the reciprocal frame volume element $\boldsymbol{A}^{\mathbf{r}} = a^1 \wedge \cdots \wedge a^r$ equals $\boldsymbol{A}_{\mathbf{r}}/ \boldsymbol{A}_{\mathbf{r}} ^2$ (cf. Eq. (285)), projection
1285.5935v1_F08694	60	■ 0.675	$\boldsymbol{A}^{\mathbf{r}}$	$\boldsymbol{A}^{\mathbf{r}}$	using either volume element gives the same result; had I used $\boldsymbol{A}^{\mathbf{r}}$, I'd have ended up with Eq. (288) with a_j
1285.5935v1_F08695	61	■ 1.000	U	U	them. If U and V are vector spaces with the same set of scalars, a function $F : U \rightarrow V$ is said to be <i>linear</i> if
1285.5935v1_F08696	61	■ 1.000	V	V	them. If U and V are vector spaces with the same set of scalars, a function $F : U \rightarrow V$ is said to be <i>linear</i> if
1285.5935v1_F08697	61	■ 1.000	$F: U \rightarrow V$	$F : U \rightarrow V$	them. If U and V are vector spaces with the same set of scalars, a function $F : U \rightarrow V$ is said to be <i>linear</i> if
1285.5935v1_F08698	61	■ 1.000	$F(\alpha u + \beta v) = \alpha F(u) + \beta F(v)$	$F(\alpha u + \beta v) = \alpha F(u) + \beta F(v)$	$F(\alpha u + \beta v) = \alpha F(u) + \beta F(v)$, so linear functions respect linear combinations. Linear functions have a very
1285.5935v1_F08699	61	■ 1.000	F	F	If F and G are linear, then so are their linear combinations, and so are their inverses if they exist. If
1285.5935v1_F08700	61	■ 1.000	G	G	If F and G are linear, then so are their linear combinations, and so are their inverses if they exist. If
1285.5935v1_F08701	61	■ 0.710	$\operatorname{Ker}(F)$	$\operatorname{Ker}(F)$	codomain the <i>range</i> .) F singles out two special subspaces: the <i>kernel</i> of F , or $\operatorname{Ker}(F)$, is a subspace of the
1285.5935v1_F08702	61	■ 0.883	range	range	domain consisting of all the vectors that F maps to 0, and the <i>range</i> of F , or $\operatorname{Range}(F)$, is a subspace of the
1285.5935v1_F08703	61	■ 0.883	$\operatorname{Range}(F)$	$\operatorname{Range}(F)$	domain consisting of all the vectors that F maps to 0, and the <i>range</i> of F , or $\operatorname{Range}(F)$, is a subspace of the
1285.5935v1_F08704	61	■ 0.995	$F^{-1}(0)$	$F^{-1}(0)$	to think of $\operatorname{Ker}(F)$ as $F^{-1}(0)$ and $\operatorname{Range}(F)$ as $F(U)$. F is one-to-one iff $\operatorname{Ker}(F) = \{0\}$, and F is onto
1285.5935v1_F08705	61	■ 0.995	$F(U) \cap F$	$F(U).F$	to think of $\operatorname{Ker}(F)$ as $F^{-1}(0)$ and $\operatorname{Range}(F)$ as $F(U)$. F is one-to-one iff $\operatorname{Ker}(F) = \{0\}$, and F is onto
1285.5935v1_F08706	61	■ 0.995	$\operatorname{Ker}(F) = \{0\}$	$\operatorname{Ker}(F) = \{0\}$	to think of $\operatorname{Ker}(F)$ as $F^{-1}(0)$ and $\operatorname{Range}(F)$ as $F(U)$. F is one-to-one iff $\operatorname{Ker}(F) = \{0\}$, and F is onto
1285.5935v1_F08707	61	■ 0.999	$\operatorname{Range}(F) = V$	$\operatorname{Range}(F) = V$	iff $\operatorname{Range}(F) = V$. The dimension of $\operatorname{Ker}(F)$ is called the <i>nullity</i> of F , or $\operatorname{null}(F)$, and the dimension of
1285.5935v1_F08708	61	■ 0.999	$\operatorname{null}(F)$	$\operatorname{null}(F)$	iff $\operatorname{Range}(F) = V$. The dimension of $\operatorname{Ker}(F)$ is called the <i>nullity</i> of F , or $\operatorname{null}(F)$, and the dimension of
1285.5935v1_F08709	61	■ 0.314	rank	rank	$\operatorname{Range}(F)$ is called the <i>rank</i> of F , or $\operatorname{rank}(F)$. If the dimension of the domain U is finite, the <i>rank-nullity</i>
1285.5935v1_F08710	61	■ 0.314	$\operatorname{rank}(F)$	$\operatorname{rank}(F)$	$\operatorname{Range}(F)$ is called the <i>rank</i> of F , or $\operatorname{rank}(F)$. If the dimension of the domain U is finite, the <i>rank-nullity</i>
1285.5935v1_F08711	61	■ 1.000	$F_{\mathbf{W}}$	$F_{\mathbf{W}}$	it's denoted $F_{\mathbf{W}}$. Since blades represent subspaces, I'll sometimes write $F_{\mathbf{A}}$ for the restriction of F to the
1285.5935v1_F08712	61	■ 1.000	$F_{\mathbf{A}}$	$F_{\mathbf{A}}$	it's denoted $F_{\mathbf{W}}$. Since blades represent subspaces, I'll sometimes write $F_{\mathbf{A}}$ for the restriction of F to the
1285.5935v1_F08713	61	■ 0.983	Id_U	Id_U	U by Id_U .
1285.5935v1_F08714	61	■ 0.742	$u * v$	$u * v$	uniquely fixed if one knows $u * v$ (or equivalently $u \rfloor v$) for all v . This has two useful consequences. The first
1285.5935v1_F08715	61	■ 0.742	$u \downharpoonleft v$	$u \rfloor v$	uniquely fixed if one knows $u * v$ (or equivalently $u \rfloor v$) for all v . This has two useful consequences. The first
1285.5935v1_F08716	62	■ 1.000	$F(u) * v$	$F(u) * v$	Theorem 40. A linear function $F : U \rightarrow V$ is completely determined by knowledge of $F(u) * v$ for all $u \in U$
1285.5935v1_F08717	62	■ 1.000	$u \in U$	$u \in U$	Theorem 40. A linear function $F : U \rightarrow V$ is completely determined by knowledge of $F(u) * v$ for all $u \in U$
1285.5935v1_F08718	62	■ 1.000	$v \in V$	$v \in V$	and $v \in V$.
1285.5935v1_F08719	62	■ 1.000	$F(u) = \sum_i \left[a^i * F(u) \right] a_i$	$F(u) = \sum_i \left[a^i * F(u) \right] a_i$	<i>Proof.</i> In the finite-dimensional case, F can be constructed explicitly: $F(u) = \sum_i \left[a^i * F(u) \right] a_i$,
1285.5935v1_F08720	62	■ 1.000	$\left\{ \mathbf{a}^{\{j\}} \right\}_{j=1, \dots, n}$	$\{a^j\}_{j=1, \dots, n}$	where $\{a_i\}_{i=1, \dots, n}$ is any frame in V and $\{a^j\}_{j=1, \dots, n}$ is its reciprocal frame. Since we know all
1285.5935v1_F08721	62	■ 1.000	$\mathbf{a}^{\{i\}} * F(u)$	$a^i * F(u)$	the $a^i * F(u)$, we know $F(u)$. We can use any frame for this construction since the set of all inner
1285.5935v1_F08722	62	■ 1.000	$F(u)$	$F(u)$	the $a^i * F(u)$, we know $F(u)$. We can use any frame for this construction since the set of all inner
1285.5935v1_F08723	62	■ 0.997	$\mathbf{a} \downharpoonleft F(\mathbf{b})$	$a \rfloor F(b)$	Any linear transformation F defines a new bilinear product on vectors by $a \rfloor F(b)$. The second consequence
1285.5935v1_F08724	62	■ 0.999	$u \circ v = u \downharpoonleft F(v)$	$u \circ v = u \rfloor F(v)$	such that $u \circ v = u \rfloor F(v)$ for all u and v .
1285.5935v1_F08725	62	■ 1.000	$F(u) = \sum_i \left(a^i \circ u \right) a_i$	$F(u) = \sum_i \left(a^i \circ u \right) a_i$	<i>Proof.</i> Again, in finite dimensions the proof is by construction: Let $F(u) = \sum_i (a^i \circ u) a_i$.
1285.5935v1_F08726	62	■ 0.806	$\mathbf{u} \downharpoonleft F(\mathbf{v}) = \mathbf{u} \circ \mathbf{v}$	$u \rfloor F(v) = u \circ v$	Since F is determined by $u \rfloor F(v) = u \circ v$ for all u and v , this function is unique.
1285.5935v1_F08727	62	■ 1.000	$\overline{F}: V \rightarrow U$	$\overline{F} : V \rightarrow U$	If $F : U \rightarrow V$ is linear, then its <i>adjoint</i> is the unique linear function $\overline{F} : V \rightarrow U$ defined by
1285.5935v1_F08728	62	■ 0.845	$\overline{F}$	$\overline{F}$	Notice that $\overline{F}$ switches domain and codomain compared to F and that $*$ is interchangeable with either $\rfloor$
1285.5935v1_F08729	62	■ 0.845	$*$	$*$	Notice that $\overline{F}$ switches domain and codomain compared to F and that $*$ is interchangeable with either $\rfloor$
1285.5935v1_F08730	62	■ 0.600	$\left\lfloor \right.$	$\lfloor$	or $\lfloor$ in this definition. The adjoint of the identity is pretty easy: $\overline{\operatorname{Id}_U} = \operatorname{Id}_U$. Theorem 40 tells me how to
1285.5935v1_F08731	62	■ 0.600	$\overline{\operatorname{Id}_U} = \operatorname{Id}_U$	$\overline{\operatorname{Id}_U} = \operatorname{Id}_U$	or $\lfloor$ in this definition. The adjoint of the identity is pretty easy: $\overline{\operatorname{Id}_U} = \operatorname{Id}_U$. Theorem 40 tells me how to
1285.5935v1_F08732	63	■ 1.000	$\left\{ \mathbf{a}_{\{i\}} \right\} \subset U$	$\{a_i\} \subset U$	(Notice that for this to make sense, $v \in V$ while the frame $\{a_i\} \subset U$.) The bilinearity of the inner product
1285.5935v1_F08733	63	■ 1.000	$G: V \rightarrow W$	$G : V \rightarrow W$	Suppose $F : U \rightarrow V$ and $G : V \rightarrow W$, and let $GF : U \rightarrow W$ denote the composition of F and G . Then
1285.5935v1_F08734	63	■ 1.000	$G F: U \rightarrow W$	$GF : U \rightarrow W$	Suppose $F : U \rightarrow V$ and $G : V \rightarrow W$, and let $GF : U \rightarrow W$ denote the composition of F and G . Then
1285.5935v1_F08735	63	■ 1.000	$F^{\{n\}}$	F^n	A special case of this arises if F is an operator on U , in which case F^n is defined for any n and
1285.5935v1_F08736	63	■ 1.000	$F^{-1}: V \rightarrow U$	$F^{-1} : V \rightarrow U$	Now suppose $F : U \rightarrow V$ is invertible, so there's an $F^{-1} : V \rightarrow U$ such that $F^{-1} F = \operatorname{Id}_U$ and $F F^{-1} = \operatorname{Id}_V$.
1285.5935v1_F08737	63	■ 1.000	$F^{-1} F = \operatorname{Id}_U$	$F^{-1} F = \operatorname{Id}_U$	Now suppose $F : U \rightarrow V$ is invertible, so there's an $F^{-1} : V \rightarrow U$ such that $F^{-1} F = \operatorname{Id}_U$ and $F F^{-1} = \operatorname{Id}_V$.
1285.5935v1_F08738	63	■ 1.000	$F F^{-1} = \operatorname{Id}_V$	$F F^{-1} = \operatorname{Id}_V$	Now suppose $F : U \rightarrow V$ is invertible, so there's an $F^{-1} : V \rightarrow U$ such that $F^{-1} F = \operatorname{Id}_U$ and $F F^{-1} = \operatorname{Id}_V$.
1285.5935v1_F08739	63	■ 0.986	$\overline{F} \overline{F^{-1}} = \overline{F^{-1} F} = \overline{\operatorname{Id}_U} = \operatorname{Id}_U$	$\overline{F} \overline{F^{-1}} = \overline{F^{-1} F} = \overline{\operatorname{Id}_U} = \operatorname{Id}_U$	In that case, since $\overline{F} \overline{F^{-1}} = \overline{F^{-1} F} = \overline{\operatorname{Id}_U} = \operatorname{Id}_U$, and similarly $\overline{F^{-1} F} = \operatorname{Id}_V$,

Identifier	Page	Conf.	LaTeX source	Rendered	Image
1205.5935v1_F00740	63	■ 0.986	$\overline{\mathrm{F}^{-1}} \ \mathrm{\bar{F}}=\mathrm{Id}_V$	$\overline{F^{-1}}\bar{F} = \mathrm{Id}_V$	In that case, since $\overline{\bar{F}F^{-1}} = \overline{F^{-1}\bar{F}} = \overline{\mathrm{Id}_U} = \mathrm{Id}_U$, and similarly $\overline{F^{-1}}\bar{F} = \mathrm{Id}_V$,
1205.5935v1_F00741	63	■ 1.000	$\operatorname{Range}(\mathrm{\bar{F}})$	$\operatorname{Range}(\bar{F})$	Now $\operatorname{Range}(\bar{F})$ and $\operatorname{Range}(\bar{F})^\perp$ are duals, so their dimensions add up to the dimension of U ; so
1205.5935v1_F00742	63	■ 1.000	$\operatorname{Range}(\mathrm{\bar{F}})^\perp$	$\operatorname{Range}(\bar{F})^\perp$	Now $\operatorname{Range}(\bar{F})$ and $\operatorname{Range}(\bar{F})^\perp$ are duals, so their dimensions add up to the dimension of U ; so
1205.5935v1_F00743	64	■ 1.000	$\mathrm{F} \ \mathrm{\bar{F}}=\mathrm{\bar{F}} \mathrm{F}$	$FF = \bar{F}F$	In the next few sections I'll be considering operators that commute with their adjoints: $F\bar{F} = \overline{F}F$. These
1205.5935v1_F00744	64	■ 0.957	$\mathrm{F}^n(\mathrm{\bar{F}})^n$	$F^n(\bar{F})^n$	<i>Proof.</i> If F is normal, then $F^n(\bar{F})^n$ is easily transformed to $(\bar{F})^nF^n$ by moving all the $\bar{F}$
1205.5935v1_F00745	64	■ 0.957	$(\mathrm{\bar{F}})^n \mathrm{F}^n$	$(\bar{F})^nF^n$	<i>Proof.</i> If F is normal, then $F^n(\bar{F})^n$ is easily transformed to $(\bar{F})^nF^n$ by moving all the $\bar{F}$
1205.5935v1_F00746	64	■ 0.828	$\mathrm{F}(u)\rfloor \mathrm{F}(v)=\mathrm{\bar{F}}(u)\rfloor \mathrm{\bar{F}}(v)$	$F(u)\rfloor F(v) = \bar{F}(u)\rfloor \bar{F}(v)$	Theorem 44. F is normal iff $F(u)\rfloor F(v) = \bar{F}(u)\rfloor \bar{F}(v)$ for any u and v .
1205.5935v1_F00747	64	■ 0.841	$40 \ \mathrm{F} \ \mathrm{\bar{F}}=\mathrm{\bar{F}} \mathrm{F}$	$40F\bar{F} = \bar{F}F$	so by Theorem 40 $F\bar{F} = \overline{F}F$ and F is normal.
1205.5935v1_F00748	64	■ 1.000	$\mathrm{F}(u)^2=\mathrm{\bar{F}}(u)^2$	$F(u)^2 = \bar{F}(u)^2$	Theorem 45. F is normal iff $F(u)^2 = \bar{F}(u)^2$ for all u .
1205.5935v1_F00749	64	■ 0.807	$\left.\mathrm{F}(u)\right.\right\rfloor \mathrm{F}(v)=\mathrm{\bar{F}}(u)\left.\right\rfloor \mathrm{\bar{F}}(v)$	$F(u)\rfloor F(v) = \bar{F}(u)\rfloor \bar{F}(v)$	$F(u)^2 = \bar{F}(u)^2$ for all u implies $F(u)\rfloor F(v) = \bar{F}(u)\rfloor \bar{F}(v)$ for all u and v . On the other hand,
1205.5935v1_F00750	64	■ 0.808	$\mathrm{F}(u)\rfloor \mathrm{F}(v)=\mathrm{\bar{F}}(u) \ \downharpoonleft \mathrm{\bar{F}}(v)$	$F(u)\rfloor F(v) = \bar{F}(u) \rfloor \bar{F}(v)$	$F(u)\rfloor F(v) = \bar{F}(u)\rfloor \bar{F}(v)$ for all u and v implies $F(u)^2 = \bar{F}(u)^2$ for all u just by considering the
1205.5935v1_F00751	64	■ 1.000	$u=v$	$u = v$	case $u = v$. So squares are equal iff inner products are equal, which takes us back to the previous
1205.5935v1_F00752	64	■ 1.000	$\operatorname{Ker}(\mathrm{F})=\operatorname{Ker}(\mathrm{\bar{F}})$	$\operatorname{Ker}(F) = \operatorname{Ker}(\bar{F})$	then $\operatorname{Ker}(F) = \operatorname{Ker}(\bar{F})$. If in addition the domain of F is finite-dimensional, $\operatorname{Range}(F) = \operatorname{Range}(\bar{F})$.
1205.5935v1_F00753	64	■ 1.000	$\operatorname{Range}(\mathrm{F})=\operatorname{Range}(\mathrm{\bar{F}})$	$\operatorname{Range}(F) = \operatorname{Range}(\bar{F})$	then $\operatorname{Ker}(F) = \operatorname{Ker}(\bar{F})$. If in addition the domain of F is finite-dimensional, $\operatorname{Range}(F) = \operatorname{Range}(\bar{F})$.
1205.5935v1_F00754	64	■ 0.999	$\operatorname{Ker}(\mathrm{F})=\operatorname{Range}(\mathrm{\bar{F}})^\perp$	$\operatorname{Ker}(F) = \operatorname{Range}(\bar{F})^\perp$	By Theorem 42, $\operatorname{Ker}(F) = \operatorname{Range}(\bar{F})^\perp$, so $\bar{F}(u) \in \operatorname{Range}(\bar{F})^\perp$. But wait a second: $\bar{F}(u) \in$
1205.5935v1_F00755	64	■ 0.999	$\mathrm{\bar{F}}(u) \ \in \operatorname{Range}(\mathrm{\bar{F}})^\perp$	$\bar{F}(u) \in \operatorname{Range}(\bar{F})^\perp$	By Theorem 42, $\operatorname{Ker}(F) = \operatorname{Range}(\bar{F})^\perp$, so $\bar{F}(u) \in \operatorname{Range}(\bar{F})^\perp$. But wait a second: $\bar{F}(u) \in$
1205.5935v1_F00756	64	■ 0.999	$\mathrm{\bar{F}}(u) \ \in$	$\bar{F}(u) \in$	By Theorem 42, $\operatorname{Ker}(F) = \operatorname{Range}(\bar{F})^\perp$, so $\bar{F}(u) \in \operatorname{Range}(\bar{F})^\perp$. But wait a second: $\bar{F}(u) \in$
1205.5935v1_F00757	64	■ 0.894	$\mathrm{\bar{F}}(u)=0$	$\bar{F}(u) = 0$	$\operatorname{Range}(\bar{F})$ by definition, and $\operatorname{Range}(\bar{F})$ is nondegenerate, so it must be that $\bar{F}(u) = 0$, so $u \in$
1205.5935v1_F00758	64	■ 0.894	$u \ \in$	$u \in$	$\operatorname{Range}(\bar{F})$ by definition, and $\operatorname{Range}(\bar{F})$ is nondegenerate, so it must be that $\bar{F}(u) = 0$, so $u \in$
1205.5935v1_F00759	64	■ 1.000	$\operatorname{Ker}(\mathrm{\bar{F}})$	$\operatorname{Ker}(\bar{F})$	$\operatorname{Ker}(\bar{F})$. Therefore $\operatorname{Ker}(F) \subset \operatorname{Ker}(\bar{F})$. The same argument with F and $\bar{F}$ interchanged shows
1205.5935v1_F00760	64	■ 1.000	$\operatorname{Ker}(\mathrm{F}) \ \subset \operatorname{Ker}(\mathrm{\bar{F}})$	$\operatorname{Ker}(F) \subset \operatorname{Ker}(\bar{F})$	$\operatorname{Ker}(\bar{F})$. Therefore $\operatorname{Ker}(F) \subset \operatorname{Ker}(\bar{F})$. The same argument with F and $\bar{F}$ interchanged shows
1205.5935v1_F00761	64	■ 1.000	$\operatorname{Ker}(\mathrm{\bar{F}}) \ \subset \operatorname{Ker}(\mathrm{F})$	$\operatorname{Ker}(\bar{F}) \subset \operatorname{Ker}(F)$	$\operatorname{Ker}(\bar{F}) \subset \operatorname{Ker}(F)$, so $\operatorname{Ker}(F) = \operatorname{Ker}(\bar{F})$.
1205.5935v1_F00762	64	■ 1.000	$\operatorname{Range}(\mathrm{F})=\operatorname{Ker}(\mathrm{\bar{F}})^\perp$	$\operatorname{Range}(F) = \operatorname{Ker}(\bar{F})^\perp$	Theorem 42 tell me that $\operatorname{Range}(F) = \operatorname{Ker}(\bar{F})^\perp$ and $\operatorname{Range}(\bar{F}) = \operatorname{Ker}(F)^\perp$. But I just showed
1205.5935v1_F00763	64	■ 1.000	$\operatorname{Range}(\mathrm{\bar{F}})=\operatorname{Ker}(\mathrm{F})^\perp$	$\operatorname{Range}(\bar{F}) = \operatorname{Ker}(F)^\perp$	Theorem 42 tell me that $\operatorname{Range}(F) = \operatorname{Ker}(\bar{F})^\perp$ and $\operatorname{Range}(\bar{F}) = \operatorname{Ker}(F)^\perp$. But I just showed
1205.5935v1_F00764	65	■ 1.000	$\mathrm{\bar{F}}=\mathrm{F}$	$\bar{F} = F$	A linear operator is <i>symmetric</i> if it equals its adjoint, $\bar{F} = F$, and <i>skew symmetric</i> or <i>skew</i> if it is the
1205.5935v1_F00765	65	■ 1.000	$\mathrm{\bar{F}}=-\mathrm{F}$	$\bar{F} = -F$	negative of its adjoint, $\bar{F} = -F$. The names come from this theorem.
1205.5935v1_F00766	65	■ 0.952	$a \ \circ b := a \ \downharpoonleft \mathrm{F}(b)$	$a \circ b := a \rfloor F(b)$	Theorem 47. The bilinear product $a \circ b := a \rfloor F(b)$ is (anti)symmetric iff F is (skew) symmetric.
1205.5935v1_F00767	65	■ 0.999	$a \ \downharpoonleft \mathrm{F}(b)=b \ \downharpoonleft \mathrm{\bar{F}}(a)$	$a \rfloor F(b) = b \rfloor \bar{F}(a)$	<i>Proof.</i> Since $a \rfloor F(b) = b \rfloor \bar{F}(a)$, it follows that $a \circ b = b \circ a$ iff $\bar{F} = F$ and $a \circ b = -b \circ a$ iff
1205.5935v1_F00768	65	■ 0.999	$a \ \circ b=b \ \circ a$	$a \circ b = b \circ a$	<i>Proof.</i> Since $a \rfloor F(b) = b \rfloor \bar{F}(a)$, it follows that $a \circ b = b \circ a$ iff $\bar{F} = F$ and $a \circ b = -b \circ a$ iff
1205.5935v1_F00769	65	■ 0.999	$a \ \circ b=-b \ \circ a$	$a \circ b = -b \circ a$	<i>Proof.</i> Since $a \rfloor F(b) = b \rfloor \bar{F}(a)$, it follows that $a \circ b = b \circ a$ iff $\bar{F} = F$ and $a \circ b = -b \circ a$ iff
1205.5935v1_F00770	65	■ 1.000	$\mathrm{\bar{F}} \mathrm{F}$	$\bar{F}F$	Also, for any linear operator F , both $\bar{F}F$ and $F\bar{F}$ are symmetric.
1205.5935v1_F00771	65	■ 1.000	$\mathrm{F} \ \mathrm{\bar{F}}$	$F\bar{F}$	Also, for any linear operator F , both $\bar{F}F$ and $F\bar{F}$ are symmetric.
1205.5935v1_F00772	65	■ 1.000	$\overline{\mathrm{F}^n}=(\mathrm{\bar{F}})^n, \ \mathrm{\bar{F}}=\mathrm{F}$	$\bar{F}^n = (\bar{F})^n, \bar{F} = F$	<i>Proof.</i> Since $\bar{F}^n = (\bar{F})^n$, $\bar{F} = F$ implies $\bar{F}^n = F^n$, so F^n is symmetric also, and $\bar{F} = -F$
1205.5935v1_F00773	65	■ 1.000	$\overline{\mathrm{F}^n}=\mathrm{F}^n$	$\bar{F}^n = F^n$	<i>Proof.</i> Since $\bar{F}^n = (\bar{F})^n$, $\bar{F} = F$ implies $\bar{F}^n = F^n$, so F^n is symmetric also, and $\bar{F} = -F$
1205.5935v1_F00774	65	■ 1.000	$\overline{\mathrm{F}^n}=(-1)^n \ \mathrm{F}^n$	$\bar{F}^n = (-1)^nF^n$	implies $\bar{F}^n = (-1)^nF^n$, so F^n is symmetric or skew as n is even or odd.
1205.5935v1_F00775	65	■ 1.000	$\left\{\lambda_i\right\}$	$\{\lambda_i\}$	$\{\lambda_i\}$, which means $F(a)$ for any a is given by
1205.5935v1_F00776	65	■ 1.000	$\mathrm{F}(a)$	$F(a)$	$\{\lambda_i\}$, which means $F(a)$ for any a is given by
1205.5935v1_F00777	65	■ 0.998	$a \ \downharpoonleft \mathrm{F}(a)=0$	$a \rfloor F(a) = 0$	As motivation, notice that if F is skew, then $a \rfloor F(a) = 0$, so F maps any vector to an orthogonal vector.
1205.5935v1_F00778	65	■ 0.964	$\mathrm{F}(a)=a \ \downharpoonleft \mathrm{A}_2$	$F(a) = a \rfloor A_2$	Well, I know something else that does that: taking the dual by a bivector. In fact, the function $F(a) = a \rfloor A_2$
1205.5935v1_F00779	66	■ 0.293	$a \ \wedge b\rfloor \mathrm{A}_2=a \ \downharpoonleft \mathrm{A}_2\left.\right\rfloor b$	$a \wedge b \rfloor A_2 = a \rfloor A_2 \rfloor b$	Therefore every antisymmetric bilinear product is of the form $a \wedge b \rfloor A_2 = a \rfloor A_2 \rfloor b$ for some A_2 .
1205.5935v1_F00780	66	■ 0.951	$\mathrm{F}(u)\rfloor \mathrm{F}(v)=u \ \downharpoonleft \mathrm{v}$	$F(u)\rfloor F(v) = u \rfloor v$	The final special linear operator is an <i>isometry</i> , which preserves inner products: $F(u)\rfloor F(v) = u \rfloor v$. (Equiv-
1205.5935v1_F00781	66	■ 0.999	$\mathrm{\bar{F}} \mathrm{F}=\mathrm{Id}$	$\bar{F}F = \mathrm{Id}$	so if F is an isometry then $\bar{F}F = \mathrm{Id}$. If F is also invertible, then its inverse has to be $\bar{F}$, so we also have
1205.5935v1_F00782	66	■ 0.539	$\mathrm{F} \ \mathrm{\bar{F}}=\mathrm{Id}$	$FF = \mathrm{Id}$	$F\bar{F} = \mathrm{Id}$. Therefore orthogonal transformations satisfy $\bar{F} = F^{-1}$ and are also normal, and as a bonus F^{-1}
1205.5935v1_F00783	66	■ 0.539	$\mathrm{\bar{F}}=\mathrm{F}^{-1}$	$\bar{F} = F^{-1}$	$F\bar{F} = \mathrm{Id}$. Therefore orthogonal transformations satisfy $\bar{F} = F^{-1}$ and are also normal, and as a bonus F^{-1}
1205.5935v1_F00784	66	■ 0.539	F^{-1}	F^{-1}	$F\bar{F} = \mathrm{Id}$. Therefore orthogonal transformations satisfy $\bar{F} = F^{-1}$ and are also normal, and as a bonus F^{-1}
1205.5935v1_F00785	66	■ 0.999	$a_1, a_2, \ldots, a_r$	$a_1, a_2, \dots, a_r$	vector u and reflects it along axes $a_1, a_2, \dots, a_r$ in succession is
1205.5935v1_F00786	67	■ 1.000	$\mathrm{A}_r=\mathrm{a}_r \cdots \mathrm{a}_2 \mathrm{a}_1$	$A_r = a_r \cdots a_2 a_1$	where $A_r = a_r \cdots a_2 a_1$ is an invertible r -versor. Thus a general isometry in finite dimensions is grade
1205.5935v1_F00787	67	■ 1.000	A_r^{-1}	A_r^{-1}	isometries generated by A_r and A_r^{-1} are inverses of each other.

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1205.5935v1_F00788	67	■ 1.000	$\backslash\lambda\mathrm{A}_{\mathrm{r}}$	λA_r	all, A_r and λA_r generate the same isometry for any $\lambda \neq 0$. (The ultimate reason for this is that a and λa
1205.5935v1_F00789	67	■ 1.000	$\backslash\lambda\mathrm{bda} \ \backslash\mathrm{neq} \ 0$	$\lambda \neq 0$	all, A_r and λA_r generate the same isometry for any $\lambda \neq 0$. (The ultimate reason for this is that a and λa
1205.5935v1_F00790	67	■ 1.000	$\backslash\lambda\mathrm{bda} \ a$	λa	all, A_r and λA_r generate the same isometry for any $\lambda \neq 0$. (The ultimate reason for this is that a and λa
1205.5935v1_F00791	67	■ 1.000	$\mathrm{A}_{\mathrm{r}}=\backslash\lambda\mathrm{bda} \ \mathrm{B}_{\mathrm{s}}$	$A_r = \lambda B_s$	Theorem 50. <i>Versors A_r and B_s represent the same isometry iff $A_r = \lambda B_s$ for some $\lambda \neq 0$.</i>
1205.5935v1_F00792	67	■ 1.000	$\mathrm{A}_{\mathrm{r}} \ u^{* \mathrm{r}} \ \mathrm{A}_{\mathrm{r}}^{\{-1\}}=\mathrm{B}_{\mathrm{s}} \ u^{* \mathrm{s}} \ \mathrm{B}_{\mathrm{s}}^{\{-1\}}$	$A_r u^{*r} A_r^{-1} = B_s u^{*s} B_s^{-1}$	prove it the other way. Suppose $A_r u^{*r} A_r^{-1} = B_s u^{*s} B_s^{-1}$ for all u . Since r and s are both even
1205.5935v1_F00793	67	■ 1.000	$\mathrm{A}_{\mathrm{r}} \ u \ \mathrm{A}_{\mathrm{r}}^{\{-1\}}=\mathrm{B}_{\mathrm{s}} \ u \ \mathrm{B}_{\mathrm{s}}^{\{-1\}}$	$A_r u A_r^{-1} = B_s u B_s^{-1}$	or both odd, I can drop the grade involutions and I'm left with $A_r u A_r^{-1} = B_s u B_s^{-1}$, which can
1205.5935v1_F00794	67	■ 1.000	$\mathrm{B}_{\mathrm{s}}^{\{-1\}} \ \mathrm{A}_{\mathrm{r}}=\backslash\lambda\mathrm{bda}+\mu\backslash\mathrm{angle}\boldsymbol{\mathrm{I}}\backslash\mathrm{rangle}_{\{-\}}$	$B_s^{-1} A_r = \lambda + \mu \langle \boldsymbol{I} \rangle_-$	Then Theorem 28 tells me that $B_s^{-1} A_r = \lambda + \mu \langle \boldsymbol{I} \rangle_-$. But $B_s^{-1} A_r$ is even, so its odd part vanishes
1205.5935v1_F00795	67	■ 1.000	$\mathrm{B}_{\mathrm{s}}^{\{-1\}} \ \mathrm{A}_{\mathrm{r}}$	$B_s^{-1} A_r$	Then Theorem 28 tells me that $B_s^{-1} A_r = \lambda + \mu \langle \boldsymbol{I} \rangle_-$. But $B_s^{-1} A_r$ is even, so its odd part vanishes
1205.5935v1_F00796	67	■ 1.000	$\mathrm{B}_{\mathrm{s}}^{\{-1\}} \ \mathrm{A}_{\mathrm{r}}=\backslash\lambda\mathrm{bda}$	$B_s^{-1} A_r = \lambda$	and I'm left with $B_s^{-1} A_r = \lambda$. Now if $\lambda = 0$, by Theorem 24 both A_r and B_s would be null,
1205.5935v1_F00797	67	■ 1.000	$\backslash\lambda\mathrm{bda}=0$	$\lambda = 0$	and I'm left with $B_s^{-1} A_r = \lambda$. Now if $\lambda = 0$, by Theorem 24 both A_r and B_s would be null,
1205.5935v1_F00798	68	■ 1.000	$\mathrm{a} \ \mathrm{b}$	ab	examine them. Consider ab where a and b are unit vectors; it represents the composition of reflections in
1205.5935v1_F00799	68	■ 0.554	$\mathrm{a}\rfloor\mathrm{b}+\mathrm{a} \wedge \mathrm{b}$	$a \rfloor b + a \wedge b$	is to expand it as $a \rfloor b + a \wedge b$ and figure out each piece separately. I'll do that by starting with Eq. (8) from
1205.5935v1_F00800	68	■ 0.570	$\mu=\mathrm{a} \ \downharpoonleft \mathrm{b}$	$\mu = a \rfloor b$	Since a and b are unit vectors, the left hand side of Eq. (321) is ± 1 . In what follows, I will set $\mu = a \rfloor b$, and
1205.5935v1_F00801	68	■ 0.997	$\mathrm{a} \ \wedge \mathrm{b}=\backslash\lambda\mathrm{bda} \ \boldsymbol{\mathrm{B}}$	$a \wedge b = \lambda \boldsymbol{B}$	I will set $a \wedge b = \lambda \boldsymbol{B}$ where $\boldsymbol{B}$ is a 2-blade. In the cases when $(a \wedge b)^2 \neq 0$, I'll choose λ so $\boldsymbol{B}$ is a unit blade;
1205.5935v1_F00802	68	■ 0.997	$(\mathrm{a} \ \wedge \mathrm{b})^{\{2\}} \ \backslash\mathrm{neq} \ 0$	$(a \wedge b)^2 \neq 0$	I will set $a \wedge b = \lambda \boldsymbol{B}$ where $\boldsymbol{B}$ is a 2-blade. In the cases when $(a \wedge b)^2 \neq 0$, I'll choose λ so $\boldsymbol{B}$ is a unit blade;
1205.5935v1_F00803	68	■ 0.984	$\mathrm{a} \ \downharpoonleft \mathrm{b}=0,(\mathrm{a} \ \wedge \mathrm{b})^{\{2\}}=\pm 1$	$a \rfloor b = 0, (a \wedge b)^2 = \pm 1$	and b be orthogonal. Since $a \rfloor b = 0$, $(a \wedge b)^2 = \pm 1$. So $ab = a \wedge b$ is already a unit blade $\boldsymbol{B}$. To see what
1205.5935v1_F00804	68	■ 0.984	$\mathrm{a} \ \mathrm{b}=\mathrm{a} \ \wedge \mathrm{b}$	$ab = a \wedge b$	and b be orthogonal. Since $a \rfloor b = 0$, $(a \wedge b)^2 = \pm 1$. So $ab = a \wedge b$ is already a unit blade $\boldsymbol{B}$. To see what
1205.5935v1_F00805	68	■ 1.000	$\boldsymbol{\mathrm{B}}^{\{2\}}$	$\boldsymbol{B}^2$	The result in the general case depends on the sign of $\boldsymbol{B}^2$.
1205.5935v1_F00806	68	■ 1.000	$\boldsymbol{\mathrm{B}}^{\{2\}}=-1$	$\boldsymbol{B}^2 = -1$	1. Suppose first that $\boldsymbol{B}^2 = -1$. Then Eq. (175) when $n = 2$ tells me that the number of negative-square
1205.5935v1_F00807	68	■ 1.000	$n=2$	$n = 2$	1. Suppose first that $\boldsymbol{B}^2 = -1$. Then Eq. (175) when $n = 2$ tells me that the number of negative-square
1205.5935v1_F00808	68	■ 1.000	$\mathrm{a}^{\{2\}}$	a^2	definite, which I call the <i>Euclidean</i> or <i>elliptic</i> case. Then a^2 and b^2 have the same sign, so $a^2 b^2 = 1$.
1205.5935v1_F00809	68	■ 1.000	$\mathrm{b}^{\{2\}}$	b^2	definite, which I call the <i>Euclidean</i> or <i>elliptic</i> case. Then a^2 and b^2 have the same sign, so $a^2 b^2 = 1$.
1205.5935v1_F00810	68	■ 1.000	$\mathrm{a}^{\{2\}} \ \mathrm{b}^{\{2\}}=1$	$a^2 b^2 = 1$	definite, which I call the <i>Euclidean</i> or <i>elliptic</i> case. Then a^2 and b^2 have the same sign, so $a^2 b^2 = 1$.
1205.5935v1_F00811	68	■ 1.000	$\mu=\backslash\cos \ (\backslash\theta / \ 2)$	$\mu = \cos(\theta/2)$	Therefore $\mu = \cos(\theta/2)$ and $\lambda = -\sin(\theta/2)$ for some θ , so
1205.5935v1_F00812	68	■ 1.000	$\backslash\lambda\mathrm{bda}=-\backslash\sin \ (\backslash\theta / \ 2)$	$\lambda = -\sin(\theta/2)$	Therefore $\mu = \cos(\theta/2)$ and $\lambda = -\sin(\theta/2)$ for some θ , so
1205.5935v1_F00813	68	■ 0.902	$\backslash\theta=\pi$	$\theta = \pi$	1.2: it's a rotation through angle θ in the plane defined by $\boldsymbol{B}$. When $\theta = \pi$, I recover the special case
1205.5935v1_F00814	68	■ 1.000	$\boldsymbol{\mathrm{B}}^{\{2\}}=1$	$\boldsymbol{B}^2 = 1$	2. Now suppose $\boldsymbol{B}^2 = 1$. In this case the inner product is indefinite, which is called the <i>hyperbolic</i> case.
1205.5935v1_F00815	68	■ 1.000	$\mu=\pm \backslash\mathrm{pm} \ \cosh \ (\backslash\phi / \ 2)$	$\mu = \pm \cosh(\phi/2)$	Therefore $\mu = \pm \cosh(\phi/2)$ and $\lambda = \mp \sinh(\phi/2)$ for some ϕ , so
1205.5935v1_F00816	68	■ 1.000	$\backslash\lambda\mathrm{bda}=\mp \ \sinh \ (\backslash\phi / \ 2)$	$\lambda = \mp \sinh(\phi/2)$	Therefore $\mu = \pm \cosh(\phi/2)$ and $\lambda = \mp \sinh(\phi/2)$ for some ϕ , so
1205.5935v1_F00817	68	■ 1.000	$\mathrm{c} \ \backslash\tanh \ \backslash\phi$	$c \tanh \phi$	the hyperbolic plane, and in special relativity it's a boost to velocity $c \tanh \phi$.) This time I couldn't
1205.5935v1_F00818	69	■ 0.998	$\mathrm{a}^{\{2\}}=-\mathrm{b}^{\{2\}}$	$a^2 = -b^2$	3. Sticking with $\boldsymbol{B}^2 = 1$, now I consider $a^2 = -b^2$. Eq. (321) becomes
1205.5935v1_F00819	69	■ 1.000	μ	μ	so μ and λ change roles: $\mu = \mp \sinh(\phi/2)$ and $\lambda = \pm \cosh(\phi/2)$ for some ϕ , so for the rotor I get
1205.5935v1_F00820	69	■ 1.000	$\mu=\mp \sinh \ (\backslash\phi / \ 2)$	$\mu = \mp \sinh(\phi/2)$	so μ and λ change roles: $\mu = \mp \sinh(\phi/2)$ and $\lambda = \pm \cosh(\phi/2)$ for some ϕ , so for the rotor I get
1205.5935v1_F00821	69	■ 1.000	$\backslash\lambda\mathrm{bda}=\pm \backslash\mathrm{pm} \ \cosh \ (\backslash\phi / \ 2)$	$\lambda = \pm \cosh(\phi/2)$	so μ and λ change roles: $\mu = \mp \sinh(\phi/2)$ and $\lambda = \pm \cosh(\phi/2)$ for some ϕ , so for the rotor I get
1205.5935v1_F00822	69	■ 1.000	$\boldsymbol{\mathrm{B}}^{\{2\}}=0$	$\boldsymbol{B}^2 = 0$	4. Finally, suppose $\boldsymbol{B}^2 = 0$, which means by Theorem 23 that the inner product is degenerate. (This
1205.5935v1_F00823	69	■ 1.000	$\pm 1=\mu^{\{2\}}$	$\pm 1 = \mu^2$	also has a name: the <i>parabolic</i> case.) Now Eq. (321) reduces to $\pm 1 = \mu^2$. This doesn't make sense if
1205.5935v1_F00824	69	■ 1.000	$\mathrm{v}^{\{2\}}$	v^2	squared of any vector in the plane is just v^2 times the square of its component along v , so they all have
1205.5935v1_F00825	69	■ 1.000	$\mu=\pm 1$	$\mu = \pm 1$	the same sign. Thus $a^2 b^2 = 1$ and $\mu = \pm 1$. Let $\boldsymbol{B}$ be any 2-blade that's convenient to use to represent
1205.5935v1_F00826	69	■ 1.000	$\mathrm{a} \ \wedge \mathrm{b}=\mp \backslash\lambda\mathrm{bda} \ \boldsymbol{\mathrm{B}}$	$a \wedge b = \mp \lambda \boldsymbol{B}$	the plane; then $a \wedge b = \mp \lambda \boldsymbol{B}$ for some λ , so the rotor becomes
1205.5935v1_F00827	69	■ 1.000	$\pm \backslash\mathrm{pm}\backslash\left(1+\frac{\lambda}{2} \backslash\lambda\mathrm{bda} \ \boldsymbol{\mathrm{B}}\right)$	$\pm \left(1 + \frac{\lambda}{2} \boldsymbol{B}\right)$	You may wonder what this rotor does. Its inverse is $\pm(1 + \frac{\lambda}{2} \boldsymbol{B})$, so for any vector u
1205.5935v1_F00828	69	■ 0.998	$\mathrm{u}^{\{2\}}$	u^2	thing really is u^2 . The verification is an interesting exercise; you find that $u \rfloor \boldsymbol{B}$ is orthogonal to u ,
1205.5935v1_F00829	69	■ 0.998	$\mathrm{u} \ \downharpoonleft \boldsymbol{\mathrm{B}}$	$u \rfloor \boldsymbol{B}$	thing really is u^2 . The verification is an interesting exercise; you find that $u \rfloor \boldsymbol{B}$ is orthogonal to u ,
1205.5935v1_F00830	69	■ 0.989	$\boldsymbol{\mathrm{B}} \ \mathrm{u} \ \boldsymbol{\mathrm{B}}$	$\boldsymbol{B} u \boldsymbol{B}$	$\boldsymbol{B} u \boldsymbol{B}$ is orthogonal to $u \rfloor \boldsymbol{B}$, $\boldsymbol{B} u \boldsymbol{B}$ is null, and the inner product of u and $\boldsymbol{B} u \boldsymbol{B}$ cancels the square of
1205.5935v1_F00831	69	■ 0.989	$\mathrm{u} \ \downharpoonleft \boldsymbol{\mathrm{B}}, \ \boldsymbol{\mathrm{B}} \ \mathrm{u} \ \boldsymbol{\mathrm{B}}$	$u \rfloor \boldsymbol{B}, \boldsymbol{B} u \boldsymbol{B}$	$\boldsymbol{B} u \boldsymbol{B}$ is orthogonal to $u \rfloor \boldsymbol{B}$, $\boldsymbol{B} u \boldsymbol{B}$ is null, and the inner product of u and $\boldsymbol{B} u \boldsymbol{B}$ cancels the square of
1205.5935v1_F00832	70	■ 1.000	$\backslash\left\{F\backslash\left(\mathrm{a}_{\mathrm{i}}\right)\backslash\right\}$	$\{F(a_i)\}$	mapped by linear function F to the space spanned by $\{F(a_i)\}$. Since blades represent subspaces, it seems
1205.5935v1_F00833	70	■ 1.000	$\mathrm{F}(\mathrm{A} \ \wedge \mathrm{B})=\mathrm{F}(\mathrm{A}) \ \wedge \mathrm{F}(\mathrm{B})$	$F(A \wedge B) = F(A) \wedge F(B)$	If I then require this formal extension of F to be linear over multivectors, I get $F(A \wedge B) = F(A) \wedge F(B)$ for
1205.5935v1_F00834	70	■ 1.000	$\mathrm{u} \ \rightarrow -\mathrm{u}$	$u \rightarrow -u$	parity operation $u \rightarrow -u$, but I extended it by making it respect not outer products but the full geometric
1205.5935v1_F00835	70	■ 1.000	$\mathrm{\mathcal{F}}=\alpha \mathrm{\mathcal{F}}_{\mathrm{1}}+\beta \mathrm{\mathcal{F}}_{\mathrm{2}}$	$\mathcal{F} = \alpha \mathcal{F}_1 + \beta \mathcal{F}_2$	is not an outermorphism. To show this, let $\mathcal{F} = \alpha \mathcal{F}_1 + \beta \mathcal{F}_2$ where $\mathcal{F}_1$ and $\mathcal{F}_2$ are outermorphisms, and
1205.5935v1_F00836	70	■ 1.000	$\mathrm{\mathcal{F}}_{\mathrm{1}}$	$\mathcal{F}_1$	is not an outermorphism. To show this, let $\mathcal{F} = \alpha \mathcal{F}_1 + \beta \mathcal{F}_2$ where $\mathcal{F}_1$ and $\mathcal{F}_2$ are outermorphisms, and
1205.5935v1_F00837	70	■ 1.000	$\mathrm{\mathcal{F}}_{\mathrm{2}}$	$\mathcal{F}_2$	is not an outermorphism. To show this, let $\mathcal{F} = \alpha \mathcal{F}_1 + \beta \mathcal{F}_2$ where $\mathcal{F}_1$ and $\mathcal{F}_2$ are outermorphisms, and
1205.5935v1_F00838	70	■ 1.000	$\mathrm{\mathcal{F}}(\mathrm{A} \ \wedge \mathrm{B})$	$\mathcal{F}(A \wedge B)$	try evaluating both $\mathcal{F}(A \wedge B)$ and $F(A) \wedge F(B)$. You'll see the problem pretty quickly. For this reason,
1205.5935v1_F00839	70	■ 1.000	$\mathrm{\mathcal{F}}(\mathrm{A}) \ \wedge \mathrm{\mathcal{F}}(\mathrm{B})$	$\mathcal{F}(A) \wedge \mathcal{F}(B)$	try evaluating both $\mathcal{F}(A \wedge B)$ and $F(A) \wedge F(B)$. You'll see the problem pretty quickly. For this reason,
1205.5935v1_F00840	70	■ 1.000	$\mathrm{\mathcal{F}}$	$\mathcal{F}$	outermorphism $\mathcal{F}$ has to be linear over everything, not just vectors, which means $\mathcal{F}(\lambda) = \mathcal{F}(\lambda 1) = \lambda \mathcal{F}(1)$.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
1205.5935v1_F00841	70	■ 1.000	$\backslash\mathrm{mathcal}\{F\}(\backslash\mathrm{lambda})=\backslash\mathrm{mathcal}\{F\}(\backslash\mathrm{lambda}\ 1)=\backslash\mathrm{lambda}\ \backslash\mathrm{mathcal}\{F\}(1)$	$\mathcal{F}(\lambda) = \mathcal{F}(\lambda 1) = \lambda \mathcal{F}(1)$	outermorphism $\mathcal{F}$ has to be linear over everything, not just vectors, which means $\mathcal{F}(\lambda) = \mathcal{F}(\lambda 1) = \lambda \mathcal{F}(1)$.
1205.5935v1_F00842	71	■ 1.000	$\backslash\mathrm{mathcal}\{F\}(1)$	$\mathcal{F}(1)$	So $\mathcal{F}(1)$ determines $\mathcal{F}$ for all scalars. But scalar multiplication is also an outer product, so $1 = 1 \wedge 1$. Thus
1205.5935v1_F00843	71	■ 1.000	$1=1\ \backslash\mathrm{wedge}\ 1$	$1 = 1 \wedge 1$	So $\mathcal{F}(1)$ determines $\mathcal{F}$ for all scalars. But scalar multiplication is also an outer product, so $1 = 1 \wedge 1$. Thus
1205.5935v1_F00844	71	■ 0.988	$\backslash\mathrm{mathcal}\{F\}(1)=\mathrm{mathcal}\{F\}(1)\ \backslash\mathrm{wedge}\ \backslash\mathrm{mathcal}\{F\}(1)$	$\mathcal{F}(1) = \mathcal{F}(1) \wedge \mathcal{F}(1)$	the outermorphism property requires $\mathcal{F}(1) = \mathcal{F}(1) \wedge \mathcal{F}(1)$. So like 1, $\mathcal{F}(1)$ must equal its outer product
1205.5935v1_F00845	71	■ 0.988	$1,\ \backslash\mathrm{mathcal}\{F\}(1)$	$1, \mathcal{F}(1)$	the outermorphism property requires $\mathcal{F}(1) = \mathcal{F}(1) \wedge \mathcal{F}(1)$. So like 1, $\mathcal{F}(1)$ must equal its outer product
1205.5935v1_F00846	71	■ 0.827	$A=A\ \backslash\mathrm{wedge}\ A$	$A = A \wedge A$	Theorem 51. <i>If $A = A \wedge A$, then $A = 0$ or 1.</i>
1205.5935v1_F00847	71	■ 0.995	$A_{\{0\}}$	A_0	so A_0 is either 0 or 1. When $r = 1$, I find
1205.5935v1_F00848	71	■ 0.936	$A_{\{1\}}=0$	$A_1 = 0$	Whether A_0 is 0 or 1, this equation requires $A_1 = 0$.
1205.5935v1_F00849	71	■ 0.976	$\backslash\mathrm{mathcal}\{F\}(1)=0$	$\mathcal{F}(1) = 0$	So $\mathcal{F}(1) = 0$ or 1. That means outermorphisms can do only two things to scalars.
1205.5935v1_F00850	71	■ 1.000	$\backslash\mathrm{mathcal}\{F\}=0$	$\mathcal{F} = 0$	Theorem 52. <i>If $\mathcal{F}$ is an outermorphism, then either $\mathcal{F} = 0$ or $\mathcal{F}(\lambda) = \lambda$.</i>
1205.5935v1_F00851	71	■ 1.000	$\backslash\mathrm{mathcal}\{F\}(\backslash\mathrm{lambda})=\backslash\mathrm{lambda}$	$\mathcal{F}(\lambda) = \lambda$	Theorem 52. <i>If $\mathcal{F}$ is an outermorphism, then either $\mathcal{F} = 0$ or $\mathcal{F}(\lambda) = \lambda$.</i>
1205.5935v1_F00852	71	■ 1.000	$\backslash\mathrm{mathcal}\{F\}(1)=1$	$\mathcal{F}(1) = 1$	<i>Proof.</i> Either $\mathcal{F}(1) = 0$ or $\mathcal{F}(1) = 1$. In the former case, for any A
1205.5935v1_F00853	71	■ 1.000	$\backslash\mathrm{overline}\{\backslash\mathrm{mathcal}\{F\}\}:\ \backslash\mathrm{mathcal}\{G\}^{\{1\}}\ \backslash\mathrm{rightarrow}\ \backslash\mathrm{mathcal}\{G\}^{\{2\}}$	$\overline{\mathcal{F}} : \mathcal{G}^1 \rightarrow \mathcal{G}^2$	they do on any other vector spaces: if $\mathcal{F} : \mathcal{G}^1 \rightarrow \mathcal{G}^2$ is linear, $\overline{\mathcal{F}} : \mathcal{G}^2 \rightarrow \mathcal{G}^1$ is a linear function given by
1205.5935v1_F00854	71	■ 1.000	$\backslash\mathrm{overline}\{\backslash\mathrm{mathcal}\{F\}\}:\ \backslash\mathrm{mathcal}\{G\}^{\{2\}}\ \backslash\mathrm{rightarrow}\ \backslash\mathrm{mathcal}\{G\}^{\{1\}}$	$\overline{\mathcal{F}} : \mathcal{G}^2 \rightarrow \mathcal{G}^1$	they do on any other vector spaces: if $\mathcal{F} : \mathcal{G}^1 \rightarrow \mathcal{G}^2$ is linear, $\overline{\mathcal{F}} : \mathcal{G}^2 \rightarrow \mathcal{G}^1$ is a linear function given by
1205.5935v1_F00855	72	■ 1.000	$[F]$	$[F]$	<i>outermorphism $[F]$ that reproduces F when applied to vectors.</i>
1205.5935v1_F00856	72	■ 1.000	$F\ \backslash\mathrm{neq}\ 0$	$F \neq 0$	I had to specify $F \neq 0$ because technically if $F = 0$ then Theorem 52 allows two extensions: one is
1205.5935v1_F00857	72	■ 1.000	$F=0$	$F = 0$	I had to specify $F \neq 0$ because technically if $F = 0$ then Theorem 52 allows two extensions: one is
1205.5935v1_F00858	72	■ 1.000	$[F]=0$	$[F] = 0$	$[F] = 0$, and the other is $[F](A) = \langle A \rangle$. Both extensions vanish on all vectors and all blades; they differ
1205.5935v1_F00859	72	■ 1.000	$[F](A)=\backslash\mathrm{angle}\ A\backslash\mathrm{rangle}$	$[F](A) = \langle A \rangle$	$[F] = 0$, and the other is $[F](A) = \langle A \rangle$. Both extensions vanish on all vectors and all blades; they differ
1205.5935v1_F00860	72	■ 1.000	$U,\ V$	U, V	Theorem 54. <i>If U, V, and W are the vector spaces of $\mathcal{G}^1, \mathcal{G}^2$, and $\mathcal{G}^3$ respectively, and $F : U \rightarrow V$ and</i>
1205.5935v1_F00861	72	■ 1.000	$\backslash\mathrm{mathcal}\{G\}^{\{1\}},\ \backslash\mathrm{mathcal}\{G\}^{\{2\}}$	$\mathcal{G}^1, \mathcal{G}^2$	Theorem 54. <i>If U, V, and W are the vector spaces of $\mathcal{G}^1, \mathcal{G}^2$, and $\mathcal{G}^3$ respectively, and $F : U \rightarrow V$ and</i>
1205.5935v1_F00862	72	■ 1.000	$\backslash\mathrm{mathcal}\{G\}^{\{3\}}$	$\mathcal{G}^3$	Theorem 54. <i>If U, V, and W are the vector spaces of $\mathcal{G}^1, \mathcal{G}^2$, and $\mathcal{G}^3$ respectively, and $F : U \rightarrow V$ and</i>
1205.5935v1_F00863	72	■ 1.000	$[G\ F]=[G][F]$	$[GF] = [G][F]$	(a) $[GF] = [G][F]$.
1205.5935v1_F00864	72	■ 0.676	$[F]^{\{-1\}}=\backslash\mathrm{left}[F^{\{-1\}}\backslash\mathrm{right}]$	$[F]^{-1} = [F^{-1}]$	(b) if F is invertible, $[F]^{-1} = [F^{-1}]$.
1205.5935v1_F00865	72	■ 1.000	$[F]\backslash\mathrm{left}[F^{\{-1\}}\backslash\mathrm{right}]=\backslash\mathrm{operatorname}\{Id\}_{\backslash\mathrm{mathcal}\{G\}^{\{2\}}\}$	$[F] [F^{-1}] = \mathrm{Id}_{\mathcal{G}^2}$	and similarly $[F] [F^{-1}] = \mathrm{Id}_{\mathcal{G}^2}$, so $[F^{-1}]$ is the inverse of $[F]$.
1205.5935v1_F00866	72	■ 1.000	$\backslash\mathrm{left}[F^{\{-1\}}\backslash\mathrm{right}]$	$[F^{-1}]$	and similarly $[F] [F^{-1}] = \mathrm{Id}_{\mathcal{G}^2}$, so $[F^{-1}]$ is the inverse of $[F]$.
1205.5935v1_F00867	73	■ 1.000	$\backslash\mathrm{overline}\{[F]\}=[\backslash\mathrm{bar}\{F\}]$	$[\overline{F}] = [\bar{F}]$	<i>Therefore $[\overline{F}] = [\bar{F}]$.</i>
1205.5935v1_F00868	73	■ 0.999	$[\backslash\mathrm{bar}\{F\}]$	$[\bar{F}]$	as long as (a) you change $[F]$ to $[\bar{F}]$ and (b) you then act on the whole thing with $[F]$.
1205.5935v1_F00869	73	■ 1.000	$\backslash\mathrm{lambda}[\backslash\mathrm{bar}\{F\}]\backslash\mathrm{left}(B_{\{s\}}\backslash\mathrm{right})$	$\lambda[\bar{F}](B_s)$	sides vanish identically, so let $r \leq s$. If $r = 0$, then $A_r = \lambda$ and both sides reduce to $\lambda[\bar{F}](B_s)$.
1205.5935v1_F00870	73	■ 1.000	$s\ \backslash\mathrm{geq}\ 1$	$s \geq 1$	$\mathbf{A}_r$ and $\mathbf{B}_s$. Next I'll prove $r = 1$ and any $s \geq 1$. It's true for $s = 1$, so assume it's true for $s - 1$,
1205.5935v1_F00871	73	■ 1.000	$s-1$	$s - 1$	$\mathbf{A}_r$ and $\mathbf{B}_s$. Next I'll prove $r = 1$ and any $s \geq 1$. It's true for $s = 1$, so assume it's true for $s - 1$,
1205.5935v1_F00872	73	■ 1.000	$\backslash\mathrm{boldsymbol}\{B\}_{\backslash\mathrm{boldsymbol}\{s\}}=b\ \backslash\mathrm{wedge}\ \backslash\mathrm{boldsymbol}\{B\}_{\backslash\mathrm{boldsymbol}\{s\}-\backslash\mathrm{mathbf}\{1\}}$	$\mathbf{B}_s = b \wedge \mathbf{B}_{s-1}$	let $\mathbf{B}_s = b \wedge \mathbf{B}_{s-1}$, and consider
1205.5935v1_F00873	73	■ 1.000	$\backslash\mathrm{boldsymbol}\{A\}_{\backslash\mathrm{boldsymbol}\{r\}}=\backslash\mathrm{boldsymbol}\{A\}_{\backslash\mathrm{boldsymbol}\{r\}-\backslash\mathrm{mathbf}\{1\}}\ \backslash\mathrm{wedge}\ a$	$\mathbf{A}_r = \mathbf{A}_{r-1} \wedge a$	Now for general $r \leq s$. Fix s and assume the result is true for $r - 1$; then let $\mathbf{A}_r = \mathbf{A}_{r-1} \wedge a$
1205.5935v1_F00874	74	■ 1.000	$\backslash\mathrm{mathcal}\{F\}=[F]$	$\mathcal{F} = [F]$	Thus $[\overline{F}]$ satisfies Eq. (338) with $\mathcal{F} = [F]$, so $[\overline{F}] = [\bar{F}]$.
1205.5935v1_F00875	74	■ 0.977	$A\backslash\mathrm{rfloor}\ F(B)=$	$A \rfloor F(B) =$	or the resulting outermorphism. That certainly makes Theorem 55 easier to read: I'll take $A \rfloor F(B) =$
1205.5935v1_F00876	74	■ 0.709	$F(\backslash\mathrm{bar}\{F\}(A)\backslash\mathrm{rfloor}\ B)$	$F(\bar{F}(A) \rfloor B)$	$F(\bar{F}(A) \rfloor B)$ any day.
1205.5935v1_F00877	74	■ 1.000	$F_{\backslash\mathrm{boldsymbol}\{A\}_{\backslash\mathrm{boldsymbol}\{r\}}\}$	$F_{\mathbf{A}_r}$	Recall that the restriction of F to the subspace represented by $\mathbf{A}_r$ is denoted $F_{\mathbf{A}_r}$. This restriction and
1205.5935v1_F00878	74	■ 1.000	$F\backslash\mathrm{left}(\backslash\mathrm{boldsymbol}\{A\}_{\backslash\mathrm{boldsymbol}\{r\}}\backslash\mathrm{right})$	$F(\mathbf{A}_r)$	$F(\mathbf{A}_r)$ are related in an important way.
1205.5935v1_F00879	74	■ 1.000	$F\backslash\mathrm{left}(\backslash\mathrm{boldsymbol}\{A\}_{\backslash\mathrm{boldsymbol}\{r\}}\backslash\mathrm{right})=0$	$F(\mathbf{A}_r) = 0$	Theorem 56. $F(\mathbf{A}_r) = 0$ iff $\mathrm{rank}(F_{\mathbf{A}_r}) < r$, which is true iff $F_{\mathbf{A}_r}$ is not one-to-one.
1205.5935v1_F00880	74	■ 1.000	$\backslash\mathrm{operatorname}\{rank\}\backslash\mathrm{left}(F_{\backslash\mathrm{boldsymbol}\{A\}_{\backslash\mathrm{boldsymbol}\{r\}}\}\backslash\mathrm{right})<r$	$\mathrm{rank}(F_{\mathbf{A}_r}) < r$	Theorem 56. $F(\mathbf{A}_r) = 0$ iff $\mathrm{rank}(F_{\mathbf{A}_r}) < r$, which is true iff $F_{\mathbf{A}_r}$ is not one-to-one.
1205.5935v1_F00881	74	■ 1.000	$\backslash\mathrm{operatorname}\{null\}\backslash\mathrm{left}(F_{\backslash\mathrm{boldsymbol}\{A\}_{\backslash\mathrm{boldsymbol}\{r\}}\}\backslash\mathrm{right})>0$	$\mathrm{null}(F_{\mathbf{A}_r}) > 0$	$\mathrm{null}(F_{\mathbf{A}_r}) > 0$, so F is not one-to-one.
1205.5935v1_F00882	74	■ 0.992	$\backslash\mathrm{boldsymbol}\{A\}_{\backslash\mathrm{boldsymbol}\{r\}}\ \backslash\mathrm{in}\ \backslash\mathrm{operatorname}\{Ker\}\ F$	$\mathbf{A}_r \in \mathrm{Ker}\ F$	So $\mathbf{A}_r \in \mathrm{Ker}\ F$ iff $\mathrm{Ker}(F_{\mathbf{A}_r}) \neq \{0\}$. Note the two different meanings of F in these two statements.
1205.5935v1_F00883	74	■ 0.992	$\backslash\mathrm{operatorname}\{Ker\}\backslash\mathrm{left}(F_{\backslash\mathrm{boldsymbol}\{A\}_{\backslash\mathrm{boldsymbol}\{r\}}\}\backslash\mathrm{right})\ \backslash\mathrm{neq}\ \{0\}$	$\mathrm{Ker}(F_{\mathbf{A}_r}) \neq \{0\}$	So $\mathbf{A}_r \in \mathrm{Ker}\ F$ iff $\mathrm{Ker}(F_{\mathbf{A}_r}) \neq \{0\}$. Note the two different meanings of F in these two statements.
1205.5935v1_F00884	74	■ 1.000	$F(A\ B)=F(A)\ F(B)$	$F(AB) = F(A)F(B)$	2. $F(AB) = F(A)F(B)$ for all A and B .
1205.5935v1_F00885	74	■ 0.960	$F(A\backslash\mathrm{rfloor}\ B)=F(A)\backslash\mathrm{rfloor}\ F(B)$	$F(A \rfloor B) = F(A) \rfloor F(B)$	3. $F(A \rfloor B) = F(A) \rfloor F(B)$ and $F(A \rfloor B) = F(A) \rfloor F(B)$ for all A and B .

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1205.5935v1_F00886	74	■ 0.960	$F(A\lfloor B)=F(A)\lfloor F(B)$	$F(A\lfloor B)=F(A)\lfloor F(B)$	3. $F(A\rfloor B)=F(A)\rfloor F(B)$ and $F(A\rfloor B)=F(A)\rfloor F(B)$ for all A and B .
1205.5935v1_F00887	74	■ 0.971	$F(a)\rfloor F(b)=a\downharpoonleft b$	$F(a)\rfloor F(b)=a\rfloor b$	<i>Proof.</i> I first assume F is an isometry. That means $F(a)\rfloor F(b)=a\rfloor b$ for all vectors; but
1205.5935v1_F00888	74	■ 0.180	$a\downharpoonleft b$	$a\rfloor b$	$a\rfloor b$ is a scalar, so $F(a\rfloor b)=a\rfloor b$, so $F(a\rfloor b)=F(a)\rfloor F(b)$. Since $F(a\wedge b)=F(a)\wedge F(b)$ by the
1205.5935v1_F00889	74	■ 0.180	$F(a\downarrow b)=a\rfloor b$	$F(a\downarrow b)=a\rfloor b$	$a\rfloor b$ is a scalar, so $F(a\rfloor b)=a\rfloor b$, so $F(a\rfloor b)=F(a)\rfloor F(b)$. Since $F(a\wedge b)=F(a)\wedge F(b)$ by the
1205.5935v1_F00890	74	■ 0.180	$F(a\rfloor b)=F(a)\downharpoonleft F(b)$	$F(a\rfloor b)=F(a)\rfloor F(b)$	$a\rfloor b$ is a scalar, so $F(a\rfloor b)=a\rfloor b$, so $F(a\rfloor b)=F(a)\rfloor F(b)$. Since $F(a\wedge b)=F(a)\wedge F(b)$ by the
1205.5935v1_F00891	74	■ 0.180	$F(a\wedge b)=F(a)\wedge F(b)$	$F(a\wedge b)=F(a)\wedge F(b)$	$a\rfloor b$ is a scalar, so $F(a\rfloor b)=a\rfloor b$, so $F(a\rfloor b)=F(a)\rfloor F(b)$. Since $F(a\wedge b)=F(a)\wedge F(b)$ by the
1205.5935v1_F00892	74	■ 1.000	$F(a\ b)=F(a)\ F(b)$	$F(ab)=F(a)F(b)$	outermorphism property, it follows that $F(ab)=F(a)F(b)$. I can extend this to $F(a_1a_2\cdots a_r)=$
1205.5935v1_F00893	74	■ 1.000	$F\left(a_{\{1}\}a_{\{2\}}\cdots a_{\{r\}}\right)=$	$F\left(a_1a_2\cdots a_r\right)=$	outermorphism property, it follows that $F(ab)=F(a)F(b)$. I can extend this to $F(a_1a_2\cdots a_r)=$
1205.5935v1_F00894	74	■ 1.000	$F\left(a_{\{1\}}\right)F\left(a_{\{2\}}\right)\cdots F\left(a_{\{r\}}\right)$	$F\left(a_1\right)F\left(a_2\right)\cdots F\left(a_r\right)$	$F(a_1)F(a_2)\cdots F(a_r)$ by induction: the result is true for $r=2$, so assume it’s true for $r-1$ and
1205.5935v1_F00895	75	■ 1.000	$F(A\ B)$	$F(AB)$	Now let’s look at $F(AB)$; by linearity it’s a sum of terms of the form $F(\boldsymbol{A}_r\boldsymbol{B}_s)$. If $r=0$ or
1205.5935v1_F00896	75	■ 1.000	$F\left(\boldsymbol{A}_{\{\boldsymbol{r}\}}\boldsymbol{B}_{\{\boldsymbol{s}\}}\right)$	$F\left(\boldsymbol{A}_r\boldsymbol{B}_s\right)$	Now let’s look at $F(AB)$; by linearity it’s a sum of terms of the form $F(\boldsymbol{A}_r\boldsymbol{B}_s)$. If $r=0$ or
1205.5935v1_F00897	75	■ 0.998	$F\left(\boldsymbol{A}_{\{\boldsymbol{r}\}}\boldsymbol{B}_{\{\boldsymbol{s}\}}\right)=F\left(\boldsymbol{A}_{\{\boldsymbol{r}\}}\boldsymbol{B}_{\{\boldsymbol{s}\}}\right)$	$F\left(\boldsymbol{A}_r\boldsymbol{B}_s\right)=F\left(\boldsymbol{A}_r\right)F\left(\boldsymbol{B}_s\right)$	$s=0$ then $F(\boldsymbol{A}_r\boldsymbol{B}_s)=F(\boldsymbol{A}_r)F(\boldsymbol{B}_s)$ by linearity and Theorem 52, so let $\boldsymbol{A}_r=a_1a_2\cdots a_r$ and
1205.5935v1_F00898	75	■ 1.000	$\boldsymbol{B}_{\{\boldsymbol{s}\}}=b_{\{1\}}b_{\{2\}}\cdots b_{\{s\}}$	$\boldsymbol{B}_s=b_1b_2\cdots b_s$	$\boldsymbol{B}_s=b_1b_2\cdots b_s$; then Eq. (348) lets me show
1205.5935v1_F00899	75	■ 1.000	$a_{\{r\}}$	a_r	and the same is true of products among the b_j ; it is the product of a_r and b_1 that does not equal an
1205.5935v1_F00900	75	■ 0.450	$F(A\rfloor B)=F(A)\downharpoonleft F(B)$	$F(A\rfloor B)=F(A)\rfloor F(B)$	Finally, assume $F(A\rfloor B)=F(A)\rfloor F(B)$ for all A and B ; then $F(a)\rfloor F(b)=F(a\rfloor b)=a\rfloor b$
1205.5935v1_F00901	75	■ 0.450	$F(a)\downharpoonleft F(b)=F(a\downarrow b)=a\rfloor b$	$F(a)\rfloor F(b)=F(a\downarrow b)=a\rfloor b$	Finally, assume $F(A\rfloor B)=F(A)\rfloor F(B)$ for all A and B ; then $F(a)\rfloor F(b)=F(a\rfloor b)=a\rfloor b$
1205.5935v1_F00902	75	■ 0.998	$F(a)=\lambda a$	$F(a)=\lambda a$	Vector a is an <i>eigenvector</i> of F with eigenvalue λ if $F(a)=\lambda a$, or equivalently $a\in\text{Ker}(F-\lambda\text{Id})$. In this
1205.5935v1_F00903	75	■ 0.998	$a\in\text{Ker}(F-\lambda\text{Id})$	$a\in\text{Ker}(F-\lambda\text{Id})$	Vector a is an <i>eigenvector</i> of F with eigenvalue λ if $F(a)=\lambda a$, or equivalently $a\in\text{Ker}(F-\lambda\text{Id})$. In this
1205.5935v1_F00904	75	■ 1.000	$\lambda_{\{1\}}$	λ_1	definition a can’t be zero, but λ can. Now suppose a_1 and a_2 are eigenvectors with eigenvalues λ_1 and λ_2
1205.5935v1_F00905	75	■ 1.000	$\lambda_{\{2\}}$	λ_2	definition a can’t be zero, but λ can. Now suppose a_1 and a_2 are eigenvectors with eigenvalues λ_1 and λ_2
1205.5935v1_F00906	75	■ 1.000	$\boldsymbol{A}=a_1\wedge a_2$	$\boldsymbol{A}=a_1\wedge a_2$	and let $\boldsymbol{A}=a_1\wedge a_2$; then
1205.5935v1_F00907	75	■ 1.000	$F(\boldsymbol{A})=\lambda\boldsymbol{A}$	$F(\boldsymbol{A})=\lambda\boldsymbol{A}$	An eigenblade is defined generally by $F(\boldsymbol{A})=\lambda\boldsymbol{A}$, regardless of what the factors of $\boldsymbol{A}$ do. Rotation
1205.5935v1_F00908	75	■ 0.999	$F(A)=\exp(-\boldsymbol{B}\theta/2)A\exp(\boldsymbol{B}\theta/2)$	$F(A)=\exp(-\boldsymbol{B}\theta/2)A\exp(\boldsymbol{B}\theta/2)$	eigenvectors. Let $F(A)=\exp(-\boldsymbol{B}\theta/2)A\exp(\boldsymbol{B}\theta/2)$, so F is a rotation in plane $\boldsymbol{B}$ through angle θ . Then
1205.5935v1_F00909	76	■ 0.999	$\boldsymbol{B}^\perp$	$\boldsymbol{B}^\perp$	reflection: every vector in the plane is an eigenvector with eigenvalue -1 .) You can check that $\boldsymbol{B}^\perp$ is also
1205.5935v1_F00910	76	■ 1.000	$F(\boldsymbol{A})=\boldsymbol{A}$	$F(\boldsymbol{A})=\boldsymbol{A}$	rotation. So $F(\boldsymbol{A})=\boldsymbol{A}$ is consistent with $F(a)=a$ for every vector in $\boldsymbol{A}$, but it’s consistent with many
1205.5935v1_F00911	76	■ 1.000	$F(a)=a$	$F(a)=a$	rotation. So $F(\boldsymbol{A})=\boldsymbol{A}$ is consistent with $F(a)=a$ for every vector in $\boldsymbol{A}$, but it’s consistent with many
1205.5935v1_F00912	76	■ 1.000	$\text{Range}(F_{\boldsymbol{A}})$	$\text{Range}(F_{\boldsymbol{A}})$	But it also hints at what $\text{Range}(F_{\boldsymbol{A}})$ is. In fact, it’s hard to see how F could map $\boldsymbol{A}$ to a multiple of itself
1205.5935v1_F00913	76	■ 1.000	$\boldsymbol{A}\wedge a=0$	$\boldsymbol{A}\wedge a=0$	prove the first part. If a lies in $\boldsymbol{A}$, then $\boldsymbol{A}\wedge a=0$, in which case
1205.5935v1_F00914	76	■ 1.000	$ F(\boldsymbol{A}) ^2=\lambda^2 \boldsymbol{A} ^2$	$ F(\boldsymbol{A}) ^2=\lambda^2 \boldsymbol{A} ^2$	Well, if $F(\boldsymbol{A})=\lambda\boldsymbol{A}$ then $ F(\boldsymbol{A}) ^2=\lambda^2 \boldsymbol{A} ^2$, so the value of λ determines how much the norm squared of $\boldsymbol{A}$
1205.5935v1_F00915	76	■ 1.000	$ \lambda $	$ \lambda $	so F multiplies the weight associated with $\boldsymbol{A}$ by $ \lambda $; and F changes the orientation of $\boldsymbol{A}$ if λ is negative.
1205.5935v1_F00916	76	■ 1.000	$F(\boldsymbol{I})$	$F(\boldsymbol{I})$	space it acts on. Finding it in geometric algebra is easy; we consider $F(\boldsymbol{I})$. This is an n -blade, so it has to
1205.5935v1_F00917	76	■ 1.000	$\det(\text{Id})=1$	$\det(\text{Id})=1$	First, it’s obvious that $\det(\text{Id})=1$. Now let F and G be linear transformations; since
1205.5935v1_F00919	77	■ 1.000	$\det(F)\neq 0$	$\det(F)\neq 0$	That tells me that $\det(F)\neq 0$ if F is invertible. I’ll use that later.
1205.5935v1_F00920	77	■ 1.000	$a_1\wedge\cdots\wedge a_n$	$a_1\wedge\cdots\wedge a_n$	$a_1\wedge\cdots\wedge a_n$ is a volume element, so
1205.5935v1_F00921	77	■ 1.000	$F(v)=A_rv^{*r}A_r^{-1}$	$F(v)=A_rv^{*r}A_r^{-1}$	• If F is orthogonal, then $F(v)=A_rv^{*r}A_r^{-1}$ for some rotor A_r , so Eq. (318) shows that
1205.5935v1_F00923	78	■ 1.000	$\{ \}^\perp$	$^\perp$	Recall that $^\perp$ is the duality transform and $^{-\perp}$ is its inverse.
1205.5935v1_F00924	78	■ 1.000	$\{ \}^{-\perp}$	$^{-\perp}$	Recall that $^\perp$ is the duality transform and $^{-\perp}$ is its inverse.
1205.5935v1_F00925	78	■ 0.987	$G(A)=\bar{F}(A^{-\perp})^\perp$	$G(A)=\bar{F}(A^{-\perp})^\perp$	Suppose $\det(F)\neq 0$ and let G be defined by $G(A)=\bar{F}(A^{-\perp})^\perp$. Then I use the result $A\rfloor\bar{F}(B)=$
1205.5935v1_F00926	78	■ 0.987	$\left.A\right\rfloor\bar{F}(B)=$	$A\rfloor\bar{F}(B)=$	Suppose $\det(F)\neq 0$ and let G be defined by $G(A)=\bar{F}(A^{-\perp})^\perp$. Then I use the result $A\rfloor\bar{F}(B)=$
1205.5935v1_F00927	78	■ 0.879	$\bar{F}(F(A)\rfloor B)$	$\bar{F}(F(A)\rfloor B)$	$\bar{F}(F(A)\rfloor B)$ from Theorem 55 to get
1205.5935v1_F00928	78	■ 1.000	$FG(A)=\det(F)A$	$FG(A)=\det(F)A$	A similar argument shows $FG(A)=\det(F)A$, so $F^{-1}=G/\det(F)$.
1205.5935v1_F00929	78	■ 1.000	$F^{-1}=G/\det(F)$	$F^{-1}=G/\det(F)$	A similar argument shows $FG(A)=\det(F)A$, so $F^{-1}=G/\det(F)$.
1205.5935v1_F00930	78	■ 1.000	$F^{-1}(a)$	$F^{-1}(a)$	calculating $F^{-1}(a)$ involves calculating $\bar{F}(a^{-\perp})$, and $a^{-\perp}$ is not a vector. (Unless we’re in a two-dimensional
1205.5935v1_F00931	78	■ 1.000	$\bar{F}(a^{-\perp})$	$\bar{F}(a^{-\perp})$	calculating $F^{-1}(a)$ involves calculating $\bar{F}(a^{-\perp})$, and $a^{-\perp}$ is not a vector. (Unless we’re in a two-dimensional
1205.5935v1_F00932	78	■ 1.000	$a^{-\perp}$	$a^{-\perp}$	calculating $F^{-1}(a)$ involves calculating $\bar{F}(a^{-\perp})$, and $a^{-\perp}$ is not a vector. (Unless we’re in a two-dimensional
1205.5935v1_F00933	78	■ 1.000	$\boldsymbol{a},\boldsymbol{b}$	$\boldsymbol{a},\boldsymbol{b}$	denoted $\boldsymbol{a},\boldsymbol{b}$, and so on, the magnitude of vector $\boldsymbol{a}$ is denoted a , unit vectors are indicated with an overhat
1205.5935v1_F00934	78	■ 1.000	$\boldsymbol{a}$	$\boldsymbol{a}$	denoted $\boldsymbol{a},\boldsymbol{b}$, and so on, the magnitude of vector $\boldsymbol{a}$ is denoted a , unit vectors are indicated with an overhat
1205.5935v1_F00935	78	■ 0.849	$\dot{}$	<i>(not rendered)</i>	$\dot{}$, and the derivative of any quantity with respect to time is indicated by an overdot $\dot{}$. Since boldface means
1205.5935v1_F00936	78	■ 1.000	$\boldsymbol{r}$	$\boldsymbol{r}$	As a particle moves over time, its position vector $\boldsymbol{r}$ sweeps out area at a rate that a picture easily shows to
1205.5935v1_F00937	78	■ 1.000	$\dot{\boldsymbol{r}}$	$\dot{\boldsymbol{r}}$	where $\boldsymbol{v}=\dot{\boldsymbol{r}}$ is the particle’s velocity vector. Unsurprisingly, the rate at which area is swept out is a 2-blade.

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1205.5935v1_F00938	78	■ 0.997	$\backslash\boldsymbol{L}=\backslash\boldsymbol{r} \times \boldsymbol{p}$	$\boldsymbol{L} = \boldsymbol{r} \times \boldsymbol{p}$	$\boldsymbol{L} = \boldsymbol{r} \times \boldsymbol{p}$, the cross product of $\boldsymbol{r}$ and $\boldsymbol{p}$; the definition given here is the dual of that vector (see Eq. (208)),
1205.5935v1_F00939	78	■ 0.997	$\backslash\boldsymbol{p}$	$\boldsymbol{p}$	$\boldsymbol{L} = \boldsymbol{r} \times \boldsymbol{p}$, the cross product of $\boldsymbol{r}$ and $\boldsymbol{p}$; the definition given here is the dual of that vector (see Eq. (208)),
1205.5935v1_F00940	79	■ 0.880	L	L	so L is conserved iff the force $\boldsymbol{F}$ is central (parallel or antiparallel to $\boldsymbol{r}$). Since L is conserved iff $\dot{A} = 0$, I
1205.5935v1_F00941	79	■ 0.880	$\backslash\boldsymbol{F}$	$\boldsymbol{F}$	so L is conserved iff the force $\boldsymbol{F}$ is central (parallel or antiparallel to $\boldsymbol{r}$). Since L is conserved iff $\dot{A} = 0$, I
1205.5935v1_F00942	79	■ 0.880	$\dot{A}=0, \mathrm{\mathit{I}}$	$\dot{A} = 0, \mathrm{I}$	so L is conserved iff the force $\boldsymbol{F}$ is central (parallel or antiparallel to $\boldsymbol{r}$). Since L is conserved iff $\dot{A} = 0$, I
1205.5935v1_F00943	79	■ 0.999	$\backslash\boldsymbol{r}=r \hat{\backslash\boldsymbol{r}}$	$\boldsymbol{r} = r\hat{\boldsymbol{r}}$	in equal times. Further, the plane in which central force motion takes place is L itself. Writing $\boldsymbol{r} = r\hat{\boldsymbol{r}}$,
1205.5935v1_F00944	79	■ 0.936	$\backslash\boldsymbol{v}=\dot{\backslash\boldsymbol{r}} \hat{\backslash\boldsymbol{r}}+r \dot{\backslash\hat{\boldsymbol{r}}}$	$\boldsymbol{v} = \dot{r}\hat{\boldsymbol{r}} + r\dot{\hat{\boldsymbol{r}}}$	which implies $\boldsymbol{v} = \dot{r}\hat{\boldsymbol{r}} + r\dot{\hat{\boldsymbol{r}}}$, I find
1205.5935v1_F00945	79	■ 1.000	$\hat{\backslash\boldsymbol{r}} \wedge \hat{\backslash\boldsymbol{r}}=0$	$\hat{\boldsymbol{r}} \wedge \hat{\boldsymbol{r}} = 0$	since $\hat{\boldsymbol{r}} \wedge \hat{\boldsymbol{r}} = 0$. But I know a bit more than that; $\hat{\boldsymbol{r}}$ is a unit vector, or $\hat{\boldsymbol{r}} \lrcorner \hat{\boldsymbol{r}} = 1$, the time derivative of which
1205.5935v1_F00946	79	■ 1.000	$\hat{\backslash\boldsymbol{r}}$	$\hat{\boldsymbol{r}}$	since $\hat{\boldsymbol{r}} \wedge \hat{\boldsymbol{r}} = 0$. But I know a bit more than that; $\hat{\boldsymbol{r}}$ is a unit vector, or $\hat{\boldsymbol{r}} \lrcorner \hat{\boldsymbol{r}} = 1$, the time derivative of which
1205.5935v1_F00947	79	■ 1.000	$\hat{\backslash\boldsymbol{r}}\rfloor \hat{\backslash\boldsymbol{r}}=1$	$\hat{\boldsymbol{r}} \lrcorner \hat{\boldsymbol{r}} = 1$	since $\hat{\boldsymbol{r}} \wedge \hat{\boldsymbol{r}} = 0$. But I know a bit more than that; $\hat{\boldsymbol{r}}$ is a unit vector, or $\hat{\boldsymbol{r}} \lrcorner \hat{\boldsymbol{r}} = 1$, the time derivative of which
1205.5935v1_F00948	79	■ 1.000	$\hat{\backslash\boldsymbol{r}}\rfloor \dot{\backslash\hat{\boldsymbol{r}}}=0$	$\hat{\boldsymbol{r}} \lrcorner \dot{\hat{\boldsymbol{r}}} = 0$	is $\hat{\boldsymbol{r}} \lrcorner \dot{\hat{\boldsymbol{r}}} = 0$. This is just the familiar fact that a constant-length vector and its time derivative must always
1205.5935v1_F00949	79	■ 0.782	$\backslash\boldsymbol{v}=\dot{\backslash\boldsymbol{r}} \hat{\backslash\boldsymbol{r}}+r \dot{\backslash\hat{\boldsymbol{r}}}$	$\boldsymbol{v} = \dot{r}\hat{\boldsymbol{r}} + r\dot{\hat{\boldsymbol{r}}}$	be perpendicular. (Incidentally, this shows that $\boldsymbol{v} = \dot{r}\hat{\boldsymbol{r}} + r\dot{\hat{\boldsymbol{r}}}$ is a decomposition into radial and tangential
1205.5935v1_F00950	79	■ 0.906	$\hat{\backslash\boldsymbol{r}} \wedge \dot{\backslash\hat{\boldsymbol{r}}}= \hat{\backslash\boldsymbol{r}} \dot{\backslash\hat{\boldsymbol{r}}}$	$\hat{\boldsymbol{r}} \wedge \dot{\hat{\boldsymbol{r}}} = \hat{\boldsymbol{r}}\dot{\hat{\boldsymbol{r}}}$	components.) If $\hat{\boldsymbol{r}} \lrcorner \dot{\hat{\boldsymbol{r}}} = 0$, then $\hat{\boldsymbol{r}} \wedge \dot{\hat{\boldsymbol{r}}} = \hat{\boldsymbol{r}}\dot{\hat{\boldsymbol{r}}}$, so
1205.5935v1_F00951	79	■ 1.000	l	l	so the scalar l , the magnitude of L , is given by
1205.5935v1_F00953	79	■ 1.000	$V=-k / r$	$V = -k/r$	form $V = -k/r$, where r is the particle's distance from some fixed origin. The particle experiences a force
1205.5935v1_F00954	79	■ 1.000	r^2	r^2	Now take a look at Eqs. (373) and (377). One is proportional to r^2 while the other is inversely proportional
1205.5935v1_F00955	79	■ 1.000	$k \hat{\backslash\boldsymbol{r}}$	$k\hat{\boldsymbol{r}}$	Well, look at that: another constant of motion. The second term in the constant, $k\hat{\boldsymbol{r}}$, is clearly a vector, and
1205.5935v1_F00957	80	■ 0.934	$\backslash\boldsymbol{v}$	$\boldsymbol{v}$	This is hardly a surprise; $\boldsymbol{v}$ is a vector in the plane defined by L , so by Theorems 16 and 17, $L\boldsymbol{v} = L \lrcorner \boldsymbol{v}$ is
1205.5935v1_F00958	80	■ 0.934	$L \backslash\boldsymbol{v}=L\lfloor\boldsymbol{v}$	$L\boldsymbol{v} = L \lrcorner \boldsymbol{v}$	This is hardly a surprise; $\boldsymbol{v}$ is a vector in the plane defined by L , so by Theorems 16 and 17, $L\boldsymbol{v} = L \lrcorner \boldsymbol{v}$ is
1205.5935v1_F00959	80	■ 1.000	$\backslash\boldsymbol{e}$	$\boldsymbol{e}$	of k , so I define a conserved vector $\boldsymbol{e}$ by
1205.5935v1_F00960	80	■ 1.000	$\hat{\backslash\boldsymbol{r}} \backslash\boldsymbol{r}=r$	$\hat{\boldsymbol{r}}\boldsymbol{r} = r$	equation, r as a function of direction. Since the expression for $\boldsymbol{e}$ has $\hat{\boldsymbol{r}}$ in it and $\hat{\boldsymbol{r}}\boldsymbol{r} = r$, it follows that I can
1205.5935v1_F00961	80	■ 1.000	$k \backslash\boldsymbol{r}$	$k\boldsymbol{r}$	get an equation for r by multiplying Eq. (380) by $k\boldsymbol{r}$, with the result
1205.5935v1_F00962	80	■ 0.965	$\backslash\boldsymbol{e} \boldsymbol{r}=e \boldsymbol{r} \cos \theta+\backslash\boldsymbol{e} \wedge \boldsymbol{r}$	$\boldsymbol{e}\boldsymbol{r} = er \cos \theta + \boldsymbol{e} \wedge \boldsymbol{r}$	while $\hat{\boldsymbol{r}}\boldsymbol{r} = r$ and $\boldsymbol{e}\boldsymbol{r} = er \cos \theta + \boldsymbol{e} \wedge \boldsymbol{r}$, so putting it all together
1205.5935v1_F00963	80	■ 0.999	e	e	which is the equation for a conic section with eccentricity e and one focus at the origin. Since the length of
1205.5935v1_F00964	80	■ 1.000	$k>0$	$k > 0$	so I'll do one at a time. First I assume $k > 0$ and I note that r equals its minimum and maximum values
1205.5935v1_F00965	80	■ 1.000	$\theta=0$	$\theta = 0$	when $\theta = 0$ and π respectively, which means that $\boldsymbol{e}$ points toward the particle's point of closest approach,
1205.5935v1_F00966	80	■ 1.000	$k<0$	$k < 0$	Now the repulsive case. If $k < 0$, we run into a problem: r has to be non-negative, so we have to have
1205.5935v1_F00967	80	■ 1.000	$1+e \cos \theta \leq 0$	$1 + e \cos \theta \leq 0$	$1 + e \cos \theta \leq 0$ for at least some values of θ , and the orbit may include only those values. This is possible iff
1205.5935v1_F00968	80	■ 1.000	$e>1$	$e > 1$	$e > 1$, with the result that the orbit is a hyperbola. In this case, r takes on its smallest value when $\theta = \pi$,
1205.5935v1_F00969	81	■ 1.000	$-2 k$	$-2k$	The second term on the left is $-2k$ times
1205.5935v1_F00975	87	■ 1.000	A^{-1}	A^{-1}	(b) A^{-1} exists iff $ A ^2 \neq 0$, in which case $A^{-1} = A^\dagger/ A ^2$ and $ A^{-1} ^2 = A ^{-2}$.
1205.5935v1_F00990	83	■ 1.000	$\boldsymbol{a}_1 \wedge \cdots \wedge \boldsymbol{a}_r$	$\boldsymbol{a}_1 \wedge \cdots \wedge \boldsymbol{a}_r$	An r -blade $\boldsymbol{A}_r$ is the outer product of r vectors, $\boldsymbol{a}_1 \wedge \cdots \wedge \boldsymbol{a}_r$. It represents the subspace spanned by
1205.5935v1_F00991	83	■ 1.000	$\backslash\boldsymbol{A}_{\backslash\boldsymbol{r}}=0$	$\boldsymbol{A}_r = 0$	$\boldsymbol{A}_r = 0$ iff the $\boldsymbol{a}_j$ are linearly dependent.
1205.5935v1_F00992	83	■ 0.999	$\left(\boldsymbol{A}_{\backslash\boldsymbol{s}}=\boldsymbol{A}_{\backslash\boldsymbol{s}}\wedge \boldsymbol{A}_r\right)$	$(\boldsymbol{A}_s = \boldsymbol{A}_{s-r} \wedge \boldsymbol{A}_r)$	or the right $(\boldsymbol{A}_s = \boldsymbol{A}_{s-r} \wedge \boldsymbol{A}_r)$. The grade- $s - r$ factors in each case may be chosen to be the same except
1205.5935v1_F00993	83	■ 1.000	$\backslash\boldsymbol{A}_{\backslash\boldsymbol{r}} B^{* r} \backslash\boldsymbol{A}_{\backslash\boldsymbol{r}}^{-1}$	$\boldsymbol{A}_r B^{*r} \boldsymbol{A}_r^{-1}$	The reflection of multivector B in subspace $\boldsymbol{A}_r$ is $\boldsymbol{A}_r B^{*r} \boldsymbol{A}_r^{-1}$.
1205.5935v1_F00994	83	■ 1.000	$\backslash\boldsymbol{A}_{\backslash\boldsymbol{r}}, \backslash\boldsymbol{A}_r^*, \backslash\boldsymbol{A}_r^\dagger, \backslash\boldsymbol{A}_r^\ddagger$	$\boldsymbol{A}_r, \boldsymbol{A}_r^*, \boldsymbol{A}_r^\dagger, \boldsymbol{A}_r^\ddagger$	$\boldsymbol{A}_r, \boldsymbol{A}_r^*, \boldsymbol{A}_r^\dagger, \boldsymbol{A}_r^\ddagger$, and $\boldsymbol{A}_r^{-1}$ (if it exists) represent the same subspace.
1205.5935v1_F00995	84	■ 1.000	$\boldsymbol{A}=\boldsymbol{a}_1 \boldsymbol{a}_2 \cdots \boldsymbol{a}_r$	$A = a_1 a_2 \cdots a_r$	If $A = a_1 a_2 \cdots a_r$, then
1205.5935v1_F00996	85	■ 0.832	$\backslash\boldsymbol{A}_{\backslash\boldsymbol{r}} \backslash\boldsymbol{B}_{\backslash\boldsymbol{s}}=\backslash\boldsymbol{A}_{\backslash\boldsymbol{r}} \downarrow \boldsymbol{B}_s$	$\boldsymbol{A}_r \boldsymbol{B}_s = \boldsymbol{A}_r \lrcorner \boldsymbol{B}_s$	If $\boldsymbol{A}_r$ is a subspace of $\boldsymbol{B}_s$, then $\boldsymbol{A}_r \boldsymbol{B}_s = \boldsymbol{A}_r \lrcorner \boldsymbol{B}_s$.
1205.5935v1_F00997	87	■ 0.906	$\boldsymbol{a} \lrcorner \boldsymbol{A}$	$\boldsymbol{a} \lrcorner A$	2. $\langle A \rangle$ and $\boldsymbol{a} \lrcorner A$ for every vector $\boldsymbol{a}$.
1205.5935v1_F00998	87	■ 1.000	$ A ^2=A^{\dagger} A=\boldsymbol{a}_1^2 \boldsymbol{a}_2^2 \cdots \boldsymbol{a}_r^2$	$ A ^2 = A^\dagger A = a_1^2 a_2^2 \cdots a_r^2$	(a) $ A ^2 = A^\dagger A = a_1^2 a_2^2 \cdots a_r^2$.
1205.5935v1_F00999	87	■ 1.000	$A^{-1}=A^{\dagger} / A ^2$	$A^{-1} = A^\dagger/ A ^2$	(b) A^{-1} exists iff $ A ^2 \neq 0$, in which case $A^{-1} = A^\dagger/ A ^2$ and $ A^{-1} ^2 = A ^{-2}$.
1205.5935v1_F01000	87	■ 1.000	$\left A^{-1}\right ^2= A ^{-2}$	$ A^{-1} ^2 = A ^{-2}$	(b) A^{-1} exists iff $ A ^2 \neq 0$, in which case $A^{-1} = A^\dagger/ A ^2$ and $ A^{-1} ^2 = A ^{-2}$.

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1205.5935v1_F01001	87	■ 1.000	$(A\ B)\ *(A\ C)=(B\ A)\ *(C\ A)= A ^{\wedge 2}\ B\ * \ C$	$(AB)\ *(AC)=(BA)\ *(CA)= A ^2B\ * \ C$	(c) $(AB)\ *(AC)=(BA)\ *(CA)= A ^2\ B\ * \ C.$
1205.5935v1_F01002	87	■ 1.000	$\left \boldsymbol{A}_{\boldsymbol{r}}\right ^{\wedge 2}=0$	$\left \boldsymbol{A}_{\boldsymbol{r}}\right ^2=0$	(a) $\left \boldsymbol{A}_{\boldsymbol{r}}\right ^2=0$ iff the inner product is degenerate on $\boldsymbol{A}_{\boldsymbol{r}}$.
1205.5935v1_F01003	87	■ 1.000	$\left \boldsymbol{A}_{\boldsymbol{r}}\right ^{\wedge 2}\ \neq\ 0$	$\left \boldsymbol{A}_{\boldsymbol{r}}\right ^2\neq 0$	(b) $\boldsymbol{A}_{\boldsymbol{r}}^{-1}$ exists iff $\left \boldsymbol{A}_{\boldsymbol{r}}\right ^2\neq 0$, in which case $\boldsymbol{A}_{\boldsymbol{r}}^{-1}=(-1)^{r(r-1)/2}\boldsymbol{A}_{\boldsymbol{r}}/\left \boldsymbol{A}_{\boldsymbol{r}}\right ^2=\boldsymbol{A}_{\boldsymbol{r}}/\boldsymbol{A}_{\boldsymbol{r}}^2.$
1205.5935v1_F01004	87	■ 1.000	$\boldsymbol{A}_{\boldsymbol{r}}^{\wedge -1}=(-1)^{r(r-1)\ / \ 2}\boldsymbol{A}_{\boldsymbol{r}}^{\wedge 1}\ /\left \boldsymbol{A}_{\boldsymbol{r}}\right ^{\wedge 2}=\boldsymbol{A}_{\boldsymbol{r}}^{\wedge -1}\ /\ \boldsymbol{A}_{\boldsymbol{r}}^{\wedge 2}$	$\boldsymbol{A}_{\boldsymbol{r}}^{-1}=(-1)^{r(r-1)/2}\boldsymbol{A}_{\boldsymbol{r}}/\left \boldsymbol{A}_{\boldsymbol{r}}\right ^2=\boldsymbol{A}_{\boldsymbol{r}}/\boldsymbol{A}_{\boldsymbol{r}}^2$	(b) $\boldsymbol{A}_{\boldsymbol{r}}^{-1}$ exists iff $\left \boldsymbol{A}_{\boldsymbol{r}}\right ^2\neq 0$, in which case $\boldsymbol{A}_{\boldsymbol{r}}^{-1}=(-1)^{r(r-1)/2}\boldsymbol{A}_{\boldsymbol{r}}/\left \boldsymbol{A}_{\boldsymbol{r}}\right ^2=\boldsymbol{A}_{\boldsymbol{r}}/\boldsymbol{A}_{\boldsymbol{r}}^2.$
1205.5935v1_F01005	88	■ 1.000	$\boldsymbol{A}^{\perp}$	$\boldsymbol{A}^{\perp}$	$\boldsymbol{A}^{\perp}$ is the orthogonal complement of $\boldsymbol{A}$.
1205.5935v1_F01006	89	■ 1.000	$\boldsymbol{a}_{\{N\}}=\boldsymbol{a}_{\{1\}}\ \wedge\ \cdots\ \wedge\ \boldsymbol{a}_{\{n\}}$	$a_N=a_1\wedge\cdots\wedge a_n$	If $\{a_i\}_{i=1,\dots,n}$ is a frame with volume element $a_N=a_1\wedge\cdots\wedge a_n$, the reciprocal frame is given by
1205.5935v1_F01007	90	■ 0.808	$F(\boldsymbol{a})=\sum_i\ \lambda_i\ \left(\boldsymbol{a}\ *\ \boldsymbol{a}^i\right)\ \boldsymbol{a}_i$	$F(a)=\sum_i\lambda_i\left(a\ * \ a^i\right)a_i$	F is symmetric iff $F(a)=\sum_i\lambda_i(a\ * \ a^i)a_i$ for some frame $\{a_i\}$ of eigenvectors with eigenvalues $\{\lambda_i\}$.
1205.5935v1_F01008	90	■ 1.000	$\operatorname{iff}\ \overline{F}F=\mathrm{Id}$	iff $\overline{F}F=\mathrm{Id}$	F is an isometry iff $\overline{F}F=\mathrm{Id}$.
1205.5935v1_F01009	90	■ 1.000	$F(\boldsymbol{a})=\boldsymbol{A}_{\boldsymbol{r}}\ \boldsymbol{a}^{* \ \boldsymbol{r}}\ \boldsymbol{A}_{\boldsymbol{r}}^{\wedge -1}$	$F(a)=A_r a^{*r} A_r^{-1}$	F is an isometry on a finite-dimensional space iff $F(a)=A_r a^{*r} A_r^{-1}$ for some invertible r -versor A_r .
1205.5935v1_F01010	90	■ 1.000	$F(\boldsymbol{B})=\boldsymbol{A}_{\boldsymbol{r}}\ \boldsymbol{B}^{* \ \boldsymbol{r}}\ \boldsymbol{A}_{\boldsymbol{r}}^{\wedge -1}$	$F(B)=A_r B^{*r} A_r^{-1}$	The extension of orthogonal F on a finite-dimensional space to all multivectors is $F(B)=A_r B^{*r} A_r^{-1}$.
1205.5935v1_F01011	91	■ 1.000	$\boldsymbol{A}\ \neq\ 0$	$\boldsymbol{A}\neq 0$	$\boldsymbol{A}\neq 0$ is an eigenblade of F if $F(\boldsymbol{A})=\lambda\boldsymbol{A}$.
1205.5935v1_F01012	91	■ 0.997	$\operatorname{det}\left(F_{\boldsymbol{A}}\right)$	$\det\left(F_{\boldsymbol{A}}\right)$	An eigenblade of F is an invariant subspace on which F is invertible. The eigenvalue is $\det(F_{\boldsymbol{A}})$.